
Randomized Transport Maps for Model-Free Policy-Gradient Mean-Field Control

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Abstract

We develop a model-free policy gradient method for discrete-time mean-field control (MFC). In MFC, the policy affects the objective both through the controlled dynamics and through the population distribution. Standard REINFORCE estimators capture the first effect but not the second. We introduce Transport REINFORCE, a transport map-based approach that perturbs a suitable transformation of the population distribution to estimate this missing mean-field contribution. The method applies to both finite and continuous state spaces. In finite state spaces, we perturb the population distribution directly on the probability simplex through a convex combination of the current population weights and random weights. In continuous state spaces, we project the population distribution onto the manifold of Gaussian mixtures, and then randomize it via a transport map that ensures the perturbed law remains within this manifold. We prove consistency of the perturbed objective and gradient as the perturbation vanishes, and derive bias and mean-square error bounds for the resulting sample-based gradient estimator. Numerical experiments on several MFC benchmarks show that Transport REINFORCE improves over standard REINFORCE.

1 INTRODUCTION

We study model-free mean-field control (MFC) in discrete time over a finite horizon. A common random-

ized policy is optimized for a population of interacting agents. Each agent’s policy, dynamics, and rewards may depend on the population distribution, which itself evolves under the policy. The learner observes sampled transitions and rewards, without analytic access to the model.

Existing learning approaches include dynamic programming on the space of population distributions (Carmona et al., 2023; Gu et al., 2021, 2023). Even for N states, approximating a value function on the resulting $(N - 1)$ -dimensional simplex can be costly. Direct policy optimization offers another approach. Policy-gradient convergence has been established for discrete-time linear-quadratic MFC (Carmona et al., 2019), while actor-critic methods and their analysis have been developed in continuous time (Frikha et al., 2025; Pham and Warin, 2025; Frikha et al., 2024). Bayraktar et al. (2026) also derive a policy-gradient formula leading to an actor-critic scheme.

The difficulty is that applying standard REINFORCE (Williams, 1992) to a representative agent while treating the population flow as fixed misses part of the MFC gradient. Changing the policy also changes the population distribution, which enters the policy, dynamics, and rewards. The closest work, Meunier et al. (2026), addresses this additional dependence in finite state spaces. Their MF-REINFORCE method randomizes the logit coordinates of the population weights and uses the resulting density score to estimate the mean-field contribution.

We introduce Transport REINFORCE, which randomizes the population law through transport maps. The perturbation combines population and random weights in finite state spaces, and perturbs Gaussian-mixture weights, means, and covariances in continuous spaces. Its explicit density score accounts for population dependence in the perturbed gradient. The trans-

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port also couples the original and perturbed laws to control their Wasserstein distance (Villani et al., 2009; Delon and Desolneux, 2020).

Our contributions are threefold.

- We derive model-free policy-gradient estimators combining policy and population scores, using simulations at prescribed population arguments. Population sensitivities are estimated through a probabilistic representation in finite state spaces and centered policy differences in continuous state spaces.
- We prove objective perturbation bounds of order $O(\lambda)$ in finite state spaces and $O(\sqrt{\lambda})$ for a fixed Gaussian-mixture representation. In the continuous case, we separate the perturbation error from the mixture approximation error. In both cases, we establish gradient consistency and bounds on the estimator’s bias and mean-square error.
- Our experimental suite compares Transport REINFORCE with REINFORCE on seven discrete and continuous benchmarks, and with MF-REINFORCE on the finite-state benchmarks, under matched simulator budgets. We also develop an explicitly solvable linear-dynamics setting with Gaussian initial conditions.

2 BACKGROUND

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a standard probability space and $T < \infty$. We set $\mathcal{T} := \{0, \dots, T\}$. The state space $(\mathcal{X}, \mathcal{B}(\mathcal{X}))$ and action space $(\mathcal{A}, \mathcal{B}(\mathcal{A}))$ are assumed to be Polish. For $t \in \llbracket 0, T-1 \rrbracket$, let

$$\begin{aligned} P_t : \mathcal{X} \times \mathcal{P}(\mathcal{X}) \times \mathcal{A} &\longrightarrow \mathcal{P}(\mathcal{X}), \\ r_t : \mathcal{X} \times \mathcal{P}(\mathcal{X}) \times \mathcal{A} &\longrightarrow \mathbb{R}, \end{aligned}$$

be respectively measurable transition kernel and running reward, and let $g : \mathcal{X} \times \mathcal{P}(\mathcal{X}) \rightarrow \mathbb{R}$ be a measurable terminal reward. A randomized mean-field feedback policy at time t is a measurable transition kernel $\pi_t : \mathcal{X} \times \mathcal{P}(\mathcal{X}) \rightarrow \mathcal{P}(\mathcal{A})$. Rather than optimizing over all such kernels, we choose a parametric family $\{\pi^\theta : \theta \in \Theta \subset \mathbb{R}^{d_\theta}\}$, where Θ represents the compact admissible set of parameters. Whenever a policy score is used, we assume that π_t^θ admits a strictly positive density p_t^θ with respect to a fixed reference measure $\nu_A \in \mathcal{P}(\mathcal{A})$: $\pi_t^\theta(\mathrm{d}a \mid x, m) = p_t^\theta(a \mid x, m)\nu_A(\mathrm{d}a)$. Given a parametrized randomized mean-field feedback policy π^θ , an initial distribution $\mu_0 \in \mathcal{P}(\mathcal{X})$ and setting $\mu_t^\theta := \mathbb{P}_{X_t^\theta}$, we consider the state and control processes, $(X_t^\theta)_{t \in \mathcal{T}}$ and $(\alpha_t^\theta)_{t=0}^{T-1}$ respectively, driven by

$$\begin{aligned} X_0^\theta &\sim \mu_0, \quad \alpha_t^\theta \sim \pi_t^\theta(\cdot \mid X_t^\theta, \mu_t^\theta), \\ X_{t+1}^\theta &\sim P_t(\cdot \mid X_t^\theta, \mu_t^\theta, \alpha_t^\theta), \end{aligned} \quad (1)$$

The finite-horizon objective we seek to maximize is then defined as

$$J(\theta) := \mathbb{E} \left[\sum_{t=0}^{T-1} r_t(X_t^\theta, \mu_t^\theta, \alpha_t^\theta) + g(X_T^\theta, \mu_T^\theta) \right]. \quad (2)$$

Appendix A establishes well-posedness of (1)-(2) under the stated assumptions.

Policy-gradient methods maximize objective (2) by using a probabilistic representation of $\nabla_\theta J(\theta)$, using a trajectory-based estimator of $\widehat{G}(\theta)$ and updating $\theta \leftarrow \theta + \gamma \widehat{G}(\theta)$ for $\gamma > 0$. To see the difficulty created by the population law, let

$$\begin{aligned} \mathcal{R}_\theta &:= \sum_{t=0}^{T-1} r_t(X_t^\theta, \mu_t^\theta, \alpha_t^\theta) + g(X_T^\theta, \mu_T^\theta), \\ \mathcal{S}_{\text{pol}}^\theta &:= \sum_{t=0}^{T-1} \nabla_\theta \log p_t^\theta(\alpha_t^\theta \mid X_t^\theta, \mu_t^\theta). \end{aligned}$$

Throughout, the policy score differentiates $\theta \mapsto \log p_t^\theta(a \mid x, m)$ with (a, x, m) held fixed, and is then evaluated at the sampled action, state and population argument. The usual REINFORCE term is $\text{RF}(\theta) = \mathbb{E}[\mathcal{R}_\theta \mathcal{S}_{\text{pol}}^\theta]$. In mean-field control, however, the whole population flow $(\mu_t^\theta)_{t \in \mathcal{T}}$ also depends on θ . Consequently, the full gradient takes the form

$$\nabla_\theta J(\theta) = \text{RF}(\theta) + \text{MF}(\theta),$$

where MF collects the effects of the population flow on rewards, policies, and state distributions which Ordinary REINFORCE misses; see Meunier et al. (2026).

3 TRANSPORT MAP RANDOMIZATION

We randomize the population argument through a transport map. For $\lambda \in [0, 1]$ and $t \in \llbracket 1, T \rrbracket$, let $T_t^\lambda : \Omega \times \mathcal{X} \times [0, 1] \rightarrow \mathcal{X}$ be jointly measurable, and let $\mathcal{G} : \mathcal{P}(\mathcal{X}) \rightarrow \mathcal{P}(\mathcal{X})$ be a measurable transformation of the law. In continuous state space, we restrict its domain and codomain to $\mathcal{P}_2(\mathbb{R}^d)$. For any $\theta \in \Theta$, we set $M_0^{\lambda, \theta} := \mu_0$, and for $t \in \llbracket 1, T \rrbracket$, define the measurable map $M_t^{\lambda, \theta} : \Omega \rightarrow \mathcal{P}(\mathcal{X})$ as

$$M_t^{\lambda, \theta}(\omega) := T_t^\lambda(\omega, \cdot, \cdot)^\# (\mathcal{G}(\mu_t^\theta) \otimes \mathcal{U}([0, 1])). \quad (3)$$

The transformation $\mathcal{G}$ supplies a finite-dimensional representation: it is the identity in finite state spaces and an encoder-decoder map $\mathcal{D} \circ \mathcal{E}$ in continuous spaces. The auxiliary uniform variable allows the transport to change mixture weights. A deterministic map of the state alone cannot, in general, split the mass of an atom. The uniform variable supplies the randomness needed for this operation.

For each fixed ω , a sample from $M_t^{\lambda,\theta}(\omega)$ is obtained by evaluating $T_t^\lambda(\omega, X, U)$ with independent draws $X \sim \mathcal{G}(\mu_t^\theta)$ and $U \sim \mathcal{U}([0, 1])$, without requiring an explicit expression for the resulting law. This sampling construction extends beyond the convex-combination and Gaussian-mixture examples below to any jointly measurable transport map that can be evaluated on sampled inputs.

The perturbation of Meunier et al. (2026) also has an encoder-decoder interpretation: it takes the form $\text{softmax}(\text{logit}(\mu_t^\theta) + \lambda U_t)$, with standard Gaussian U_t . Their noise is added to the encoded coordinates. In (3), randomness enters through the transport map on the state space. This representation gives a coupling between the original and perturbed laws, which can be used to study distances between these laws.

Using the probability-valued random variable $M_t^{\lambda,\theta}$, we then consider the state and control processes $(X_t^{\lambda,\theta})_{t \in \mathcal{T}}$ and $(\alpha_t^{\lambda,\theta})_{t=0}^{T-1}$ driven by

$$\begin{aligned} X_0^{\lambda,\theta} &\sim \mu_0, & \alpha_t^{\lambda,\theta} &\sim \pi_t^\theta(\cdot \mid X_t^{\lambda,\theta}, M_t^{\lambda,\theta}), \\ X_{t+1}^{\lambda,\theta} &\sim P_t(\cdot \mid X_t^{\lambda,\theta}, M_t^{\lambda,\theta}, \alpha_t^{\lambda,\theta}), \end{aligned} \quad (4)$$

with return and policy score

$$\begin{aligned} \mathcal{R}_\theta^\lambda &:= \sum_{t=0}^{T-1} r_t(X_t^{\lambda,\theta}, M_t^{\lambda,\theta}, \alpha_t^{\lambda,\theta}) + g(X_T^{\lambda,\theta}, M_T^{\lambda,\theta}), \\ \mathcal{S}_{\text{pol}}^{\lambda,\theta} &:= \sum_{t=0}^{T-1} \nabla_\theta \log p_t^\theta(\alpha_t^{\lambda,\theta} \mid X_t^{\lambda,\theta}, M_t^{\lambda,\theta}), \end{aligned} \quad (5)$$

and objective $J^\lambda(\theta) := \mathbb{E}[\mathcal{R}_\theta^\lambda]$.

Since T_t^0 is the identity, $J^0 = J$ for $\mathcal{G} = \text{Id.}$; Gaussian-mixture projection instead gives J_K^λ in (14). Appendix A gives a canonical construction of the control problem.

3.1 Finite-state space

Let $\mathcal{X} = \{x_1, \dots, x_N\}$, with $N \geq 2$. We identify $\mathcal{P}(\mathcal{X})$ with $\Delta_N = \{p \in [0, 1]^N : \sum_i p_i = 1\}$ and write $\mu(i) := \mu(\{x_i\}) = \mu(x_i)$ for the density with respect to the counting measure. Since $\mu = \sum_{i=1}^N \mu(i) \delta_{x_i}$, randomizing the population law amounts to randomizing its weights. In the canonical construction of Appendix A.1, take $E = \mathbb{R}^{N-1}$ and $R_t = U_t$, where $(U_t)_{t \in \llbracket 1, T \rrbracket}$ are independent random vectors with density ρ with respect to Lebesgue measure. Define

$$\varphi : \mathbb{R}^{N-1} \rightarrow \overset{\circ}{\Delta}_N, \quad \varphi(u) := \frac{(e^{u_1}, \dots, e^{u_{N-1}}, 1)}{1 + \sum_{j=1}^{N-1} e^{u_j}}, \quad (6)$$

and set $Q_t := \varphi(U_t)$. For $q \in \Delta_N$, let $F_q : [0, 1] \rightarrow \mathcal{X}$ be the inverse distribution map, with $F_q(0) = x_1$ and

$F_q(v) = x_{\min\{i: v \leq \sum_{j=1}^i q_j\}}$ for $v > 0$. For $\lambda \in (0, 1)$, define

$$T_t^\lambda(\omega, x, v) := \begin{cases} x, & v \leq 1 - \lambda, \\ F_{Q_t(\omega)}((v - 1 + \lambda)/\lambda), & v > 1 - \lambda, \end{cases}$$

and set $T_t^0(\omega, x, v) = x$. The transport retains the state with probability $1 - \lambda$ and otherwise draws from Q_t . Consequently,

$$\begin{aligned} M_t^{\lambda,\theta}(\omega) &:= T_t^\lambda(\omega, \cdot, \cdot) \# (\mu_t^\theta \otimes \mathcal{U}([0, 1])) \\ &= (1 - \lambda) \mu_t^\theta + \lambda Q_t(\omega), \quad t \in \llbracket 1, T \rrbracket. \end{aligned} \quad (7)$$

The assumptions and proofs below are given in Appendix B.2.

Proposition 3.1 (Perturbation estimate). *For every $\lambda \in [0, 1]$, $\theta \in \Theta$, and $t \in \mathcal{T}$,*

$$d_{\text{TV}}(M_t^{\lambda,\theta}(\omega), \mu_t^\theta) \leq \lambda, \quad \mathbb{P}(d\omega)\text{-a.s.}$$

Theorem 3.2 (Perturbation consistency). *Under Assumption B.1, there exists $C_T > 0$, independent of θ and λ , such that, for $\lambda \in [0, 1]$,*

$$\sup_{\theta \in \Theta} |J^\lambda(\theta) - J(\theta)| \leq C_T \lambda.$$

Moreover, under Assumption B.3, there exists $C_T^\nabla > 0$ such that

$$\sup_{\theta \in \Theta} \|\nabla_\theta J^\lambda(\theta) - \nabla_\theta J(\theta)\| \leq C_T^\nabla \lambda.$$

Corollary 3.3 (Near-optimality of perturbed maximizers). *Under Assumptions B.1 and B.3, suppose that Θ is compact. Then, for every $\lambda \in [0, 1]$, J^λ attains its maximum on Θ at some θ_λ^* , and*

$$J(\theta_\lambda^*) \geq \max_{\theta \in \Theta} J(\theta) - 2C_T \lambda.$$

Moreover, every accumulation point of (θ_λ^*) as $\lambda \rightarrow 0$ maximizes J .

Perturbation density and score. Use simplex coordinates $(q_1, \dots, q_{N-1})$, with $q_N = 1 - \sum_{i=1}^{N-1} q_i$, and set $\varphi^{-1}(q) := (\log(q_i/q_N))_{i=1}^{N-1}$. Densities on Δ_N are taken with respect to Lebesgue measure in these coordinates. By the change-of-variable formula, Q_t has density

$$f_Q(q) = \frac{\rho(\varphi^{-1}(q))}{\prod_{i=1}^N q_i}, \quad q \in \overset{\circ}{\Delta}_N.$$

If ρ is strictly positive and $\mathcal{C}^1$, define, for $u \in \mathbb{R}^{N-1}$, $a_i(u) := \partial_{u_i} \log \rho(u)$. Set $H := (H_1, \dots, H_{N-1})$, and define, for $k \in \llbracket 1, N-1 \rrbracket$, $H_k : \overset{\circ}{\Delta}_N \rightarrow \mathbb{R}$ by

$$\begin{aligned} H_k(q) &:= \partial_{q_k} \log f_Q(q) \\ &= \frac{a_k(\varphi^{-1}(q)) - 1}{q_k} + \frac{1 + \sum_{i=1}^{N-1} a_i(\varphi^{-1}(q))}{q_N}. \end{aligned}$$

Write $z_t^\theta := (\mu_t^\theta(1), \dots, \mu_t^\theta(N-1))$, and define, for $z \in \Delta_N$ and $q \in \overset{\circ}{\Delta}_N$,

$$\Psi_\lambda(z, q) := (1 - \lambda)z + \lambda q \in \overset{\circ}{\Delta}_N. \quad (8)$$

Set $Z_t^{\lambda, \theta} := \Psi_\lambda(z_t^\theta, Q_t)$. By (7), $Z_t^{\lambda, \theta}$ is the coordinate vector associated to $M_t^{\lambda, \theta}$. Fix $z \in \Delta_N$, set $Y_\lambda(z) := (1 - \lambda)z + \lambda \overset{\circ}{\Delta}_N$, and let $q^\lambda(y | z) := (y - (1 - \lambda)z) / \lambda$ be the inverse coordinate map on $Y_\lambda(z)$. By Lemma B.5, $\Psi_\lambda(z, Q_t)$ admits a density $h^\lambda(\cdot | z)$, zero outside $Y_\lambda(z)$ and given, for $y \in Y_\lambda(z)$, by

$$h^\lambda(y | z) = \lambda^{-(N-1)} f_Q(q^\lambda(y | z)). \quad (9)$$

In particular, $Z_t^{\lambda, \theta}$ has density $h^\lambda(\cdot | z_t^\theta)$. Its base-coordinate score, with y held fixed, is

$$s^\lambda(y, z) := \nabla_z \log h^\lambda(y | z) = \frac{\lambda - 1}{\lambda} H(q^\lambda(y | z)). \quad (10)$$

The score with respect to the policy parameter is

$$\nabla_\theta \log h^\lambda(y | z_t^\theta) = (D_t^\theta)^\top s^\lambda(y, z_t^\theta),$$

where $D_t^\theta := \nabla_{\theta} z_t^\theta \in \mathbb{R}^{(N-1) \times d_\theta}$.

The support $Y_\lambda(z_t^\theta)$ of $h^\lambda(\cdot | z_t^\theta)$, and hence the integration domain, moves with θ . For the score-function representation to have no boundary terms, the perturbation density f_Q must vanish on the simplex boundary $\partial \Delta_N$. This holds when ρ satisfies Assumption B.4: a Gaussian randomizer does, whereas a uniform distribution on the simplex does not. The logit-softmax parametrization avoids this requirement, since its integration domain does not depend on θ .

Policy-gradient representation. Define the population score along a perturbed trajectory by

$$\begin{aligned} \mathcal{S}_{\text{mf}}^{\lambda, \theta} &:= \sum_{t=1}^T \nabla_\theta \log h^\lambda(Z_t^{\lambda, \theta} | z_t^\theta) \\ &= \sum_{t=1}^T (D_t^\theta)^\top s^\lambda(Z_t^{\lambda, \theta}, z_t^\theta) = \frac{\lambda - 1}{\lambda} \sum_{t=1}^T (D_t^\theta)^\top H(Q_t), \end{aligned} \quad (11)$$

where the last equality uses $q^\lambda(Z_t^{\lambda, \theta} | z_t^\theta) = Q_t$ in (10).

Theorem 3.4 (Model-free policy-gradient representation). *Under Assumptions B.1, B.3 and B.4, for every $\lambda \in (0, 1)$, and using the quantities defined in (5) and (11),*

$$\nabla_\theta J^\lambda(\theta) = \mathbb{E}[\mathcal{R}_\theta^\lambda(\mathcal{S}_{\text{pol}}^{\lambda, \theta} + \mathcal{S}_{\text{mf}}^{\lambda, \theta})].$$

The estimator follows Algorithm 1: μ_t^θ is replaced by the empirical flow $\hat{\mu}_t^M$ of M interacting particles, and

D_t^θ by the estimate $\hat{D}_t^{\eta, M, n}$ in (66), computed from n auxiliary trajectories at scale η . With the returns $\hat{\mathcal{R}}^{(b)}$ and policy scores $\hat{\mathcal{S}}_{\text{pol}}^{(b)}$ of B independent trajectories perturbed by $Q_t^{(b)}$, the estimator of $\nabla_\theta J^\lambda(\theta)$ is

$$\hat{G}_{B, M, n}^{\lambda, \eta}(\theta) := \frac{1}{B} \sum_{b=1}^B \hat{\mathcal{R}}^{(b)} (\hat{\mathcal{S}}_{\text{pol}}^{(b)} + \hat{\mathcal{S}}_{\text{mf}}^{(b)}). \quad (12)$$

A deterministic scalar baseline may reduce variance without changing the expectation, since both scores have mean zero.

Theorem 3.5 (Bias and mean-square error). *Under Assumptions B.1, B.3, B.4, and B.8, if $0 < \lambda, \eta < 1$, integers $B, M, n \geq 1$, and $n\eta^2 \geq 1$, there exists $C > 0$, independent of $\theta, \lambda, \eta, B, M, n$, such that*

$$\begin{aligned} &\|\mathbb{E}[\hat{G}_{B, M, n}^{\lambda, \eta}(\theta)] - \nabla_\theta J(\theta)\| \\ &\leq C \left(\lambda + \eta + \frac{1}{\eta \sqrt{n}} + \frac{1}{\sqrt{M}} \right), \\ &\mathbb{E}[\|\hat{G}_{B, M, n}^{\lambda, \eta}(\theta) - \nabla_\theta J(\theta)\|^2] \\ &\leq C \left(\lambda^2 + \frac{1 + \lambda^{-2}}{B} + \eta^2 + \frac{1}{n\eta^2} + \frac{1}{M} \right). \end{aligned}$$

Minimizing the λ -dependent terms gives $\lambda_\star \asymp B^{-1/4}$, and balancing the auxiliary terms gives $\eta_\star \asymp n^{-1/4}$. These are choices based on the MSE bound, with unknown constants.

3.2 Continuous-state space

Projection onto Gaussian mixtures. Let $\mathcal{X} = \mathbb{R}^d$ and let the action space $\mathcal{A}$ be a non-empty closed subset of $\mathbb{R}^q$. Population laws belong to $\mathcal{P}_2(\mathbb{R}^d)$, and the policies take values in $\mathcal{P}_2(\mathcal{A})$. To obtain a tractable perturbation density, we represent the population law by a Gaussian mixture and randomize its parameters. Fix $K \geq 1$ and a unit vector $e \in \mathbb{R}^d$. Define $\mathbb{M}_K^e := \{(m_1, \dots, m_K) \in (\mathbb{R}^d)^K : \langle e, m_1 \rangle < \dots < \langle e, m_K \rangle\}$, $\mathcal{Z}_K := \Delta_K^\circ \times \mathbb{M}_K^e \times (\mathbb{S}_{++}^d)^K$ and $q_K := (K-1) + Kd + K \frac{d(d+1)}{2}$.

We use the first $K-1$ weights as coordinates, with $p_K = 1 - \sum_{j=1}^{K-1} p_j$. Together with the means and the independent covariance entries, they identify $\mathcal{Z}_K$ with an open subset of $\mathbb{R}^{q_K}$. The ordering fixes component labels and restricts fitted mixtures to distinct projected means. Randomization keeps these labels but may reorder the means, so the decoder is defined on the labeled space $\tilde{\mathcal{Z}}_K := \Delta_K^\circ \times (\mathbb{R}^d)^K \times (\mathbb{S}_{++}^d)^K$, with no ordering constraint. The encoder assigns mixture parameters to each population law:

$$\mathcal{E}_K : \mathcal{P}_2(\mathbb{R}^d) \rightarrow \mathcal{Z}_K, \quad \mu \mapsto (p_K(\mu), m_K(\mu), \Sigma_K(\mu)),$$

where $p_K(\mu) = (p_{j,K}(\mu))_{j=1}^K$, $m_K(\mu) = (m_{j,K}(\mu))_{j=1}^K$, and $\Sigma_K(\mu) = (\Sigma_{j,K}(\mu))_{j=1}^K$ are the fitted weights, means, and covariance matrices, respectively. The decoder reconstructs the corresponding mixture:

$$\mathcal{D}_K : \tilde{\mathcal{Z}}_K \rightarrow \mathcal{P}_2(\mathbb{R}^d), \quad (p, m, \Sigma) \mapsto \sum_{j=1}^K p_j \mathcal{N}(m_j, \Sigma_j).$$

Thus the population approximation is

$$\mathcal{G}_K : \mathcal{P}_2(\mathbb{R}^d) \rightarrow \mathcal{P}_2(\mathbb{R}^d), \quad \mathcal{G}_K(\mu) := \mathcal{D}_K(\mathcal{E}_K(\mu)).$$

Throughout, Σ_j denotes a positive definite covariance matrix and $\Gamma_j = \Sigma_j^{1/2}$ its unique symmetric positive-definite square root. We assume a measurable encoder with bounded, nondegenerate fitted parameters along the population flow. Gradient results additionally require smooth dependence of these parameters on θ ; Appendix B.3.1 states the precise conditions required to ensure the needed regularity. One choice is a constrained likelihood fit, approximated on empirical laws by constrained Expectation-Maximization iterations (Dempster et al., 1977).

The initial population argument is kept exact: $M_{K,0}^{\lambda,\theta} = \mu_0$. For each $t \in \llbracket 1, T \rrbracket$, write $z_{K,t}^\theta := \mathcal{E}_K(\mu_t^\theta)$, with components $(p_{K,t}^\theta, m_{K,t}^\theta, \Sigma_{K,t}^\theta)$, and $\nu_{K,t}^\theta := \mathcal{D}_K(z_{K,t}^\theta)$. The vectors $p_{K,t}^\theta$, $m_{K,t}^\theta$, and $\Sigma_{K,t}^\theta$ collect the component weights, means, and covariance matrices.

Randomized transport. Let $R_t = (Q_t, A_t, B_t)$, $t \in \llbracket 1, T \rrbracket$, be i.i.d. randomizers with law ρ , independent of θ and of the simulation randomness. Their coordinates lie in $\Delta_K^\circ \times (\mathbb{R}^d)^K \times (\mathbb{S}^d)^K$. We take $Q_t = \varphi(U_t)$, using the map in (6) with N replaced by K . Assume that $B_{j,t} \in \mathcal{B}_\lambda := \{B \in \mathbb{S}^d : (1 - \lambda)I_d + \lambda B \succ 0\}$ almost surely. When $K = 1$, $Q_t = 1$ and the free weight coordinates are absent. For $z = (p, m, (\Sigma_j)_{j=1}^K)$ and $r = (q, a, (b_j)_{j=1}^K)$, define

$$\Psi_\lambda(z, r) = \left((1 - \lambda)p + \lambda q, (1 - \lambda)m + \lambda a, \right. \\ \left. (\Sigma_j^{1/2}((1 - \lambda)I_d + \lambda b_j)^2 \Sigma_j^{1/2})_{j=1}^K \right), \quad (13)$$

and set $Z_{K,t}^{\lambda,\theta} := \Psi_\lambda(z_{K,t}^\theta, R_t)$, with components $(p_{K,t}^{\lambda,\theta}, m_{K,t}^{\lambda,\theta}, \Sigma_{K,t}^{\lambda,\theta})$. It takes values in $\tilde{\mathcal{Z}}_K$, since positive definiteness follows from the support condition on B_t . Set $M_{K,t}^{\lambda,\theta}(\omega) := \mathcal{D}_K(Z_{K,t}^{\lambda,\theta}(\omega))$, $\Gamma_{j,K,t}^{\lambda,\theta}(\omega) := (\Sigma_{j,K,t}^{\lambda,\theta}(\omega))^{1/2}$, and $\Gamma_{j,K,t}^\theta := (\Sigma_{j,K,t}^\theta)^{1/2}$.

Proposition 3.6 (Transport representation). *For every $\lambda \in [0, 1]$, the construction in Appendix B.3.2 gives a jointly measurable map $T_{K,t}^{\lambda,\theta} : \Omega \times \mathbb{R}^d \times [0, 1] \rightarrow \mathbb{R}^d$*

such that

$$M_{K,t}^{\lambda,\theta}(\omega) = T_{K,t}^{\lambda,\theta}(\omega, \cdot, \cdot) \# (\nu_{K,t}^\theta \otimes \mathcal{U}([0, 1])) \\ = \sum_{j=1}^K p_{j,K,t}^{\lambda,\theta}(\omega) \mathcal{N}(m_{j,K,t}^{\lambda,\theta}(\omega), \Sigma_{j,K,t}^{\lambda,\theta}(\omega)),$$

The perturbed process $(X_t^{K,\lambda,\theta}, \alpha_t^{K,\lambda,\theta})$, its return $\mathcal{R}_K^{\lambda,\theta}$ and its policy score $\mathcal{S}_{\text{pol}}$ are defined by (4)–(5), with $M_{K,t}^{\lambda,\theta}$ in place of $M_t^{\lambda,\theta}$, and

$$J_K^\lambda(\theta) := \mathbb{E}[\mathcal{R}_K^{\lambda,\theta}]. \quad (14)$$

For $\lambda = 0$, the transport is the identity, so $M_{K,t}^{0,\theta} = \nu_{K,t}^\theta$ for $t \in \llbracket 1, T \rrbracket$. We write $J_K := J_K^0$ for the projected objective. The assumptions and proofs of the following results are in Appendix B.3.

Proposition 3.7 (Perturbation estimate). *Under Assumption B.11, there exists $C_K > 0$, uniform in t, θ , and $\lambda \in [0, 1]$, such that*

$$(\mathbb{E}[\mathcal{W}_2^2(M_{K,t}^{\lambda,\theta}, \mu_t^\theta)])^{1/2} \leq \mathcal{W}_2(\nu_{K,t}^\theta, \mu_t^\theta) + C_K \sqrt{\lambda}.$$

Theorem 3.8 (Objective perturbation consistency). *Under Assumptions B.11 and B.12, there exists $C_{K,T} > 0$, independent of θ and λ , such that*

$$\sup_{\theta \in \Theta} |J_K^\lambda(\theta) - J_K(\theta)| \leq C_{K,T} \sqrt{\lambda}, \quad \lambda \in [0, 1].$$

Projection error. The preceding limits hold for fixed K . To compare with the original objective, define $\varepsilon_K := \sup_{\theta \in \Theta} \max_{1 \leq t \leq T} \mathcal{W}_2(\nu_{K,t}^\theta, \mu_t^\theta)$. Under Assumption B.12, the same stability argument gives

$$\sup_{\theta \in \Theta} |J_K(\theta) - J(\theta)| \leq C_T^{\text{proj}} \varepsilon_K, \\ \sup_{\theta \in \Theta} |J_K^\lambda(\theta) - J(\theta)| \leq C_T^{\text{proj}} \varepsilon_K + C_{K,T} \sqrt{\lambda}, \quad (15)$$

where $C_T^{\text{proj}} > 0$ depends only on T and the Lipschitz constant; see Appendix B.3.4. A rate for ε_K as K grows requires additional approximation assumptions.

Proposition 3.9 (Optimality gap). *Under Assumptions B.11 and B.12, let $\theta_{K,\lambda}^* \in \Theta$ be a maximizer of J_K^λ over Θ . Then*

$$0 \leq \sup_{\theta \in \Theta} J(\theta) - J(\theta_{K,\lambda}^*) \leq 2(C_T^{\text{proj}} \varepsilon_K + C_{K,T} \sqrt{\lambda}).$$

Thus a maximizer of J_K^λ is near-optimal for the original problem. The loss has two sources: the mixture projection error ε_K , controlled by K , and a perturbation cost of order $\sqrt{\lambda}$, controlled by the randomization scale. The proof is in Appendix B.3.4.

Theorem 3.10 (Gradient perturbation consistency). *Under the preceding Assumptions and Assumption B.13, J_K^λ and J_K are differentiable, there exists $C_{K,T}^\nabla < \infty$, independent of θ and λ , and there exists $\lambda_0 \in (0, 1)$, such that, for $\lambda \in [0, \lambda_0]$,*

$$\sup_{\theta \in \Theta} \|\nabla_\theta J_K^\lambda(\theta) - \nabla_\theta J_K(\theta)\| \leq C_{K,T}^\nabla \lambda.$$

Perturbation density and policy gradient.

Densities on $\tilde{\mathcal{Z}}_K$ are taken with respect to Lebesgue measure in the free weight coordinates, mean coordinates, and independent covariance entries. Fix $\lambda \in (0, 1)$ and a base point $z = (p, m, \Sigma) \in \tilde{\mathcal{Z}}_K$, and set

$$Y_\lambda(z) := ((1 - \lambda)p + \lambda \Delta_K^\circ) \times (\mathbb{R}^d)^K \times (\mathbb{S}_{++}^d)^K.$$

Inverting $\Psi_\lambda(z, \cdot)$ at $y = (\tilde{p}, \tilde{m}, \tilde{\Sigma}) \in Y_\lambda(z)$ gives the inverse coordinates $(q^\lambda, a^\lambda, b^\lambda)(y | z)$: for $j \in \llbracket 1, K \rrbracket$,

$$\begin{aligned} (q^\lambda, a^\lambda) &= \lambda^{-1}[(\tilde{p}, \tilde{m}) - (1 - \lambda)(p, m)], \\ b_j^\lambda &= \lambda^{-1}[(\Sigma_j^{-1/2} \tilde{\Sigma}_j \Sigma_j^{-1/2})^{1/2} - (1 - \lambda)I_d]. \end{aligned} \quad (16)$$

Lemma 3.11 (Perturbation density). *Under conditions (1)–(2) of Assumption B.14, the random variable $\Psi_\lambda(z, R_t)$ admits a density $h^\lambda(\cdot | z) : \tilde{\mathcal{Z}}_K \rightarrow [0, \infty)$. For $y \in Y_\lambda(z)$, write $(q, a, b) = (q^\lambda, a^\lambda, b^\lambda)(y | z)$. Then*

$$h^\lambda(y | z) = \frac{f_Q(q) f_A(a) f_B(b)}{\mathcal{J}^\lambda(z, b)},$$

where $\mathcal{J}^\lambda : \tilde{\mathcal{Z}}_K \times (\mathcal{B}_\lambda)^K \rightarrow (0, \infty)$ is the Jacobian factor given in (49). The density is zero outside $Y_\lambda(z)$.

The formula for $\mathcal{J}^\lambda$ and the proof are given in Lemma B.16 in Appendix B.3.6. For $K = 1$, the factor f_Q and the free weight coordinates are omitted. Where the density is positive and differentiable in z , define its base-coordinate score by

$$s^\lambda(y, z) := \nabla_z \log h^\lambda(y | z) \in \mathbb{R}^{q_K}, \quad (17)$$

which is the score with respect to the base coordinates z , with y held fixed. The score with respect to the policy parameter, $D_t^\theta = \nabla_\theta z_{K,t}^\theta \in \mathbb{R}^{q_K \times d_\theta}$, is

$$\nabla_\theta \log h^\lambda(y | z_{K,t}^\theta) = (D_t^\theta)^\top s^\lambda(y, z_{K,t}^\theta).$$

This general construction specializes to the fully solvable linear dynamics framework developed in Section C, where restricting to a single Gaussian component ($K = 1$), the parameter score in (17) reduces exactly to the explicit mean-field score of Corollary C.7.

Along a perturbed trajectory, define the population score

$$\mathcal{S}_{\text{mf}} := \sum_{t=1}^T \nabla_\theta \log h^\lambda(Z_{K,t}^{\lambda, \theta} | z_{K,t}^\theta) = \sum_{t=1}^T (D_t^\theta)^\top s^\lambda(Z_{K,t}^{\lambda, \theta}, z_{K,t}^\theta).$$

Theorem 3.12 (Model-free policy-gradient representation). *Let $\lambda \in (0, 1)$. Under Assumptions B.14 and B.17, the following holds:*

$$\nabla_\theta J_K^\lambda(\theta) = \mathbb{E}[\mathcal{R}_K^{\lambda, \theta}(\mathcal{S}_{\text{pol}} + \mathcal{S}_{\text{mf}})].$$

The fitted coordinates are $\hat{z}_{K,t}^M = \mathcal{E}_K(\hat{\mu}_t^M)$, $\hat{D}_t^{\eta, M, n}$ is the centered difference (67) (Jia et al., 2026) over $n = 2d_\theta n_0$ particles, and $\hat{Z}_t^{(b)} = \Psi_\lambda(\hat{z}_{K,t}^M, R_t^{(b)})$. With the returns $\hat{\mathcal{R}}^{(b)}$ and policy scores $\hat{\mathcal{S}}_{\text{pol}}^{(b)}$ of B independent main trajectories, the estimator is

$$\hat{G}_{B, M, n}^{\lambda, \eta}(\theta) := \frac{1}{B} \sum_{b=1}^B \hat{\mathcal{R}}^{(b)} (\hat{\mathcal{S}}_{\text{pol}}^{(b)} + \hat{\mathcal{S}}_{\text{mf}}^{(b)}). \quad (18)$$

Let a_L bound the root mean-square error of the coordinates fitted from L particles. Then a_M controls the population fitting error, while a_{n_0}/η controls the fitting error in the centered differences. Unlike empirical frequencies in finite state space, mixture coordinates require a non-linear fit, so the rate $a_L = L^{-1/2}$ needs an additional assumption on its stability.

Theorem 3.13 (Bias and mean-square error). *Under the previous assumptions and Assumptions B.18, let $n_0 = n/(2d_\theta)$. For admissible scales $0 < \lambda \leq \lambda_0$, $0 < \eta \leq \eta_0$, with $a_{n_0} \leq \eta$, there exists $C < \infty$, independent of $\theta, \lambda, \eta, B, M, n_0$ such that*

$$\begin{aligned} &\left\| \mathbb{E}[\hat{G}_{B, M, n}^{\lambda, \eta}(\theta)] - \nabla_\theta J_K(\theta) \right\| \\ &\leq C \left(\lambda + a_M + \eta^2 + \frac{a_{n_0}}{\eta} \right), \\ &\mathbb{E} \left[\left\| \hat{G}_{B, M, n}^{\lambda, \eta}(\theta) - \nabla_\theta J_K(\theta) \right\|^2 \right] \\ &\leq C \left(\lambda^2 + a_M^2 + \eta^4 + \frac{a_{n_0}^2}{\eta^2} + \frac{1 + \lambda^{-2}}{B} \right). \end{aligned}$$

If Assumption B.18 (2) holds with $a_N = N^{-1/2}$, i.e. $\sup_\theta \mathbb{E}[\|\hat{z}_K^{N, \theta} - z^\theta\|^2] \leq C/N$, then for $n_0 \eta^2 \geq 1$, we can balance the scale-dependent terms in the MSE bound to get $\lambda_\star \asymp B^{-1/4}$ and $\eta_\star \asymp n_0^{-1/6}$.

4 POLICY GRADIENT ALGORITHM

Each policy update has two stages. The offline stage estimates the population flow and its sensitivities at the current parameter θ . The online stage holds these estimates fixed and uses B perturbed trajectories to estimate the gradient. Both stages are repeated after each update; Appendix D gives the sampling details and computational cost. Let $(e_\ell)_{\ell=1}^{d_\theta}$ denote the standard basis of $\mathbb{R}^{d_\theta}$.

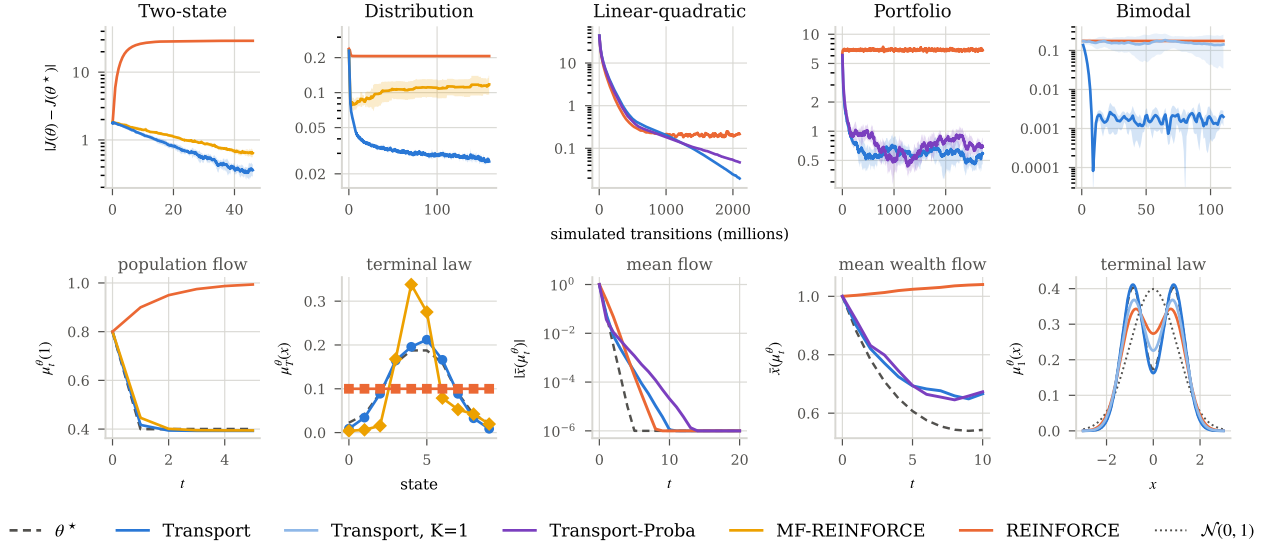


Figure 1: Top: validation gap against simulated transitions (mean and standard deviation over five paired seeds). Bottom: the final policy’s population flow or terminal law for the first seed. Transport and Transport-Proba are shown at their best scale. Dashed black curves show the reference policy.

Table 1: Final optimality gap $|J(\hat{\theta}) - J(\theta^*)|$ (mean $\pm$ std. over five paired seeds; smaller is better). Transport is shown at $\lambda_* = B^{-1/4}$ and at its best tested scale in parentheses. Transport-Proba uses its best tested scale. Scale choices and evaluation of the reference policies are detailed in Appendix E.1.

Benchmark	$J(\theta^*)$	REINFORCE	MF-REINFORCE	Transport		Transport-Proba
				at λ_*	best scale	best scale
Two-state	-2.64	29.0 ± 0.0	0.637 ± 0.093	1.36 ± 0.13	0.359 ± 0.091 ($\lambda_*/4$)	—
Distribution	-0.05699	0.206 ± 0.002	0.118 ± 0.024	0.0296 ± 0.0012	0.0255 ± 0.0011 ($\lambda_*/2$)	—
Linear-quadratic	-7.224	0.216 ± 0.002	—	0.0390 ± 0.0027	0.0196 ± 0.0012 ($\lambda_*/4$)	0.0469 ± 0.0028 ($\lambda_*/2$)
Portfolio	-13.15	6.87 ± 0.07	—	1.22 ± 0.07	0.586 ± 0.152 ($\lambda_*/4$)	0.692 ± 0.156 ($\lambda_*/2$)
Bimodal ($K = 2$)	0	0.175 ± 0.009	—	0.0903 ± 0.0053	0.00199 ± 0.00094 ($\lambda_*/8$)	—

5 NUMERICAL EXPERIMENTS

Setup. We evaluate four finite-state benchmarks (two-state control, cybersecurity, distribution planning, advertising) and three continuous-state benchmarks (linear-quadratic (LQ) control, portfolio, bimodal allocation). The results for Cybersecurity and advertising are presented in Appendix E.3. Baselines are REINFORCE, MF-REINFORCE (Meunier et al., 2026) on finite states, and, on LQ control and portfolio, Transport-Proba, which replaces the centered differences by a likelihood-ratio estimate of the moment sensitivities, as per Appendix C. Methods share the policy class, initialization, number of updates and simulator budget. Results use five paired seeds. We report $|J(\hat{\theta}) - J(\theta^*)|$, where $\hat{\theta}$ is the final parameter returned by training and θ^* maximizes the original, unperturbed objective J over the benchmark’s admissible policy class. Thus the reference θ^* does not depend on λ , although the learned parameter does. Distribu-

tion planning uses a numerical reference. Appendix E gives models, evaluation and settings. The code is available at: <https://anonymous.4open.science/r/RL-MFC-TransportREINFORCE-44A3/>.

Optimality gap. For each seed s , we compute the paired validation gap $\hat{\Delta}_s = |\hat{J}_s(\hat{\theta}_s) - \hat{J}_s(\theta^*)|$ and report its mean and standard deviation. Transport has the smallest mean gap in every row of Table 1 after scale selection. The largest absolute gains occur in two-state control and the portfolio, where REINFORCE ignores how the policy changes the population; in two-state control it moves almost all the mass to state 1 and incurs the population penalties (Figure 1). The comparison is benchmark-dependent: cybersecurity gives similar outcomes for Transport and REINFORCE, while MF-REINFORCE has the higher mean objective on advertising (Appendix E.3.1). At matched budgets, a Transport run takes 2 to 16 times the wall-clock time of REINFORCE, since each update also estimates sen-

Algorithm 1 Transport REINFORCE

Require: initial parameter θ ; budgets M, n, B ; scales λ, η ; step size γ ; randomizer law ρ ; mixture size K (continuous case)

- 1: **for** each policy update **do**
- 2: **Offline stage: population and sensitivity**
- 3: Estimate $(\hat{\mu}_t^M)_{t=0}^T$ using M interacting particles under θ
- 4: **if** finite state space **then**
- 5: Compute $(\hat{D}_t^{\eta, M, n})_{t=1}^T$ recursively by (66), using n rollouts per horizon at scale η
- 6: **else**
- 7: Fit K -component mixtures to $(\hat{\mu}_t^M)_t$
- 8: For each $\ell = 1, \dots, d_\theta$, fit $\hat{z}_t^{\theta \pm \eta e_\ell}$ from independent $n/(2d_\theta)$ -particle systems under $\theta \pm \eta e_\ell$; set, for $t = 1, \dots, T$,
- 9: $\hat{D}_t^{\eta, M, n} e_\ell \leftarrow (\hat{z}_t^{\theta + \eta e_\ell} - \hat{z}_t^{\theta - \eta e_\ell}) / (2\eta)$
- 10: **end if**
- 11: **Online stage: gradient estimation**
- 12: Hold the offline estimates fixed and simulate B fresh perturbed trajectories at scale λ , as in Section 3.1 or 3.2
- 13: Compute $\hat{G}$ using (12) or (18)
- 14: $\theta \leftarrow \theta + \gamma \hat{G}$
- 15: **end for**

sitivities and population scores (Table 7).

Mixture size. A single Gaussian retains the moments needed by linear-quadratic control and the portfolio. Bimodal allocation instead depends on the shape of the terminal law: its mean and variance are constant in θ , so the $K = 1$ projected objective is constant. With $K = 2$, the law is represented exactly. At the same scale, the gap decreases from 0.1425 ± 0.1239 to 0.0020 ± 0.0009 (Table 5).

Perturbation scale. Smaller λ reduces perturbation bias but increases the variance of the population score. Consequently, the best training scale depends on both approximation and optimization error. Figure 2 shows the resulting trade-off on four benchmarks: the smallest gap is reached at $\lambda_*/4$ or $\lambda_*/2$. On two-state control, λ_* gives a larger gap than MF-REINFORCE, whereas $\lambda_*/4$ gives a smaller one. On bimodal allocation, the mean optimization error is below 10^{-4} across the grid, and the smallest tested λ performs best. Thus $B^{-1/4}$ provides a scale for tuning, with a benchmark-dependent constant.

Bias and mean-square error. Table 2 compares predicted and measured power. Sampling follows n^{-1} , M^{-1} and B^{-1} , perturbation bias is nearly linear in λ at small scales, and exact-flow sensitivity follows η^4 .

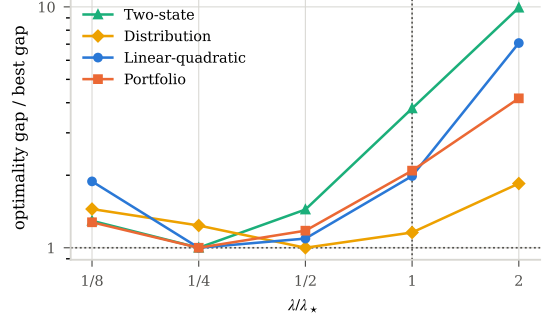


Figure 2: Final gap against λ/λ_* , divided by the smallest gap on the grid (mean over five paired seeds).

Table 2: Predicted powers p_0 and ranges of fitted log-log slopes p across the six benchmarks in Table 9. The exact-flow η sweep uses the two continuous benchmarks. Fitting details are in Appendix E.3.6.

Error	Swept	p_0	Measured p
Sensitivity MSE	η (exact)	4	[3.99, 4.02]
Sensitivity MSE	n	-1	[-1.23, -0.51]
Gradient bias	λ	1	[1.00, 1.28]
Gradient variance	B	-1	[-1.03, -0.90]
Population MSE	M	-1	[-1.07, -0.98]

6 CONCLUSION

We introduced Transport REINFORCE, a model-free policy-gradient method for discrete-time mean-field control. Randomizing the population argument through a transport map gives the perturbed law a tractable density, whose score recovers the mean-field contribution missed by REINFORCE. We established objective and gradient consistency for the finite-state and projected continuous-state problems, with estimator bias, mean-square error, and objective projection error bounds. At matched simulator budgets, the method attains the smallest mean optimality gap on the five main benchmarks after scale selection, and the measured error rates agree with the predicted ones.

Several limitations remain. The bounds only fix the order $B^{-1/4}$ of the perturbation scale; the best constant is benchmark-dependent, between $\lambda_*/8$ and $\lambda_*/2$, and is selected after training. In continuous state spaces, Gaussian mixtures are dense in $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$ (Delon and Desolneux, 2020), but our guarantees concern J_K for a chosen K . Deriving rates for ε_K remains future work. Centered differences require $2d_\theta$ auxiliary particle systems. Finally, Transport is comparable to REINFORCE on cybersecurity and below MF-REINFORCE on advertising. Future work includes adaptive scale selection and sensitivity estimators whose cost does not grow with d_θ .

Potential extensions include non-exchangeable mean-field systems (De Crescenzo et al., 2026; Mekkaoui and Pham, 2026) and graphon interactions (Bayraktar et al., 2023; Coppini et al., 2025; Cao and Laurière, 2025). This replaces $\mathcal{P}(\mathcal{X})$ by $\mathcal{P}_\nu(I \times \mathcal{X})$, the space of probability laws with prescribed label marginal ν on a compact label space I (Mekkaoui et al., 2026); transport perturbations would then act on the state coordinate while preserving labels. Another direction is multi-agent reinforcement learning (Yang et al., 2018), particularly cooperative systems approximated by mean-field control (Gu et al., 2021), where population randomization could account for the effect of policy updates on the collective dynamics.

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AI use statement

In this work, we used generative AI tools for implementing methods numerically, providing feedback on research methodology and experiments, and assisting in the writing of proofs. The proofs rely on standard mathematical techniques; AI assistance in this context was limited to helping locate references for inequalities used to bound or control certain quantities. We have not used generative AI tools for helping develop theoretical models or conceptual frameworks, formulating mathematical claims, providing critical ingredients for proving mathematical claims, proposing or refining hypotheses, assisting with translation, or interpreting results; generating synthetic data sets, cleaning and reformatting datasets, and supporting qualitative and thematic data analysis are not applicable to this work. Additionally, we used generative AI tools for creating and modifying scientific figures, creating and editing software code, brainstorming, sourcing and searching for information, and editing the research paper to improve readability. We have reviewed all AI-assisted work. All AI-assisted proofs were checked line by line by the authors. All LLM-generated code, including the code used to generate the scientific figures, was verified and tested by the authors. All edits made

to improve readability were reviewed, and all information provided by the LLM was verified and fact-checked. We take responsibility for the final content of this work, including text, claims, or artifacts produced with the aid of generative AI.

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CHECKLIST

1. For all models and algorithms presented, check if you include:
 - (a) A clear description of the mathematical setting, assumptions, algorithm, and/or model. [\[Yes, see Sections 2, 3 and 4.\]](#)
 - (b) An analysis of the properties and complexity (time, space, sample size) of any algorithm. [\[Yes, see Appendix D.\]](#)
 - (c) (Optional) Anonymized source code, with specification of all dependencies, including external libraries. [\[Yes, see Section 5.\]](#)
2. For any theoretical claim, check if you include:
 - (a) Statements of the full set of assumptions of all theoretical results. [\[Yes, see Appendix B.\]](#)
 - (b) Complete proofs of all theoretical results. [\[Yes, see Appendix B.\]](#)
 - (c) Clear explanations of any assumptions. [\[Yes, see Appendix B.\]](#)
3. For all figures and tables that present empirical results, check if you include:
 - (a) The code, data, and instructions needed to reproduce the main experimental results (either in the supplemental material or as a URL). [\[Yes, see Section 5.\]](#)
 - (b) All the training details (e.g., data splits, hyperparameters, how they were chosen). [\[Yes, see Appendix E.\]](#)
 - (c) A clear definition of the specific measure or statistics and error bars (e.g., with respect to the random seed after running experiments multiple times). [\[Yes, see Appendix E.\]](#)
 - (d) A description of the computing infrastructure used. (e.g., type of GPUs, internal cluster, or cloud provider). [\[Yes, see Appendix E.\]](#)
4. If you are using existing assets (e.g., code, data, models) or curating/releasing new assets, check if you include:
 - (a) Citations of the creator If your work uses existing assets. [\[Yes, see Section 5 and Appendix E.\]](#)
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5. If you used crowdsourcing or conducted research with human subjects, check if you include:
- (a) The full text of instructions given to participants and screenshots. [\[Not Applicable.\]](#)
 - (b) Descriptions of potential participant risks, with links to Institutional Review Board (IRB) approvals if applicable. [\[Not Applicable.\]](#)
 - (c) The estimated hourly wage paid to participants and the total amount spent on participant compensation. [\[Not Applicable.\]](#)

A PROBLEM FORMULATION

We present in this section the formal setup used throughout the paper: the canonical construction of the controlled process, the well-posedness of the control problem, and the decomposition of the policy gradient.

Notations. For a Polish space E , let $\mathcal{B}(E)$ be its Borel σ -algebra and $\mathcal{P}(E)$ its Borel probability measures, equipped with the weak topology. For a measurable map $F : E \rightarrow E'$, $F\#\mu$ denotes the pushforward of $\mu \in \mathcal{P}(E)$ by F ; in particular, $\mathcal{L}(X) = X\#\mathbb{P}$ whenever $X : \Omega \rightarrow E$ is measurable and $\mathbb{P} \in \mathcal{P}(\Omega)$. We write $\mathcal{P}_2(\mathbb{R}^d)$ for probability measures with finite second moment. Let $\mathcal{C}_c^1(D)$ denote the space of continuously differentiable functions with compact support over $D \subset \mathbb{R}^m$. We use $|\cdot|$ for absolute values, $\|\cdot\|$ for Euclidean norms and operator norms of matrices, $\|\cdot\|_F$ for the Frobenius norm, and $\|v\|_1 = \sum_i |v_i|$. Define

$$d_{\text{TV}}(\mu, \nu) := \sup_{A \in \mathcal{B}(E)} |\mu(A) - \nu(A)|, \quad \mathcal{W}_2^2(\mu, \nu) := \inf_{\gamma \in \Pi(\mu, \nu)} \int_{\mathbb{R}^d \times \mathbb{R}^d} \|x - y\|^2 \gamma(dx, dy),$$

where the second definition applies to $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$ and $\Pi(\mu, \nu)$ is their set of couplings. On a finite space, $d_{\text{TV}}(\mu, \nu) = \frac{1}{2} \|\mu - \nu\|_1$ (Levin and Peres, 2026, Proposition 4.2). Let $\Delta_N = \{p \in [0, 1]^N : \sum_i p_i = 1\}$ with relative interior $\overset{\circ}{\Delta}_N := \{p \in \Delta_N : p_i > 0 \forall i\}$ and relative boundary $\partial\Delta_N := \Delta_N \setminus \overset{\circ}{\Delta}_N = \{p \in \Delta_N : p_i = 0 \text{ for at least one } i\}$. Finally, $\mathbb{S}^d$ denotes the set of symmetric matrices over $\mathbb{R}^{d \times d}$ while $\mathbb{S}_{++}^d$ denotes the symmetric positive-definite ones. We denote $\mathcal{N}(m, \Sigma)$ for the Gaussian law with mean m and covariance Σ , and $\mathcal{U}([0, 1])$ for the uniform law on $[0, 1]$. Given two matrices $A, B \in \mathbb{S}^d$, we say that $A \succ B$ if $A - B \in \mathbb{S}_{++}^d$. Given an open set $O \subset \mathbb{R}^p$ and a differentiable function $f : O \rightarrow \mathbb{R}$, we denote by $\nabla f : O \rightarrow \mathbb{R}^p$ its gradient. More generally, if $f : O_1 \times O_2 \rightarrow \mathbb{R}$, we denote by $\nabla_x f$ and $\nabla_y f$ its gradient maps with respect to the first and second variables, respectively.

A.1 Canonical construction of the control problem

Fix a finite integer $T \geq 1$ and set $\mathcal{T} := \{0, \dots, T\}$. Throughout, $\mathcal{X}$ and $\mathcal{A}$ are non-empty Polish spaces, $\mathcal{P}(\mathcal{X})$ is endowed with the topology of weak convergence and its Borel σ -algebra, and the kernels $P_t : \mathcal{X} \times \mathcal{P}(\mathcal{X}) \times \mathcal{A} \rightarrow \mathcal{P}(\mathcal{X})$ and $\pi_t : \mathcal{X} \times \mathcal{P}(\mathcal{X}) \rightarrow \mathcal{P}(\mathcal{A})$, $t \in \llbracket 0, T-1 \rrbracket$, are measurable. By Kallenberg (1997, Lemma 2.22 (kernels and randomization)), there exist measurable maps

$$F_t : \mathcal{X} \times \mathcal{P}(\mathcal{X}) \times \mathcal{A} \times [0, 1] \rightarrow \mathcal{X}, \quad \tilde{F}_t : \mathcal{X} \times \mathcal{P}(\mathcal{X}) \times [0, 1] \rightarrow \mathcal{A},$$

such that $F_t(x, m, a, \cdot)\#\mathcal{U}([0, 1]) = P_t(x, m, a)$ and $\tilde{F}_t(x, m, \cdot)\#\mathcal{U}([0, 1]) = \pi_t(x, m)$ for all $(t, x, m, a) \in \llbracket 0, T-1 \rrbracket \times \mathcal{X} \times \mathcal{P}(\mathcal{X}) \times \mathcal{A}$. Let also E be a Polish space and $\rho \in \mathcal{P}(E)$, used below to randomize the measure argument.

Canonical space. Let $\mu_0 \in \mathcal{P}(\mathcal{X})$. Set

$$\Omega := \mathcal{X} \times [0, 1]^T \times [0, 1]^T \times E^T,$$

with canonical element $\omega = (x_0, \epsilon_{1:T}, \tilde{\epsilon}_{1:T}, e_{1:T})$, endowed with its product Borel σ -algebra $\mathcal{F}$ and the product measure

$$\mathbb{P}(d\omega) := \mu_0(dx_0) \otimes \mathcal{U}([0, 1])^{\otimes T} (d\epsilon_{1:T}) \otimes \mathcal{U}([0, 1])^{\otimes T} (d\tilde{\epsilon}_{1:T}) \otimes \rho^{\otimes(T)}(de_{1:T}).$$

The coordinate maps $\xi(\omega) = x_0$, $\epsilon_t(\omega) = \epsilon_t$, $\tilde{\epsilon}_t(\omega) = \tilde{\epsilon}_t$, $R_t(\omega) = e_t$ are then mutually independent, with $\xi \sim \mu_0$, $\epsilon_t, \tilde{\epsilon}_t \sim \mathcal{U}([0, 1])$ and $R_t \sim \rho$.

State and control processes. Define $X_t : \Omega \rightarrow \mathcal{X}$ for $t \in \mathcal{T}$ and $\alpha_t : \Omega \rightarrow \mathcal{A}$ for $t \in \llbracket 0, T-1 \rrbracket$ recursively by

$$\begin{cases} X_0 = \xi, \\ \alpha_t = \tilde{F}_t(X_t, \mathbb{P}_{X_t}, \tilde{\epsilon}_{t+1}), \\ X_{t+1} = F_t(X_t, \mathbb{P}_{X_t}, \alpha_t, \epsilon_{t+1}), \end{cases} \quad t \in \llbracket 0, T-1 \rrbracket. \quad (19)$$

This is well defined since $\mathbb{P}_{X_t}$ is determined by the construction up to time t . Let $\mathcal{F}_t := \sigma(\xi, \epsilon_{1:t}, \tilde{\epsilon}_{1:t})$. Since $\tilde{\epsilon}_{t+1}$ is independent of $\mathcal{F}_t$ and ϵ_{t+1} is independent of $\mathcal{F}_t \vee \sigma(\tilde{\epsilon}_{t+1})$, we get, $\mathbb{P}$ -a.s.,

$$\mathcal{L}(\alpha_t | \mathcal{F}_t) = \pi_t(X_t, \mathbb{P}_{X_t}), \quad \mathcal{L}(X_{t+1} | \mathcal{F}_t \vee \sigma(\tilde{\epsilon}_{t+1})) = P_t(X_t, \mathbb{P}_{X_t}, \alpha_t).$$

Randomized measure argument. Fix $\theta \in \Theta$ and let $(X_t^\theta)_{t \in \mathcal{T}}$ denote the unperturbed process associated with π^θ , with population flow $\mu_t^\theta := \mathbb{P}_{X_t^\theta}$. We set $M_0^{\lambda, \theta} := \mu_0^\theta = \mu_0$. For $\lambda \in [0, 1)$ and $t \in \llbracket 1, T \rrbracket$, define the $\mathcal{P}(\mathcal{X})$ -valued random variable

$$\Omega \ni \omega \mapsto M_t^{\lambda, \theta}(\omega) := T_t^\lambda(\omega, \cdot, \cdot) \# (\mathcal{G}(\mu_t^\theta) \otimes \mathcal{U}([0, 1])),$$

where $T_t^\lambda(\omega, x, u) := \Phi^\lambda(R_t(\omega), x, u)$ for some measurable maps $\Phi^\lambda : E \times \mathcal{X} \times [0, 1] \rightarrow \mathcal{X}$ and $\mathcal{G} : \mathcal{P}(\mathcal{X}) \rightarrow \mathcal{P}(\mathcal{X})$. For every bounded continuous function $f : \mathcal{X} \rightarrow \mathbb{R}$,

$$\int_{\mathcal{X}} f(y) M_t^{\lambda, \theta}(\omega)(dy) = \int_{\mathcal{X}} \int_{[0, 1]} f(\Phi^\lambda(R_t(\omega), x, u)) \mathcal{U}([0, 1])(du) \mathcal{G}(\mu_t^\theta)(dx)$$

is $\sigma(R_t)$ -measurable by Fubini's theorem. Since the Borel σ -algebra of $\mathcal{P}(\mathcal{X})$ is generated by the maps $m \mapsto \int_{\mathcal{X}} f dm$, $M_t^{\lambda, \theta}$ is a $\sigma(R_t)$ -measurable random probability measure.

Perturbed state and control processes. For fixed $\theta \in \Theta$, apply the preceding randomization construction to the policy π^θ and denote by $\tilde{F}_t^\theta : \mathcal{X} \times \mathcal{P}(\mathcal{X}) \times [0, 1] \rightarrow \mathcal{A}$ a measurable map satisfying $\tilde{F}_t^\theta(x, m, \cdot) \# \mathcal{U}([0, 1]) = \pi_t^\theta(\cdot | x, m)$. Define $X_t^{\lambda, \theta} : \Omega \rightarrow \mathcal{X}$ for $t \in \mathcal{T}$ and $\alpha_t^{\lambda, \theta} : \Omega \rightarrow \mathcal{A}$ for $t \in \llbracket 0, T - 1 \rrbracket$ recursively by

$$\begin{cases} X_0^{\lambda, \theta} = \xi, \\ \alpha_t^{\lambda, \theta} = \tilde{F}_t^\theta(X_t^{\lambda, \theta}, M_t^{\lambda, \theta}, \tilde{\epsilon}_{t+1}), \\ X_{t+1}^{\lambda, \theta} = F_t(X_t^{\lambda, \theta}, M_t^{\lambda, \theta}, \alpha_t^{\lambda, \theta}, \epsilon_{t+1}) \end{cases} \quad t \in \llbracket 0, T - 1 \rrbracket.$$

As for the unperturbed process, conditionally on $(X_t^{\lambda, \theta}, M_t^{\lambda, \theta})$, the action $\alpha_t^{\lambda, \theta}$ has law $\pi_t^\theta(\cdot | X_t^{\lambda, \theta}, M_t^{\lambda, \theta})$ and, given also $\alpha_t^{\lambda, \theta}$, $X_{t+1}^{\lambda, \theta}$ has law $P_t(\cdot | X_t^{\lambda, \theta}, M_t^{\lambda, \theta}, \alpha_t^{\lambda, \theta})$. For $t \geq 1$, $X_t^{\lambda, \theta}$ depends on the perturbation randomizers only through $R_1, \dots, R_{t-1}$, whereas $M_t^{\lambda, \theta}$ depends only on R_t ; the two are therefore independent. In general, $M_t^{\lambda, \theta} \neq \mathcal{L}(X_t^{\lambda, \theta})$, since $M_t^{\lambda, \theta}$ is a randomized population argument supplied to the policy and transition kernel rather than the law induced by the perturbed state process.

Simulation access. Given $(t, x, m, a) \in \llbracket 0, T - 1 \rrbracket \times \mathcal{X} \times \mathcal{P}(\mathcal{X}) \times \mathcal{A}$, the simulator returns the reward $r_t(x, m, a)$ and an independent draw from $P_t(\cdot | x, m, a)$. At the terminal time, given $(x, m) \in \mathcal{X} \times \mathcal{P}(\mathcal{X})$, it returns $g(x, m)$. The transition kernels and reward functions are otherwise unknown. The policy π_t^θ and its score $\nabla_\theta \log p_t^\theta$ are known and can be evaluated at prescribed population arguments. Population laws are estimated from particles whenever they are not available exactly.

A.2 Well-posedness of the control problem

The state space $\mathcal{X}$ is either a finite subset of $\mathbb{R}^d$ or $\mathbb{R}^d$ itself. The action space $\mathcal{A}$ is a non-empty closed subset of $\mathbb{R}^q$. Policies are parametrized by $\theta \in \Theta$: we denote by $(X_t^\theta, \alpha_t^\theta)$ the processes defined by (19) with $\pi_t = \pi_t^\theta$, set $\mu_t^\theta := \mathbb{P}_{X_t^\theta}$, and define the objective $J : \Theta \rightarrow \mathbb{R}$ as

$$J(\theta) := \mathbb{E} \left[\sum_{t=0}^{T-1} r_t(X_t^\theta, \mu_t^\theta, \alpha_t^\theta) + g(X_T^\theta, \mu_T^\theta) \right].$$

We first give the finite-state argument, then treat the moment and continuity requirements in continuous state space.

A.2.1 Finite state space

Let $\mathcal{X} = \{x_1, \dots, x_N\}$ and identify its probability laws with Δ_N . For fixed θ , the policy-averaged kernel $K_t^\theta : \mathcal{X} \times \Delta_N \rightarrow \Delta_N$ and reward $R_t^\theta : \mathcal{X} \times \Delta_N \rightarrow \mathbb{R}$ are

$$\begin{aligned} K_t^\theta(x_j | x_i, m) &:= \int_{\mathcal{A}} P_t(\{x_j\} | x_i, m, a) \pi_t^\theta(da | x_i, m), \\ R_t^\theta(x_i, m) &:= \int_{\mathcal{A}} r_t(x_i, m, a) \pi_t^\theta(da | x_i, m). \end{aligned}$$

Assumption A.1 (Finite-state well-posedness). *The initial law $\mu_0 \in \Delta_N$ is independent of θ , and the policies and transition kernels are measurable. The rewards satisfy $|r_t(x_i, m, a)| \leq \bar{r}$ and $|g(x_i, m)| \leq \bar{g}$ uniformly in their arguments. For every $t, i, j \in \llbracket 0, T-1 \rrbracket \times \llbracket 1, N \rrbracket \times \llbracket 1, N \rrbracket$, the maps $(\theta, m) \mapsto K_t^\theta(x_j \mid x_i, m)$ and $(\theta, m) \mapsto R_t^\theta(x_i, m)$ are continuous on $\Theta \times \Delta_N$, and $m \mapsto g(x_i, m)$ is continuous on Δ_N .*

Proposition A.2. *Under Assumption A.1, the original process and its finite-state perturbation are uniquely defined in law. Their objectives J and J^λ , $\lambda \in [0, 1)$, are continuous on Θ and bounded in absolute value by $T\bar{r} + \bar{g}$. In particular, they attain their maxima when Θ is non-empty and compact.*

Proof. Starting from μ_0 , compute successively

$$\mu_{t+1}^\theta(j) = \sum_{i=1}^N \mu_t^\theta(i) K_t^\theta(x_j \mid x_i, \mu_t^\theta), \quad j \in \llbracket 1, N \rrbracket.$$

Each right-hand side is nonnegative, and summing over j gives one. This recursion therefore determines a unique flow in Δ_N . Substituting that flow into the policy and transition kernels determines the joint law of states and actions by the canonical construction. To check continuity, let $\theta_n \rightarrow \theta$. The claim is immediate at $t = 0$, since $\mu_0^{\theta_n} = \mu_0^\theta = \mu_0$. Suppose that $\mu_t^{\theta_n}(i) \rightarrow \mu_t^\theta(i)$ for every i . By the continuity of K_t^θ and the fact that the sum is finite, for every j ,

$$\mu_{t+1}^{\theta_n}(j) = \sum_{i=1}^N \mu_t^{\theta_n}(i) K_t^{\theta_n}(x_j \mid x_i, \mu_t^{\theta_n}) \longrightarrow \sum_{i=1}^N \mu_t^\theta(i) K_t^\theta(x_j \mid x_i, \mu_t^\theta) = \mu_{t+1}^\theta(j).$$

Thus induction gives continuity of $\theta \mapsto \mu_t^\theta$ for every t . Consequently,

$$J(\theta) = \sum_{t=0}^{T-1} \sum_{i=1}^N \mu_t^\theta(i) R_t^\theta(x_i, \mu_t^\theta) + \sum_{i=1}^N \mu_T^\theta(i) g(x_i, \mu_T^\theta)$$

is continuous and has the stated bound.

For the perturbed process, fix a realization $q = (q_1, \dots, q_T) \in (\Delta_N)^T$ of the randomizers and write

$$M_0^{\lambda, \theta}(q) := \mu_0, \quad M_t^{\lambda, \theta}(q) := (1 - \lambda) \mu_t^\theta + \lambda q_t, \quad t \geq 1.$$

These supplied population arguments are continuous in θ by the result just proved. Conditional on q , define the state-law recursion

$$\begin{cases} \tilde{\mu}_0^{\lambda, \theta, q} &:= \mu_0, \\ \tilde{\mu}_{t+1}^{\lambda, \theta, q}(j) &:= \sum_{i=1}^N \tilde{\mu}_t^{\lambda, \theta, q}(i) K_t^\theta(x_j \mid x_i, M_t^{\lambda, \theta}(q)). \end{cases}$$

The same induction as above, now using continuity of both $\theta \mapsto M_t^{\lambda, \theta}(q)$ and $(\theta, m) \mapsto K_t^\theta(\cdot \mid x_i, m)$, shows that $\tilde{\mu}_t^{\lambda, \theta_n, q} \rightarrow \tilde{\mu}_t^{\lambda, \theta, q}$ for every t . Hence the conditional objective

$$\mathcal{J}^\lambda(\theta, q) := \sum_{t=0}^{T-1} \sum_{i=1}^N \tilde{\mu}_t^{\lambda, \theta, q}(i) R_t^\theta(x_i, M_t^{\lambda, \theta}(q)) + \sum_{i=1}^N \tilde{\mu}_T^{\lambda, \theta, q}(i) g(x_i, M_T^{\lambda, \theta}(q))$$

is continuous in θ . Moreover, $|\mathcal{J}^\lambda(\theta, q)| \leq T\bar{r} + \bar{g}$ for every (θ, q) . Since $J^\lambda(\theta) = \mathbb{E}_q[\mathcal{J}^\lambda(\theta, q)]$ and the law of q is independent of θ , dominated convergence gives continuity of J^λ . The final claim follows from compactness. $\square$

A.2.2 Continuous state space

We now take $\mathcal{X} = \mathbb{R}^d$. Equip $\mathcal{P}_2(\mathcal{X})$ and $\mathcal{P}_2(\mathcal{A})$ with the Wasserstein distance $\mathcal{W}_2$. For any $m \in \mathcal{P}_2(\mathcal{X})$, we also denote $M_2(m) := \int_{\mathcal{X}} \|x\|^2 m(dx)$. The following assumptions ensure finite second moments and a finite, continuous objective.

Assumption A.3. *The parameter set $\Theta \subset \mathbb{R}^{d_\theta}$ is non-empty, and the initial law $\mu_0 \in \mathcal{P}_2(\mathcal{X})$ is independent of θ . For each $t \in \llbracket 0, T-1 \rrbracket$, the following conditions hold.*

1. Continuity of the kernels. For every sequence $(\theta_n, x_n, m_n, a_n) \rightarrow (\theta, x, m, a)$ in $\Theta \times \mathcal{X} \times \mathcal{P}_2(\mathcal{X}) \times \mathcal{A}$,

$$\mathcal{W}_2(\pi_t^{\theta_n}(\cdot | x_n, m_n), \pi_t^\theta(\cdot | x, m)) \rightarrow 0, \quad \mathcal{W}_2(P_t(\cdot | x_n, m_n, a_n), P_t(\cdot | x, m, a)) \rightarrow 0.$$

2. Continuity of the rewards. The maps $r_t : \mathcal{X} \times \mathcal{P}_2(\mathcal{X}) \times \mathcal{A} \rightarrow \mathbb{R}$ and $g : \mathcal{X} \times \mathcal{P}_2(\mathcal{X}) \rightarrow \mathbb{R}$ are jointly continuous.

3. Quadratic growth. There exists $C \geq 0$ such that, for all $(\theta, x, m, a) \in \Theta \times \mathcal{X} \times \mathcal{P}_2(\mathcal{X}) \times \mathcal{A}$,

$$\begin{aligned} \int_{\mathcal{A}} \|b\|^2 \pi_t^\theta(\mathrm{d}b | x, m) &\leq C(1 + \|x\|^2 + M_2(m)), \\ \int_{\mathcal{X}} \|y\|^2 P_t(\mathrm{d}y | x, m, a) &\leq C(1 + \|x\|^2 + \|a\|^2 + M_2(m)), \\ |r_t(x, m, a)| &\leq C(1 + \|x\|^2 + \|a\|^2 + M_2(m)), \\ |g(x, m)| &\leq C(1 + \|x\|^2 + M_2(m)). \end{aligned}$$

Proposition A.4. Under Assumption A.3, $\mu_t^\theta \in \mathcal{P}_2(\mathcal{X})$ for all $(\theta, t) \in \Theta \times \mathcal{T}$, and

$$\sup_{\theta \in \Theta} \left(\max_{t \in \mathcal{T}} \mathbb{E}[\|X_t^\theta\|^2] + \max_{t \in \llbracket 0, T-1 \rrbracket} \mathbb{E}[\|\alpha_t^\theta\|^2] \right) < \infty.$$

Moreover, for every $t \in \mathcal{T}$, the map $\theta \mapsto \mu_t^\theta$ is continuous from Θ to $(\mathcal{P}_2(\mathcal{X}), \mathcal{W}_2)$, and the objective $J : \Theta \rightarrow \mathbb{R}$ is finite, bounded and continuous on Θ .

Proof. Fix $\theta \in \Theta$. The canonical construction yields, for $t \in \llbracket 0, T-1 \rrbracket$ and $B \in \mathcal{B}(\mathcal{X})$,

$$\mu_{t+1}^\theta(B) = \int_{\mathcal{X}} \int_{\mathcal{A}} P_t(B | x, \mu_t^\theta, a) \pi_t^\theta(\mathrm{d}a | x, \mu_t^\theta) \mu_t^\theta(\mathrm{d}x). \quad (20)$$

Starting from $\mu_0^\theta = \mu_0$, equation (20) determines the probability flow $(\mu_t^\theta)_{t \in \mathcal{T}}$.

The joint law of the states and actions is then

$$\mu_0(\mathrm{d}x_0) \prod_{t=0}^{T-1} [\pi_t^\theta(\mathrm{d}a_t | x_t, \mu_t^\theta) P_t(\mathrm{d}x_{t+1} | x_t, \mu_t^\theta, a_t)].$$

We prove by induction that each μ_t^θ has finite second moment. Whenever $\mu_t^\theta \in \mathcal{P}_2(\mathcal{X})$, write $u_t^\theta := M_2(\mu_t^\theta)$. Assumption A.3 gives

$$\mathbb{E}[\|\alpha_t^\theta\|^2] \leq C(1 + 2u_t^\theta), \quad u_{t+1}^\theta \leq C(1 + 2u_t^\theta + C(1 + 2u_t^\theta)).$$

Thus $\mu_{t+1}^\theta \in \mathcal{P}_2(\mathcal{X})$ and the second inequality bounds u_{t+1}^θ . Since $u_0^\theta = M_2(\mu_0)$ is independent of θ , C is uniform in t and θ , and T is finite, induction gives the uniform second-moment bounds. Moreover,

$$\mathbb{E}[|r_t(X_t^\theta, \mu_t^\theta, \alpha_t^\theta)|] \leq C(1 + 2u_t^\theta + \mathbb{E}[\|\alpha_t^\theta\|^2]) < \infty, \quad \mathbb{E}[|g(X_T^\theta, \mu_T^\theta)|] \leq C(1 + 2u_T^\theta) < \infty.$$

Hence $J(\theta)$ is finite and the uniform bound over $\theta \in \Theta$ follows from the uniform constant C .

To prove continuity, let $\theta_n \rightarrow \theta$ in Θ . We prove $\mu_t^{\theta_n} \rightarrow \mu_t^\theta$ in $\mathcal{W}_2$ by induction on t . The case $t = 0$ is immediate because all initial laws equal μ_0 . Suppose the claim holds at some $t < T$.

We use the characterization of convergence in the Wasserstein space in Villani et al. (2009, Definition 6.6 and Theorem 6.7). Fix a continuous function $f : \mathcal{X} \rightarrow \mathbb{R}$ satisfying $|f(y)| \leq C_f(1 + \|y\|^2)$ for some $C_f < \infty$. Applying this characterization successively to P_t and π_t^θ , together with their continuity and quadratic-growth bounds, shows that the map

$$\Theta \times \mathcal{X} \times \mathcal{P}_2(\mathcal{X}) \ni (\theta, x, m) \longmapsto h(\theta, x, m) := \int_{\mathcal{A}} \int_{\mathcal{X}} f(y) P_t(\mathrm{d}y | x, m, a) \pi_t^\theta(\mathrm{d}a | x, m) \in \mathbb{R}$$

is continuous and satisfies

$$|h(\theta, x, m)| \leq C_f \left(1 + C(1 + C)(1 + \|x\|^2 + M_2(m)) \right). \quad (21)$$

Passage to the limit follows by uniform convergence on compact sets and uniform integrability. Indeed, by (20),

$$\int_{\mathcal{X}} f d\mu_{t+1}^{\theta_n} - \int_{\mathcal{X}} f d\mu_{t+1}^{\theta} = \int_{\mathcal{X}} (h(\theta_n, x, \mu_t^{\theta_n}) - h(\theta, x, \mu_t^{\theta})) \mu_t^{\theta_n}(dx) + \left(\int_{\mathcal{X}} h(\theta, x, \mu_t^{\theta}) \mu_t^{\theta_n}(dx) - \int_{\mathcal{X}} h(\theta, x, \mu_t^{\theta}) \mu_t^{\theta}(dx) \right).$$

The second term vanishes by the induction hypothesis and the characterization, since $h(\theta, \cdot, \mu_t^{\theta})$ is continuous with quadratic growth by (21). For the first term, fix $R > 0$. On the compact set $\mathcal{X} \cap \{\|x\| \leq R\}$, $h(\theta_n, \cdot, \mu_t^{\theta_n}) \rightarrow h(\theta, \cdot, \mu_t^{\theta})$ uniformly, because h is uniformly continuous on the compact set $(\{\theta_n\}_n \cup \{\theta\}) \times (\mathcal{X} \cap \{\|x\| \leq R\}) \times (\{\mu_t^{\theta_n}\}_n \cup \{\mu_t^{\theta}\})$. On $\{\|x\| > R\}$, the bound (21) and $\sup_n M_2(\mu_t^{\theta_n}) < \infty$ give $|h(\theta_n, x, \mu_t^{\theta_n}) - h(\theta, x, \mu_t^{\theta})| \leq K(1 + \|x\|^2)$ for some K independent of n . Hence

$$\limsup_{n \rightarrow \infty} \left| \int_{\mathcal{X}} (h(\theta_n, x, \mu_t^{\theta_n}) - h(\theta, x, \mu_t^{\theta})) \mu_t^{\theta_n}(dx) \right| \leq K \limsup_{n \rightarrow \infty} \int_{\{\|x\| > R\}} (1 + \|x\|^2) \mu_t^{\theta_n}(dx),$$

and the right-hand side vanishes as $R \rightarrow \infty$ by the uniform integrability part of the characterization. Therefore $\int f d\mu_{t+1}^{\theta_n} \rightarrow \int f d\mu_{t+1}^{\theta}$. Since f is arbitrary, the characterization yields $\mu_{t+1}^{\theta_n} \rightarrow \mu_{t+1}^{\theta}$ in $\mathcal{W}_2$, which completes the induction.

Finally, for $t \in \llbracket 0, T-1 \rrbracket$,

$$\mathbb{E}[r_t(X_t^{\theta}, \mu_t^{\theta}, \alpha_t^{\theta})] = \int_{\mathcal{X}} \int_{\mathcal{A}} r_t(x, \mu_t^{\theta}, a) \pi_t^{\theta}(da | x, \mu_t^{\theta}) \mu_t^{\theta}(dx).$$

Since r_t is continuous with quadratic growth by Assumption A.3, the argument above, with $r_t(x, m, a)$ in place of $\int_{\mathcal{X}} f(y) P_t(dy | x, m, a)$, shows that $(\theta, x, m) \mapsto \int_{\mathcal{A}} r_t(x, m, a) \pi_t^{\theta}(da | x, m)$ is continuous and bounded by $C(1 + C)(1 + \|x\|^2 + M_2(m))$. The same splitting then gives $\mathbb{E}[r_t(X_t^{\theta_n}, \mu_t^{\theta_n}, \alpha_t^{\theta_n})] \rightarrow \mathbb{E}[r_t(X_t^{\theta}, \mu_t^{\theta}, \alpha_t^{\theta})]$. The terminal term is handled identically with $g(\cdot, \mu_T^{\theta_n})$ in place of $h(\theta_n, \cdot, \mu_t^{\theta_n})$. Hence $J(\theta_n) \rightarrow J(\theta)$, and J is continuous on Θ . $\square$

B PROOFS

B.1 Common definitions

Under Assumption A.3 (or Assumption B.1 below), the following maps are well defined. For $t \in \llbracket 0, T-1 \rrbracket$ and $\theta \in \Theta$, define the policy-averaged transition kernel and reward by

$$\begin{aligned} K_t^{\theta} : \mathcal{X} \times \mathcal{P}_2(\mathcal{X}) &\longrightarrow \mathcal{P}_2(\mathcal{X}), & K_t^{\theta}(dx' | x, m) &:= \int_{\mathcal{A}} P_t(dx' | x, m, a) \pi_t^{\theta}(da | x, m), \\ R_t^{\theta} : \mathcal{X} \times \mathcal{P}_2(\mathcal{X}) &\longrightarrow \mathbb{R}, & R_t^{\theta}(x, m) &:= \int_{\mathcal{A}} r_t(x, m, a) \pi_t^{\theta}(da | x, m). \end{aligned}$$

Thus $K_t^{\theta}(\cdot | x, m)$ is the conditional law of the next state given the current state x and population argument m , after averaging over the randomized action. Define also

$$\begin{aligned} \Phi_t^{\theta} : \mathcal{P}_2(\mathcal{X}) \times \mathcal{P}_2(\mathcal{X}) &\longrightarrow \mathcal{P}_2(\mathcal{X}), & \Phi_t^{\theta}(\nu, m) &:= \int_{\mathcal{X}} K_t^{\theta}(\cdot | x, m) \nu(dx), \\ \bar{R}_t^{\theta} : \mathcal{P}_2(\mathcal{X}) \times \mathcal{P}_2(\mathcal{X}) &\longrightarrow \mathbb{R}, & \bar{R}_t^{\theta}(\nu, m) &:= \int_{\mathcal{X}} R_t^{\theta}(x, m) \nu(dx), \\ \bar{g} : \mathcal{P}_2(\mathcal{X}) \times \mathcal{P}_2(\mathcal{X}) &\longrightarrow \mathbb{R}, & \bar{g}(\nu, m) &:= \int_{\mathcal{X}} g(x, m) \nu(dx). \end{aligned} \quad (22)$$

The first argument is the state law being integrated, while the second is the population argument supplied to the policy, transition kernel, and rewards.

B.2 Finite-state space

This appendix gives the assumptions and proofs for Section 3.1. Simplex derivatives and densities use the coordinates $(m_1, \dots, m_{N-1})$, with $m_N = 1 - \sum_{k < N} m_k$.

B.2.1 Perturbation estimate

Proof of Proposition 3.1. At $t = 0$, the distance is zero because $M_0^{\lambda, \theta} = \mu_0$. For $t \geq 1$, since $M_t^{\lambda, \theta}(\omega) = (1 - \lambda)\mu_t^\theta + \lambda Q_t(\omega)$ for any $\omega \in \Omega$, and $d_{\text{TV}}(\mu, \nu) \leq 1$ for any $\mu, \nu \in \mathcal{P}(\mathcal{X})$, it follows that

$$d_{\text{TV}}(M_t^{\lambda, \theta}(\omega), \mu_t^\theta) = \lambda d_{\text{TV}}(Q_t(\omega), \mu_t^\theta) \leq \lambda, \text{ for any } \omega \in \Omega.$$

□

B.2.2 Perturbation consistency

Assumption B.1 (Boundedness and Lipschitz continuity). *The maps r_t and g are uniformly bounded in all their arguments by $\bar{R}$ and $\bar{g}$, respectively. For all admissible t, θ, i, m, m' , we assume that the following bounds hold with constants independent of these arguments:*

$$\begin{cases} d_{\text{TV}}(K_t^\theta(\cdot | i, m), K_t^\theta(\cdot | i, m')) & \leq L_K d_{\text{TV}}(m, m'), \\ |R_t^\theta(i, m) - R_t^\theta(i, m')| & \leq L_R d_{\text{TV}}(m, m'), \\ |g(i, m) - g(i, m')| & \leq L_g d_{\text{TV}}(m, m'). \end{cases}$$

Lemma B.2 (Stability of the state law). *Under Assumption B.1, the law $\nu_t^{\lambda, \theta} = \mathcal{L}(X_t^{\lambda, \theta})$ satisfies*

$$d_{\text{TV}}(\nu_t^{\lambda, \theta}, \mu_t^\theta) \leq L_K t \lambda, \quad t \in \llbracket 1, T \rrbracket.$$

Proof. For $t \in \llbracket 0, T - 1 \rrbracket$, define the averaged kernel $\bar{K}_t^{\lambda, \theta} : \mathcal{X} \rightarrow \mathcal{P}(\mathcal{X})$ by $\bar{K}_t^{\lambda, \theta}(\cdot | x) := \mathbb{E}[K_t^\theta(\cdot | x, M_t^{\lambda, \theta})]$. By definition of K_t^θ , conditionally on $(X_t^{\lambda, \theta}, M_t^{\lambda, \theta}) = (x, m)$, the next state has law $K_t^\theta(\cdot | x, m)$. For $t \geq 1$, $X_t^{\lambda, \theta}$ depends on the perturbation randomizers only through $U_1, \dots, U_{t-1}$, whereas $M_t^{\lambda, \theta}$ depends on the randomizer U_t . Hence $X_t^{\lambda, \theta}$ and $M_t^{\lambda, \theta}$ are independent. At $t = 0$, $M_0^{\lambda, \theta} = \mu_0$ is deterministic. Therefore,

$$\mathcal{L}(X_{t+1}^{\lambda, \theta} | X_t^{\lambda, \theta} = x) = \bar{K}_t^{\lambda, \theta}(\cdot | x).$$

It follows that $\nu_{t+1}^{\lambda, \theta} = \nu_t^{\lambda, \theta} \bar{K}_t^{\lambda, \theta}$. On the other hand, the unperturbed population flow satisfies

$$\mu_{t+1}^\theta = \mu_t^\theta K_t^\theta(\cdot | \cdot, \mu_t^\theta).$$

For a probability measure ν and a Markov kernel K , we use the notation $(\nu K)(i) := \sum_{j=1}^N K(i | j) \nu(j)$. We bound $d_{\text{TV}}(\nu_{t+1}^{\lambda, \theta}, \mu_{t+1}^\theta)$ by inserting the intermediate law $\nu_t^{\lambda, \theta} K_t^\theta(\cdot | \cdot, \mu_t^\theta)$. By the triangle inequality,

$$d_{\text{TV}}(\nu_{t+1}^{\lambda, \theta}, \mu_{t+1}^\theta) \leq \underbrace{d_{\text{TV}}(\nu_t^{\lambda, \theta} \bar{K}_t^{\lambda, \theta}, \nu_t^{\lambda, \theta} K_t^\theta(\cdot | \cdot, \mu_t^\theta))}_{\text{(I)}} + \underbrace{d_{\text{TV}}(\nu_t^{\lambda, \theta} K_t^\theta(\cdot | \cdot, \mu_t^\theta), \mu_t^\theta K_t^\theta(\cdot | \cdot, \mu_t^\theta))}_{\text{(II)}}. \quad (23)$$

Bounding (I): Fix $j \in \llbracket 1, N \rrbracket$. Since $\bar{K}_t^{\lambda, \theta}(\cdot | j) = \mathbb{E}[K_t^\theta(\cdot | j, M_t^{\lambda, \theta})]$ and $\mu \mapsto d_{\text{TV}}(\mu, \nu) = \sup_A |\mu(A) - \nu(A)|$ is a supremum of functionals affine in μ (hence convex), Jensen's inequality gives

$$\begin{aligned} d_{\text{TV}}(\bar{K}_t^{\lambda, \theta}(\cdot | j), K_t^\theta(\cdot | j, \mu_t^\theta)) &= d_{\text{TV}}(\mathbb{E}[K_t^\theta(\cdot | j, M_t^{\lambda, \theta})], K_t^\theta(\cdot | j, \mu_t^\theta)) \\ &\leq \mathbb{E}[d_{\text{TV}}(K_t^\theta(\cdot | j, M_t^{\lambda, \theta}), K_t^\theta(\cdot | j, \mu_t^\theta))] \\ &\leq L_K \mathbb{E}[d_{\text{TV}}(M_t^{\lambda, \theta}, \mu_t^\theta)] \leq L_K \lambda, \end{aligned}$$

using Assumption B.1 in the second-to-last step and the perturbation estimate of Proposition 3.1 in the last step. Since $\nu_t^{\lambda, \theta}$ is a probability measure on $\llbracket 1, N \rrbracket$, for any A

$$\begin{aligned} |\nu_t^{\lambda, \theta} \bar{K}_t^{\lambda, \theta}(A) - \nu_t^{\lambda, \theta} K_t^\theta(\cdot | \cdot, \mu_t^\theta)(A)| &= \left| \sum_{j=1}^N \nu_t^{\lambda, \theta}(j) (\bar{K}_t^{\lambda, \theta}(A | j) - K_t^\theta(A | j, \mu_t^\theta)) \right| \\ &\leq \sum_{j=1}^N \nu_t^{\lambda, \theta}(j) d_{\text{TV}}(\bar{K}_t^{\lambda, \theta}(\cdot | j), K_t^\theta(\cdot | j, \mu_t^\theta)) \leq L_K \lambda, \end{aligned}$$

and taking the supremum over A yields (I) $\leq L_K \lambda$.

Bounding (II): Here the same kernel row $K_t^\theta(\cdot | \cdot, \mu_t^\theta)$ is applied to the two laws $\nu_t^{\lambda, \theta}$ and μ_t^θ , so it suffices to show the general contraction property

$$d_{\text{TV}}(\nu K, \mu K) \leq d_{\text{TV}}(\nu, \mu) \quad (24)$$

for any probability measures ν, μ on $\llbracket 1, N \rrbracket$ and any Markov kernel K on $\llbracket 1, N \rrbracket$. Let $A^* = \{j : \nu(j) > \mu(j)\}$; since $\sum_j (\nu(j) - \mu(j)) = 0$, the standard identity (Levin and Peres, 2026, Proposition 4.2 and Remark 4.3) $d_{\text{TV}}(\nu, \mu) = \nu(A^*) - \mu(A^*) = \sum_{j \in A^*} (\nu(j) - \mu(j)) = -\sum_{j \notin A^*} (\nu(j) - \mu(j))$ holds. Then, for any A ,

$$\begin{aligned} (\nu K)(A) - (\mu K)(A) &= \sum_{j=1}^N (\nu(j) - \mu(j)) K(A | j) \\ &= \sum_{j \in A^*} (\nu(j) - \mu(j)) K(A | j) + \sum_{j \notin A^*} (\nu(j) - \mu(j)) K(A | j) \\ &\leq \sum_{j \in A^*} (\nu(j) - \mu(j)) \cdot 1 + \sum_{j \notin A^*} (\nu(j) - \mu(j)) \cdot 0 = d_{\text{TV}}(\nu, \mu), \end{aligned}$$

where the inequality uses $K(A | j) \leq 1$ on A^* (where $\nu(j) - \mu(j) \geq 0$) and $K(A | j) \geq 0$ off A^* (where $\nu(j) - \mu(j) \leq 0$). Applying the same argument to $(\mu K)(A) - (\nu K)(A)$ gives the reverse bound, so $|(\nu K)(A) - (\mu K)(A)| \leq d_{\text{TV}}(\nu, \mu)$ for every A , and (24) follows by taking the supremum over A . Applying (24) with $K = K_t^\theta(\cdot | \cdot, \mu_t^\theta)$ gives (II) $\leq d_{\text{TV}}(\nu_t^{\lambda, \theta}, \mu_t^\theta)$. Consequently, combining the bounds on (I) and (II) in (23),

$$d_{\text{TV}}(\nu_{t+1}^{\lambda, \theta}, \mu_{t+1}^\theta) \leq d_{\text{TV}}(\nu_t^{\lambda, \theta}, \mu_t^\theta) + L_K \lambda.$$

The result follows by induction from $\nu_0^{\lambda, \theta} = \mu_0$. $\square$

Objective estimate in Theorem 3.2. Fix $t \in \llbracket 0, T-1 \rrbracket$. We add and subtract $\mathbb{E}[R_t^\theta(X_t^{\lambda, \theta}, \mu_t^\theta)]$. By the Lipschitz property in Assumption B.1 and Proposition 3.1, $|R_t^\theta(X_t^{\lambda, \theta}, M_t^{\lambda, \theta}) - R_t^\theta(X_t^{\lambda, \theta}, \mu_t^\theta)| \leq L_R d_{\text{TV}}(M_t^{\lambda, \theta}, \mu_t^\theta) \leq L_R \lambda$ point-wise. Moreover, since $|R_t^\theta| \leq \bar{R}$ and $\nu_t^{\lambda, \theta} = \mathcal{L}(X_t^{\lambda, \theta})$,

$$\left| \mathbb{E}[R_t^\theta(X_t^{\lambda, \theta}, \mu_t^\theta)] - \sum_i \mu_t^\theta(i) R_t^\theta(i, \mu_t^\theta) \right| = \left| \sum_i (\nu_t^{\lambda, \theta}(i) - \mu_t^\theta(i)) R_t^\theta(i, \mu_t^\theta) \right| \leq 2\bar{R} d_{\text{TV}}(\nu_t^{\lambda, \theta}, \mu_t^\theta),$$

using $\sum_i |\nu(i) - \mu(i)| = 2d_{\text{TV}}(\nu, \mu)$. Hence

$$\left| \mathbb{E}[R_t^\theta(X_t^{\lambda, \theta}, M_t^{\lambda, \theta})] - \sum_i \mu_t^\theta(i) R_t^\theta(i, \mu_t^\theta) \right| \leq 2\bar{R} d_{\text{TV}}(\nu_t^{\lambda, \theta}, \mu_t^\theta) + L_R \lambda.$$

The terminal contribution is bounded by $2\bar{g} d_{\text{TV}}(\nu_T^{\lambda, \theta}, \mu_T^\theta) + L_g \lambda$. Summing and applying the lemma yields

$$|J^\lambda(\theta) - J(\theta)| \leq C_T \lambda, \quad C_T = \bar{R} L_K T(T-1) + L_R T + 2\bar{g} L_K T + L_g.$$

$\square$

For the gradient estimate, use the coordinate simplex $\mathcal{Z}_N := \left\{u \in [0, 1]^{N-1} : \sum_{k=1}^{N-1} u_k \leq 1\right\}$. Write $z_t^\theta = (\mu_t^\theta(1), \dots, \mu_t^\theta(N-1))$, $z^\theta = (z_t^\theta)_{t=0}^T \in \mathbb{R}^{(T+1) \times (N-1)}$, and $D^\theta = \nabla_\theta z^\theta$. The time- t block is $D_t^\theta \in \mathbb{R}^{(N-1) \times d_\theta}$. For a prescribed coordinate flow $z = (z_t)_{t=0}^T$, let $m_t = (z_t, 1 - \sum_{k < N} z_t(k))$ and define $v_t : \Theta \times \mathcal{Z}_N^{T+1} \rightarrow \Delta_N$ and $F_N : \Theta \times \mathcal{Z}_N^{T+1} \rightarrow \mathbb{R}$ by

$$\begin{aligned} v_0^\theta(z) &= \mu_0, & v_{t+1}^\theta(z) &= v_t^\theta(z) K_t^\theta(\cdot \mid \cdot, m_t), \\ F_N(\theta, z) &= \sum_{t=0}^{T-1} \sum_i v_t^\theta(z)(i) R_t^\theta(i, m_t) + \sum_i v_T^\theta(z)(i) g(i, m_T). \end{aligned} \quad (25)$$

Here, the product of a probability vector and a transition kernel is defined component-wise by

$$[v_t^\theta(z) K_t^\theta(\cdot \mid \cdot, m_t)](j) := \sum_{i=1}^N v_t^\theta(z)(i) K_t^\theta(x_j \mid x_i, m_t), \quad j \in \llbracket 1, N \rrbracket.$$

Equivalently,

$$F_N(\theta, z) = \int_{\mathcal{X}^{T+1} \times \mathcal{A}^T} \left[\sum_{t=0}^{T-1} r_t(x_t, m_t, a_t) + g(x_T, m_T) \right] \mu_0(dx_0) \prod_{t=0}^{T-1} \left[\pi_t^\theta(da_t \mid x_t, m_t) P_t(dx_{t+1} \mid x_t, m_t, a_t) \right].$$

For fixed z , the flow $(m_t)_{t=0}^T$ is fixed, so only the policy factors depend on θ . Thus differentiating with respect to θ produces only the policy score terms. Thus F_N is the objective for a prescribed population flow. For $\omega \in \Omega$, write $q_t(\omega) = (Q_t(1)(\omega), \dots, Q_t(N-1)(\omega))$ and $q(\omega) = (q_t(\omega))_{t=1}^T$. With $z^{\lambda, \theta}(\omega) = (1-\lambda)z^\theta + \lambda q(\omega)$, and setting $z_0^{\lambda, \theta} = z_0$, we hence have that $J(\theta) = F_N(\theta, z^\theta)$ and $J^\lambda(\theta) = \mathbb{E}[F_N(\theta, z^{\lambda, \theta})]$.

Assumption B.3 (Gradient regularity). *There is an open set $\Theta^\circ \supset \Theta$ such that the following conditions hold. Derivatives in m are taken in the coordinates $(m_1, \dots, m_{N-1})$, with $m_N = 1 - \sum_{k=1}^{N-1} m_k$.*

- (1) For every $t \in \llbracket 0, T-1 \rrbracket$, $i, j \in \llbracket 1, N \rrbracket$ and ν_A -a.e. $a \in \mathcal{A}$, the map $(\theta, m) \mapsto p_t^\theta(a \mid x_i, m)$ is $\mathcal{C}^1$ on $\Theta^\circ \times \overset{\circ}{\Delta}_N$, and the maps $m \mapsto P_t(\{x_j\} \mid x_i, m, a)$, $m \mapsto r_t(x_i, m, a)$ and $m \mapsto g(x_i, m)$ are $\mathcal{C}^1$ on $\overset{\circ}{\Delta}_N$. These maps are continuous on their full domains $\Theta^\circ \times \Delta_N$ and Δ_N , respectively, and their first derivatives extend continuously from the interior to these domains.
- (2) The functions P_t , r_t , g and their first derivatives are bounded and Lipschitz in m , uniformly in $(t, i, j, a) \in \llbracket 0, T-1 \rrbracket \times \llbracket 1, N \rrbracket \times \llbracket 1, N \rrbracket \times \mathcal{A}$.
- (3) The function p_t^θ , its first derivatives in (θ, m) , and the Lipschitz constants in m of these derivatives are bounded by a measurable map $\ell : \mathcal{A} \rightarrow [0, \infty)$, uniformly in $(t, i, \theta) \in \llbracket 0, T-1 \rrbracket \times \llbracket 1, N \rrbracket \times \Theta^\circ$, $m \in \Delta_N$, for some $\ell \in L^1(\nu_A)$.
- (4) The initial law μ_0 is independent of θ .

Gradient estimate in Theorem 3.2. Starting from the constant map $\theta \mapsto \mu_0$, the population recursion defined in B.2.2 as $\mu_{t+1}^\theta(j) = \sum_{i=1}^N \mu_t^\theta(i) K_t^\theta(j \mid i, \mu_t^\theta)$ and Assumption B.3 give $\mathcal{C}^1$ maps $\mu_t : \Theta^\circ \rightarrow \Delta_N$ by induction for any $t \in \llbracket 1, T \rrbracket$. Differentiating this finite sum by the product and chain rules gives

$$\nabla_\theta \mu_{t+1}^\theta(j) = \sum_{i=1}^N K_t^\theta(j \mid i, \mu_t^\theta) \nabla_\theta \mu_t^\theta(i) + \sum_{i=1}^N \mu_t^\theta(i) \left[\nabla_\theta K_t^\theta(j \mid i, \mu_t^\theta) + (D_t^\theta)^\top \nabla_m K_t^\theta(j \mid i, \mu_t^\theta) \right] \quad (26)$$

Here the last term is the chain rule through μ_t^θ , whose first $N-1$ coordinates have derivative D_t^θ .

Since $\mu_t^\theta(N) = 1 - \sum_{i < N} \mu_t^\theta(i)$, and by Cauchy-Schwarz

$$\sum_{i=1}^N \|\nabla_\theta \mu_t^\theta(i)\| \leq 2 \sum_{i < N} \|\nabla_\theta \mu_t^\theta(i)\| \leq 2\sqrt{N-1} \|D_t^\theta\|_F,$$

where we recall that $\|D_t^\theta\|_F^2 = \sum_{i=1}^{N-1} \|\nabla_\theta \mu_t^\theta(i)\|^2$. The derivatives of K_t^θ are uniformly bounded by Assumption B.3, and $\sum_i \mu_t^\theta(i) = 1$. Thus, summing (26) over $j < N$ gives $\|D_{t+1}^\theta\|_F \leq C_{N,T}(1 + \|D_t^\theta\|_F)$. Because μ_0 does not depend on θ , $D_0^\theta = 0$. Induction over the finite horizon therefore yields

$$\sup_{\theta \in \Theta} \max_{t \in \mathcal{T}} \|D_t^\theta\|_F \leq C_{N,T}. \quad (27)$$

We next record the regularity of F_N . Recall from (25) that

$$F_N(\theta, z) = \sum_{t=0}^{T-1} \sum_{i=1}^N v_t^\theta(z)(i) R_t^\theta(i, m_t) + \sum_{i=1}^N v_T^\theta(z)(i) g(i, m_T),$$

where m_t is affine in z_t , with constant Jacobian $A_N \in \mathbb{R}^{N \times (N-1)}$. By the product and chain rules,

$$\begin{aligned} \nabla_\theta F_N(\theta, z) &= \sum_{t=0}^{T-1} \sum_i \left(\nabla_\theta v_t^\theta(z)(i) R_t^\theta(i, m_t) + v_t^\theta(z)(i) \nabla_\theta R_t^\theta(i, m_t) \right) + \sum_i \nabla_\theta v_T^\theta(z)(i) g(i, m_T), \\ \nabla_{z_t} F_N(\theta, z) &= \sum_{s=0}^{T-1} \sum_i \nabla_{z_t} v_s^\theta(z)(i) R_s^\theta(i, m_s) + \sum_i \nabla_{z_t} v_T^\theta(z)(i) g(i, m_T) + \sum_i v_t^\theta(z)(i) \nabla_m \tilde{R}_t(i, m_t), \end{aligned}$$

for $t \in \llbracket 1, T \rrbracket$, where $\tilde{R}_t = R_t^\theta$ for $t < T$ and $\tilde{R}_T = g$. By Assumption B.3 and dominated convergence, K_t^θ , R_t^θ , g and their first derivatives in (θ, m) are bounded and Lipschitz in m , uniformly in θ . Differentiating the recursion $v_{t+1}^\theta(z) = v_t^\theta(z) K_t^\theta(\cdot | \cdot, m_t)$ expresses $\nabla_\theta v_{t+1}^\theta$ and $\nabla_{z_t} v_{t+1}^\theta$ through v_t^θ , ∇v_t^θ , K_t^θ and its derivatives. Since $v_t^\theta(z) \in \Delta_N$ and $v_0^\theta = \mu_0$ does not depend on (θ, z) , induction over the finite horizon shows that v_t^θ , $\nabla_\theta v_t^\theta$ and $\nabla_{z_t} v_t^\theta$ are bounded and Lipschitz in z , uniformly in θ . Each term in the two gradients above is a product of such factors, which gives, uniformly in θ and in probability flows z, z' ,

$$\begin{aligned} \|\nabla_\theta F_N(\theta, z)\| + \|\nabla_{z_t} F_N(\theta, z)\| &\leq C_{N,T}, \\ \|\nabla_\theta F_N(\theta, z) - \nabla_\theta F_N(\theta, z')\| + \|\nabla_{z_t} F_N(\theta, z) - \nabla_{z_t} F_N(\theta, z')\| &\leq C_{N,T} \|z - z'\|. \end{aligned} \quad (28)$$

We use the Euclidean norm $\|z - z'\|^2 := \sum_{t=0}^T \sum_{k=1}^{N-1} |z_t(k) - z'_t(k)|^2$. Under this identification, $D^\theta = \nabla_\theta \in \mathbb{R}^{(T+1)(N-1)}$ is the Jacobian of the coordinate flow.

For each realization of q , set $z^{\lambda, \theta} = (1 - \lambda)z^\theta + \lambda q$. Since each q_t and μ_t^θ is a probability vector,

$$\|z^{\lambda, \theta} - z^\theta\| \leq C_{N,T} \lambda, \quad \nabla_\theta z^{\lambda, \theta} = (1 - \lambda) D^\theta. \quad (29)$$

The chain rule for $J(\theta) = F_N(\theta, z^\theta)$ gives

$$\nabla_\theta J(\theta) = \nabla_\theta F_N(\theta, z^\theta) + (D^\theta)^\top \nabla_{z_t} F_N(\theta, z^\theta).$$

For the perturbed composition the same rule gives the integrand $\nabla_\theta F_N(\theta, z^{\lambda, \theta}) + (1 - \lambda)(D^\theta)^\top \nabla_{z_t} F_N(\theta, z^{\lambda, \theta})$. The derivatives in this expression are uniformly bounded by (27) and (28). Thus they provide an integrable bound independent of q . Differentiation under the expectation defining J^λ is justified by dominated convergence, and hence

$$\nabla_\theta J^\lambda(\theta) = \mathbb{E} \left[\nabla_\theta F_N(\theta, z^{\lambda, \theta}) + (1 - \lambda)(D^\theta)^\top \nabla_{z_t} F_N(\theta, z^{\lambda, \theta}) \right]. \quad (30)$$

Subtracting these identities, and adding and subtracting $(1 - \lambda)(D^\theta)^\top \nabla_{z_t} F_N(\theta, z^\theta)$, yields

$$\begin{aligned} \nabla_\theta J^\lambda(\theta) - \nabla_\theta J(\theta) &= \mathbb{E} \left[\nabla_\theta F_N(\theta, z^{\lambda, \theta}) - \nabla_\theta F_N(\theta, z^\theta) \right. \\ &\quad \left. + (1 - \lambda)(D^\theta)^\top (\nabla_{z_t} F_N(\theta, z^{\lambda, \theta}) - \nabla_{z_t} F_N(\theta, z^\theta)) - \lambda(D^\theta)^\top \nabla_{z_t} F_N(\theta, z^\theta) \right]. \end{aligned}$$

Applying (27), (28), and (29), and using $1 - \lambda \leq 1$, gives

$$\sup_{\theta \in \Theta} \|\nabla_\theta J^\lambda(\theta) - \nabla_\theta J(\theta)\| \leq C_T^\nabla \lambda, \quad C_T^\nabla := C_{N,T}.$$

This proves the gradient estimate. $\square$

Proof of Corollary 3.3. By the proof of Theorem 3.2, J and J^λ are differentiable on $\Theta^\circ \supset \Theta$, hence continuous on the compact set Θ . Maximizers θ^* of J and θ_λ^* of J^λ therefore exist, and the objective estimate gives

$$J(\theta_\lambda^*) \geq J^\lambda(\theta_\lambda^*) - C_T \lambda \geq J^\lambda(\theta^*) - C_T \lambda \geq J(\theta^*) - 2C_T \lambda.$$

If $\theta_{\lambda_n}^* \rightarrow \bar{\theta}$ with $\lambda_n \rightarrow 0$, continuity of J gives $J(\bar{\theta}) = \lim_n J(\theta_{\lambda_n}^*) \geq J(\theta^*)$, so $\bar{\theta}$ maximizes J . $\square$

B.2.3 Density and score for the affine perturbation of the weights

Assumption B.4 (Assumptions on the law ρ of the randomizer $(U_t)_{t \in \llbracket 1, T \rrbracket}$). *We will assume the following:*

(1) *The random variables $(U_t)_{t \in \llbracket 1, T \rrbracket}$ are independent and identically distributed over $\mathbb{R}^{N-1}$, with density ρ with respect to the Lebesgue measure, with $\rho \in \mathcal{C}^1(\mathbb{R}^{N-1}; (0, +\infty))$.*

(2) *There exist constants $C_\rho > 0$ and $\beta > N$ such that, for every $u \in \mathbb{R}^{N-1}$,*

$$\rho(u)(1 + \|\nabla \log \rho(u)\|_1) \leq C_\rho e^{-\beta \|u\|_1}.$$

This condition concerns only ρ and its score. Since $\rho > 0$, its left-hand side equals $\rho(u) + \|\nabla \rho(u)\|_1$. Every nondegenerate Gaussian density satisfies it: its density has quadratic exponential decay and its score grows at most linearly.

Lemma B.5 (Law of the perturbed population). *For every $z \in \mathcal{Z}_N$ and $\lambda \in (0, 1)$, the random variable $\Psi_\lambda(z, Q_t)$ admits the density $h^\lambda(\cdot | z)$ defined in (9) with respect to Lebesgue measure in simplex coordinates:*

$$h^\lambda(y | z) = \frac{1}{\lambda^{N-1}} f_Q(q^\lambda(y | z)), \quad y \in Y_\lambda(z),$$

and the density is zero elsewhere. In particular, for every $t \in \llbracket 1, T \rrbracket$ and $\theta \in \Theta$, the random variable $M_t^{\lambda, \theta}$ defined in (7) has density $h^\lambda(\cdot | z_t^\theta)$ in these coordinates.

Proof. Fix $z \in \mathcal{Z}_N$ and $\lambda \in (0, 1)$. The affine map $\Psi_\lambda(z, \cdot)$ defined in (8) is a $\mathcal{C}^1$ -diffeomorphism from the simplex interior onto $Y_\lambda(z)$, with inverse $q^\lambda(\cdot | z)$ and Jacobian determinant λ^{N-1} .

Since $Q_t = \varphi(U_t)$ and U_t has density ρ , the change-of-variables formula gives

$$f_Q(q) = \frac{\rho(\varphi^{-1}(q))}{\prod_{i=1}^N q_i}.$$

Applying the change-of-variables formula again gives

$$h^\lambda(y | z) = \frac{1}{\lambda^{N-1}} f_Q(q^\lambda(y | z)), \quad y \in Y_\lambda(z),$$

with density zero elsewhere. Taking $z = z_t^\theta$ gives the result for $M_t^{\lambda, \theta}$. $\square$

Lemma B.6 (Extension at the simplex boundary). *Under Assumption B.4, identify the simplex with its first $N - 1$ coordinates and define*

$$\tilde{f}_Q : \mathbb{R}^{N-1} \longrightarrow [0, \infty), \quad \tilde{f}_Q(q) := \begin{cases} f_Q(q), & q \in \mathcal{Z}_N^\circ, \\ 0, & q \notin \mathcal{Z}_N^\circ. \end{cases}$$

Then $\tilde{f}_Q \in \mathcal{C}_c^1(\mathbb{R}^{N-1})$ and both its value and its gradient vanish on $\partial \mathcal{Z}_N$.

Proof. For $q \in \mathcal{Z}_N^\circ$, set $q_N = 1 - \sum_{i=1}^{N-1} q_i$, $u = \varphi^{-1}(q)$, $r = \|u\|_1$ and $u_N = 0$. Writing $S = \sum_{i=1}^N e^{u_i}$ gives $q_i = e^{u_i}/S$. If $a = \max_{1 \leq i \leq N} u_i \geq 0$, then $S \leq Ne^a$, $a - u_i \leq r$ and $Na - \sum_{i=1}^{N-1} u_i \leq (N-1)r$. Hence

$$\max_{1 \leq i \leq N} \frac{1}{q_i} \leq Ne^r, \quad \frac{1}{\prod_{i=1}^N q_i} = S^N e^{-\sum_{i=1}^{N-1} u_i} \leq N^N e^{(N-1)r}.$$

Using $f_Q(q) = \rho(u) / \prod_{i=1}^N q_i$ and the formula for $H_k = \partial_{q_k} \log f_Q$, these bounds give

$$\begin{aligned} |f_Q(q)| + \|\nabla_q f_Q(q)\|_1 &\leq C_N e^{Nr} \rho(u) (1 + \|\nabla \log \rho(u)\|_1) \\ &\leq C_N C_\rho e^{-(\beta-N)r}, \end{aligned}$$

where C_N depends only on N .

As $q \in \mathcal{Z}_N^\circ$ approaches any point of $\partial \mathcal{Z}_N$, we have $\min_{1 \leq i \leq N} q_i \rightarrow 0$. The bound on $\max_{1 \leq i \leq N} q_i^{-1}$ therefore gives

$$r = \|\varphi^{-1}(q)\|_1 \geq \log \frac{1}{N \min_{1 \leq i \leq N} q_i} \rightarrow \infty.$$

Since $\beta > N$, the preceding estimate shows that $f_Q(q) \rightarrow 0$ and $\nabla_q f_Q(q) \rightarrow 0$ as q approaches the boundary from the interior. In particular, $\tilde{f}_Q$ is continuous at every boundary point.

It remains to prove differentiability there. Define

$$\partial \mathcal{Z}_N := \left\{ z \in \mathcal{Z}_N : z_k = 0 \text{ for some } k \text{ or } \sum_{k=1}^{N-1} z_k = 1 \right\}, \quad \mathcal{Z}_N^\circ := \left\{ z \in (0, \infty)^{N-1} : \sum_{k=1}^{N-1} z_k < 1 \right\}.$$

Fix $\bar{q} \in \partial \mathcal{Z}_N$ and let $q \in \mathcal{Z}_N^\circ$. The segment from $\bar{q}$ to q lies in the interior except at $\bar{q}$. Applying the one-dimensional mean value theorem along this segment gives a point ξ_q in its open part such that

$$\tilde{f}_Q(q) - \tilde{f}_Q(\bar{q}) = \nabla_q f_Q(\xi_q) \cdot (q - \bar{q}).$$

Since ξ_q lies between $\bar{q}$ and q , $\|\xi_q - \bar{q}\| \leq \|q - \bar{q}\|$. Thus $\xi_q \rightarrow \bar{q}$ as $q \rightarrow \bar{q}$, and the boundary limit of the interior gradient gives $\nabla_q f_Q(\xi_q) \rightarrow 0$. By the Cauchy-Schwarz inequality,

$$\frac{|\tilde{f}_Q(q) - \tilde{f}_Q(\bar{q})|}{\|q - \bar{q}\|} \leq \|\nabla_q f_Q(\xi_q)\| \rightarrow 0.$$

For $q \notin \mathcal{Z}_N^\circ$, the numerator is zero by definition. Thus the difference quotient tends to zero from every direction, proving that $\nabla \tilde{f}_Q(\bar{q}) = 0$.

The gradient is continuous across the boundary: it tends to zero from the interior and is zero outside. Therefore $\tilde{f}_Q \in \mathcal{C}^1(\mathbb{R}^{N-1})$. Finally, its support is contained in the closed bounded set $\mathcal{Z}_N$, so $\tilde{f}_Q \in \mathcal{C}_c^1(\mathbb{R}^{N-1})$. $\square$

Lemma B.7 (Stein identity on the simplex). *Suppose Assumption B.4 holds and let $Q = \varphi(U)$, with $U \sim \rho$. Let $\kappa \in (0, 1)$, $c \in \Delta_N$, and let $h : \Delta_N \rightarrow \mathbb{R}$ be continuous, $\mathcal{C}^1$ on $\mathring{\Delta}_N$ in the coordinates $(m_1, \dots, m_{N-1})$, with bounded gradient. Then, for every $k \in \llbracket 1, N-1 \rrbracket$,*

$$\mathbb{E}[h((1-\kappa)c + \kappa Q) H_k(Q)] = -\kappa \mathbb{E}[\partial_k h((1-\kappa)c + \kappa Q)]. \quad (31)$$

In particular, $\mathbb{E}[H(Q)] = 0$.

Proof. Define $g : \Delta_N \rightarrow \mathbb{R}$ by $g(q) := h((1-\kappa)c + \kappa q)$. Since $\kappa > 0$, for $q \in \mathring{\Delta}_N$ every coordinate of $(1-\kappa)c + \kappa q$ is strictly positive, hence $(1-\kappa)c + \kappa q \in \mathring{\Delta}_N$, even if $c \in \partial \Delta_N$. Therefore g is continuous on Δ_N , $\mathcal{C}^1$ on $\mathring{\Delta}_N$, and

$$\partial_k g(q) = \kappa \partial_k h((1-\kappa)c + \kappa q) \quad (32)$$

is bounded. Using Lemma B.6, since $H_k = \partial_k \log f_Q$ on $\mathring{\Delta}_N$, we have $H_k f_Q = \partial_k f_Q = \partial_k \tilde{f}_Q$ there, and consequently

$$\mathbb{E}[g(Q) H_k(Q)] = \int_{\mathring{\Delta}_N} g(q) \partial_k \tilde{f}_Q(q) dq, \quad (33)$$

where the integrand is bounded, because g is continuous on the compact set Δ_N and $\tilde{f}_Q \in \mathcal{C}_c^1(\mathbb{R}^{N-1})$.

Splitting the integral. Throughout, points of $\overset{\circ}{\Delta}_N$ are written in the coordinates $q = (q_1, \dots, q_{N-1})$, with $q_N = 1 - \sum_{i=1}^{N-1} q_i$. Fix k and separate the k -th coordinate from the others: write $q_{-k} := (q_i)_{i \neq k, i \leq N-1}$, and denote by (s, q_{-k}) the point whose k -th coordinate is s and whose other coordinates are q_{-k} . A point (s, q_{-k}) lies in $\overset{\circ}{\Delta}_N$ if and only if

$$q_{-k} \in S_{-k} := \left\{ q_{-k} : q_i > 0 \text{ for all } i \neq k, \sum_{i \neq k} q_i < 1 \right\} \quad \text{and} \quad 0 < s < b(q_{-k}) := 1 - \sum_{i \neq k} q_i.$$

By Fubini's theorem, which applies because the integrand in (33) is bounded on a bounded set,

$$\mathbb{E}[g(Q)H_k(Q)] = \int_{S_{-k}} \left(\int_0^{b(q_{-k})} g(s, q_{-k}) \partial_k \tilde{f}_Q(s, q_{-k}) ds \right) dq_{-k}. \quad (34)$$

Integration by parts in one variable. Fix $q_{-k} \in S_{-k}$ and write $b = b(q_{-k})$. The functions $s \mapsto g(s, q_{-k})$ and $s \mapsto \tilde{f}_Q(s, q_{-k})$ are continuous on $[0, b]$ and C^1 on $(0, b)$, with bounded derivatives. Both endpoints lie on $\partial\Delta_N$: at $s = 0$, the k -th coordinate is $q_k = 0$, and at $s = b$, the last coordinate is $q_N = 1 - b - \sum_{i \neq k} q_i = 0$. A point of Δ_N with a zero coordinate lies on $\partial\Delta_N$. By Assumption B.4, $\tilde{f}_Q = 0$ on $\partial\Delta_N$, so $\tilde{f}_Q(0, q_{-k}) = \tilde{f}_Q(b, q_{-k}) = 0$. Integrating by parts on $[0, b]$ therefore gives

$$\begin{aligned} \int_0^b g(s, q_{-k}) \partial_k \tilde{f}_Q(s, q_{-k}) ds &= \left[g(s, q_{-k}) \tilde{f}_Q(s, q_{-k}) \right]_{s=0}^{s=b} - \int_0^b \partial_k g(s, q_{-k}) \tilde{f}_Q(s, q_{-k}) ds \\ &= - \int_0^b \partial_k g(s, q_{-k}) \tilde{f}_Q(s, q_{-k}) ds. \end{aligned}$$

Conclusion. Substituting this into (34), recombining the two integrals by Fubini's theorem, and using $\tilde{f}_Q = f_Q$ on $\overset{\circ}{\Delta}_N$ and (32),

$$\mathbb{E}[g(Q)H_k(Q)] = - \int_{\overset{\circ}{\Delta}_N} \partial_k g(q) f_Q(q) dq = -\kappa \mathbb{E}[\partial_k h((1 - \kappa)c + \kappa Q)].$$

Since $g(Q) = h((1 - \kappa)c + \kappa Q)$, this is (31). Taking h constant gives $\mathbb{E}[H_k(Q)] = 0$ for every $k \in \llbracket 1, N - 1 \rrbracket$, hence $\mathbb{E}[H(Q)] = 0$. $\square$

B.2.4 Model-free policy-gradient representation

We now turn to the computation of the policy gradient representation.

Proof of Theorem 3.4. Fix $\lambda \in (0, 1)$ and write $Q = (Q_1, \dots, Q_T)$. Recall that the return $\mathcal{R}_\theta^\lambda$ and the policy score $\mathcal{S}_{\text{pol}}^{\lambda, \theta}$ are defined in (5), and the population score $\mathcal{S}_{\text{mf}}^{\lambda, \theta}$ is defined in (11). Throughout the proof, $\nabla_\theta F_N(\theta, z)$ denotes the derivative with respect to the first argument, with z held fixed.

Recall that $z_0^{\lambda, \theta} = z_0$ and $z_t^{\lambda, \theta} = (1 - \lambda)z_t^\theta + \lambda q_t$ for $t \in \llbracket 1, T \rrbracket$, where q_t denotes the first $N - 1$ coordinates of Q_t . Since $D_0^\theta = 0$, equation (30) gives

$$\nabla_\theta J^\lambda(\theta) = \mathbb{E}[\nabla_\theta F_N(\theta, z^{\lambda, \theta})] + (1 - \lambda) \sum_{t=1}^T (D_t^\theta)^\top \mathbb{E}[\nabla_{z_t} F_N(\theta, z^{\lambda, \theta})]. \quad (35)$$

We identify the two terms separately.

Policy-score term. Fix a prescribed coordinate flow z and write $m_t = (z_t, 1 - \sum_{k=1}^{N-1} z_t(k))$. For a trajectory $(x, a) = (x_0, \dots, x_T, a_0, \dots, a_{T-1})$, set

$$\mathcal{R}(x, a; m) := \sum_{t=0}^{T-1} r_t(x_t, m_t, a_t) + g(x_T, m_T).$$

The integral representation of the prescribed-flow objective is

$$F_N(\theta, z) = \int_{\mathcal{X}^{T+1} \times \mathcal{A}^T} \mathcal{R}(x, a; m) \mu_0(dx_0) \prod_{t=0}^{T-1} [p_t^\theta(a_t | x_t, m_t) \nu_A(da_t) P_t(dx_{t+1} | x_t, m_t, a_t)].$$

With z fixed, the return and transition kernels are independent of θ , only the policy densities depend on θ . Assumption B.3 justifies differentiation under the integral, giving

$$\begin{aligned} \nabla_\theta F_N(\theta, z) &= \int_{\mathcal{X}^{T+1} \times \mathcal{A}^T} \mathcal{R}(x, a; m) \left[\sum_{s=0}^{T-1} \nabla_\theta \log p_s^\theta(a_s | x_s, m_s) \right] \mu_0(dx_0) \\ &\quad \times \prod_{t=0}^{T-1} \left[p_t^\theta(a_t | x_t, m_t) \nu_A(da_t) P_t(dx_{t+1} | x_t, m_t, a_t) \right]. \end{aligned}$$

We now evaluate this partial derivative at $z = z^{\lambda, \theta}$. Conditionally on Q , the product of kernels above is the trajectory law of the perturbed process. Hence

$$\nabla_\theta F_N(\theta, z)|_{z=z^{\lambda, \theta}} = \mathbb{E}[\mathcal{R}_\theta^\lambda \mathcal{S}_{\text{pol}}^{\lambda, \theta} | Q].$$

Taking expectations yields

$$\mathbb{E}[\nabla_\theta F_N(\theta, z^{\lambda, \theta})] = \mathbb{E}[\mathcal{R}_\theta^\lambda \mathcal{S}_{\text{pol}}^{\lambda, \theta}]. \quad (36)$$

Population-score term. Fix $t \in \llbracket 1, T \rrbracket$, and condition on $Q_{-t} := (Q_s)_{s \neq t}$. For fixed Q_{-t} , define $h_t : \Delta_N \rightarrow \mathbb{R}$ by

$$h_t(m) := F_N(\theta, z_0^{\lambda, \theta}, \dots, z_{t-1}^{\lambda, \theta}, (m_1, \dots, m_{N-1}), z_{t+1}^{\lambda, \theta}, \dots, z_T^{\lambda, \theta})$$

Derivatives of h_t are taken in the coordinates $(m_1, \dots, m_{N-1})$, with $m_N = 1 - \sum_{k=1}^{N-1} m_k$. Thus, for every $k \in \llbracket 1, N-1 \rrbracket$,

$$\begin{aligned} h_t((1-\lambda)\mu_t^\theta + \lambda Q_t) &= F_N(\theta, z^{\lambda, \theta}), \\ \partial_{m_k} h_t((1-\lambda)\mu_t^\theta + \lambda Q_t) &= \partial_{z_t(k)} F_N(\theta, z^{\lambda, \theta}). \end{aligned}$$

The required regularity follows from Assumption B.3 and (28). Since Q_t is independent of Q_{-t} , Lemma B.7, applied component-wise, gives

$$\mathbb{E}[F_N(\theta, z^{\lambda, \theta}) H(Q_t) | Q_{-t}] = -\lambda \mathbb{E}[\nabla_{z_t} F_N(\theta, z^{\lambda, \theta}) | Q_{-t}].$$

Taking expectations,

$$\mathbb{E}[F_N(\theta, z^{\lambda, \theta}) H(Q_t)] = -\lambda \mathbb{E}[\nabla_{z_t} F_N(\theta, z^{\lambda, \theta})]. \quad (37)$$

Moreover, since $H(Q_t)$ is $\sigma(Q)$ -measurable,

$$\mathbb{E}[\mathcal{R}_\theta^\lambda H(Q_t)] = \mathbb{E}[\mathbb{E}[\mathcal{R}_\theta^\lambda | Q] H(Q_t)] = \mathbb{E}[F_N(\theta, z^{\lambda, \theta}) H(Q_t)].$$

Combining this with (37) yields

$$-\frac{1-\lambda}{\lambda} (D_t^\theta)^\top \mathbb{E}[\mathcal{R}_\theta^\lambda H(Q_t)] = (1-\lambda) (D_t^\theta)^\top \mathbb{E}[\nabla_{z_t} F_N(\theta, z^{\lambda, \theta})].$$

Summing over $t \in \llbracket 1, T \rrbracket$ gives

$$\mathbb{E}[\mathcal{R}_\theta^\lambda \mathcal{S}_{\text{mf}}^{\lambda, \theta}] = (1-\lambda) \sum_{t=1}^T (D_t^\theta)^\top \mathbb{E}[\nabla_{z_t} F_N(\theta, z^{\lambda, \theta})]. \quad (38)$$

Finally, combining (36) and (38) with (35) gives $\nabla_\theta J^\lambda(\theta) = \mathbb{E}[\mathcal{R}_\theta^\lambda (\mathcal{S}_{\text{pol}}^{\lambda, \theta} + \mathcal{S}_{\text{mf}}^{\lambda, \theta})]$. $\square$

B.2.5 Bias and mean-square error

Assumption B.8 (Score regularity). *The policy scores and the population score satisfy*

$$\sup_{t,\theta,i,m} \int_{\mathcal{A}} \|\nabla_{\theta} \log p_t^{\theta}(a \mid x_i, m)\|^2 \pi_t^{\theta}(\mathrm{d}a \mid x_i, m) < \infty,$$

$$\mathbb{E}[\|H(Q_t)\|^2] < \infty.$$

Proof. Fix θ . Throughout, C may change from line to line and depend on the fixed dimensions, horizon, and regularity bounds, but not on $\theta, \lambda, \eta, B, M, n$. Write $\mathcal{Z}_N = \left\{u \in [0, 1]^{N-1} : \sum_{k=1}^{N-1} u_k \leq 1\right\}$.

For $s \in \llbracket 1, T \rrbracket$, write

$$z_s^{\theta} := (\mu_s^{\theta}(1), \dots, \mu_s^{\theta}(N-1)), \quad \widehat{z}_s := (\widehat{\mu}_s^M(1), \dots, \widehat{\mu}_s^M(N-1)),$$

and set $z^{\theta} = (z_s^{\theta})_{s=1}^T$ and $\widehat{z} = (\widehat{z}_s)_{s=1}^T$. Here $\widehat{\mu}_s^M = \frac{1}{M} \sum_{\ell=1}^M \delta_{X_s^{\ell}}$ is the empirical population law. We use the Euclidean norm for vectors and stacked coordinate flows. In particular, $\|\widehat{z} - z^{\theta}\|^2 = \sum_{s=1}^T \sum_{k=1}^{N-1} |\widehat{\mu}_s^M(k) - \mu_s^{\theta}(k)|^2$. The time-zero population argument remains fixed at μ_0 .

Conditional mean and its stability. For deterministic $c = (c_t)_{t=1}^T \in \mathcal{Z}_N^T$, $D \in (\mathbb{R}^{(N-1) \times d_{\theta}})^T$, and $\kappa \in [0, 1]$, set $z^{\kappa} = (z_0, (1 - \kappa)c_1 + \kappa q_1, \dots, (1 - \kappa)c_T + \kappa q_T)$, where q_s consists of the $N-1$ coordinates of Q_s . Define

$$\Gamma^{\kappa} : \mathcal{Z}_N^T \times (\mathbb{R}^{(N-1) \times d_{\theta}})^T \rightarrow \mathbb{R}^{d_{\theta}}, \quad \Gamma^{\kappa}(c, D) := \mathbb{E}[\nabla_{\theta} F_N(\theta, z^{\kappa})] + (1 - \kappa) \sum_{s=1}^T D_s^{\top} \mathbb{E}[\nabla_{z_s} F_N(\theta, z^{\kappa})].$$

For $s \in \llbracket 1, T \rrbracket$, let $\bar{c}_s = (c_s, 1 - \sum_{k=1}^{N-1} c_s(k))$ and set

$$m_0^{\kappa} = \mu_0, \quad m_s^{\kappa} = (1 - \kappa)\bar{c}_s + \kappa Q_s.$$

Define

$$X_0^{\kappa} \sim \mu_0, \quad \alpha_s^{\kappa} \sim \pi_s^{\theta}(\cdot \mid X_s^{\kappa}, m_s^{\kappa}), \quad X_{s+1}^{\kappa} \sim P_s(\cdot \mid X_s^{\kappa}, m_s^{\kappa}, \alpha_s^{\kappa}), \quad s \in \llbracket 0, T-1 \rrbracket.$$

Its return and policy score are

$$\mathcal{R}^{\kappa} := \sum_{s=0}^{T-1} r_s(X_s^{\kappa}, m_s^{\kappa}, \alpha_s^{\kappa}) + g(X_T^{\kappa}, m_T^{\kappa}), \quad \mathcal{S}_{\text{pol}}^{\kappa} := \sum_{s=0}^{T-1} \nabla_{\theta} \log p_s^{\theta}(\alpha_s^{\kappa} \mid X_s^{\kappa}, m_s^{\kappa}),$$

where each policy derivative holds the action, state and population argument fixed.

For $\kappa > 0$, the policy-score identity (36) and the Stein identity (37) give

$$\mathbb{E}[\mathcal{R}^{\kappa} \mathcal{S}_{\text{pol}}^{\kappa}] = \mathbb{E}[\nabla_{\theta} F_N(\theta, z^{\kappa})], \quad \mathbb{E}[\mathcal{R}^{\kappa} H(Q_s)] = -\kappa \mathbb{E}[\nabla_{z_s} F_N(\theta, z^{\kappa})].$$

Multiplying the second identity by $-\frac{1-\kappa}{\kappa} D_s^{\top}$ and summing over s shows that

$$\mathbb{E}\left[\mathcal{R}^{\kappa} \left(\mathcal{S}_{\text{pol}}^{\kappa} - \frac{1-\kappa}{\kappa} \sum_{s=1}^T D_s^{\top} H(Q_s)\right)\right] = \Gamma^{\kappa}(c, D).$$

Consequently, $\Gamma^{\kappa}(c, D)$ is the expectation of the score estimator using population coordinates c and sensitivity matrices D .

Since $\Gamma^{\kappa}(c, D) - \Gamma^{\kappa}(c, D^{\theta}) = (1 - \kappa) \sum_s (D_s - D_s^{\theta})^{\top} \mathbb{E}[\nabla_{z_s} F_N(\theta, z^{\kappa})]$, the derivative bounds (28) imply

$$\|\Gamma^{\kappa}(c, D) - \Gamma^{\kappa}(c, D^{\theta})\| \leq C \sum_{s=1}^T \|D_s - D_s^{\theta}\|_{\text{F}}.$$

Moreover, the time-zero block of $z^\kappa - (z_0, z^\theta)$ vanishes and, for $s \geq 1$,

$$z_s^\kappa - z_s^\theta = (1 - \kappa)(c_s - z_s^\theta) + \kappa(q_s - z_s^\theta).$$

Since $q_s, z_s^\theta \in \mathcal{Z}_N$ have Euclidean norm at most one, $\|q_s - z_s^\theta\| \leq 2$, and the triangle inequality gives

$$\|z^\kappa - (z_0, z^\theta)\| \leq \|c - z^\theta\| + 2\kappa\sqrt{T}.$$

Write $\bar{z}^\theta = (z_0, z^\theta)$. Using $\Gamma^0(z^\theta, D^\theta) = \nabla_\theta J(\theta)$, we obtain

$$\begin{aligned} \Gamma^\kappa(c, D^\theta) - \nabla_\theta J(\theta) &= \mathbb{E}[\nabla_\theta F_N(\theta, z^\kappa) - \nabla_\theta F_N(\theta, \bar{z}^\theta)] \\ &\quad + (1 - \kappa) \sum_{s=1}^T (D_s^\theta)^\top \mathbb{E}[\nabla_{z_s} F_N(\theta, z^\kappa) - \nabla_{z_s} F_N(\theta, \bar{z}^\theta)] - \kappa \sum_{s=1}^T (D_s^\theta)^\top \nabla_{z_s} F_N(\theta, \bar{z}^\theta). \end{aligned}$$

The first two terms are controlled by the Lipschitz continuity in z of the derivatives of F_N , while the last term is bounded by $C\kappa$. Indeed, (28) and (27) give

$$\begin{aligned} \|\Gamma^\kappa(c, D^\theta) - \nabla_\theta J(\theta)\| &\leq C\mathbb{E}[\|z^\kappa - \bar{z}^\theta\|] + C\kappa \\ &\leq C(\|c - z^\theta\| + \kappa) \end{aligned}$$

Combining these estimates yields

$$\|\Gamma^\kappa(c, D) - \nabla_\theta J(\theta)\| \leq C\left(\kappa + \|c - z^\theta\| + \sum_{s=1}^T \|D_s - D_s^\theta\|_F\right). \quad (39)$$

Population-flow error. Let $\mathcal{F}_t^{\text{pop}} := \sigma(X_s^\ell : s \in \llbracket 0, t \rrbracket, \ell \in \llbracket 1, M \rrbracket)$ where σ refers to the generated sigma-algebra. The particles use independent simulation randomness at each time and rollout step. Conditionally on $\mathcal{F}_t^{\text{pop}}$, the variables $(X_{t+1}^\ell)_{\ell \in \llbracket 1, M \rrbracket}$ are therefore independent, with laws $K_t^\theta(\cdot \mid X_t^\ell, \hat{\mu}_t^M)$. Therefore, $\mathbb{E}[\hat{\mu}_{t+1}^M \mid \mathcal{F}_t^{\text{pop}}] = \Phi_t^\theta(\hat{\mu}_t^M, \hat{\mu}_t^M)$, with Φ_t^θ from (22).

For each $j \in \llbracket 1, N \rrbracket$, writing $p_{\ell j} := K_t^\theta(\{x_j\} \mid X_t^\ell, \hat{\mu}_t^M)$,

$$\hat{\mu}_{t+1}^M(j) - \Phi_t^\theta(\hat{\mu}_t^M, \hat{\mu}_t^M)(j) = \frac{1}{M} \sum_{\ell=1}^M (\mathbf{1}_{\{X_{t+1}^\ell = x_j\}} - p_{\ell j}).$$

Conditionally on $\mathcal{F}_t^{\text{pop}}$, the summands are independent and centered, with variances $p_{\ell j}(1 - p_{\ell j})$. Since the squared Euclidean norm is the sum of the squared coordinates,

$$\mathbb{E}[\|\hat{\mu}_{t+1}^M - \Phi_t^\theta(\hat{\mu}_t^M, \hat{\mu}_t^M)\|^2 \mid \mathcal{F}_t^{\text{pop}}] = \frac{1}{M^2} \sum_{j=1}^N \sum_{\ell=1}^M p_{\ell j}(1 - p_{\ell j}) \leq \frac{1}{M^2} \sum_{\ell=1}^M \sum_{j=1}^N p_{\ell j} = \frac{1}{M},$$

where we used $1 - p_{\ell j} \leq 1$ and $\sum_{j=1}^N p_{\ell j} = 1$.

For $m, m' \in \Delta_N$, insert the intermediate law $\Phi_t^\theta(m, m')$ to obtain

$$\begin{aligned} d_{\text{TV}}(\Phi_t^\theta(m, m), \Phi_t^\theta(m', m')) &\leq d_{\text{TV}}(\Phi_t^\theta(m, m), \Phi_t^\theta(m, m')) + d_{\text{TV}}(\Phi_t^\theta(m, m'), \Phi_t^\theta(m', m')) \\ &\leq \sum_{i=1}^N m(i) d_{\text{TV}}(K_t^\theta(\cdot \mid x_i, m), K_t^\theta(\cdot \mid x_i, m')) + d_{\text{TV}}(m, m') \\ &\leq (1 + L_K) d_{\text{TV}}(m, m') \end{aligned}$$

The first term uses the Lipschitz bound of Assumption B.1 on the kernel rows. For the second term, the population argument is fixed at m' , so the total-variation contraction (24) applies.

Since $\mu_{t+1}^\theta = \Phi_t^\theta(\mu_t^\theta, \mu_t^\theta)$, write

$$\hat{\mu}_{t+1}^M - \mu_{t+1}^\theta = \underbrace{\hat{\mu}_{t+1}^M - \Phi_t^\theta(\hat{\mu}_t^M, \hat{\mu}_t^M)}_{\xi_{t+1}} + \underbrace{\Phi_t^\theta(\hat{\mu}_t^M, \hat{\mu}_t^M) - \Phi_t^\theta(\mu_t^\theta, \mu_t^\theta)}_{\zeta_t}.$$

Since $\mathbb{E}[\xi_{t+1} \mid \mathcal{F}_t^{\text{pop}}] = 0$ and ζ_t is $\mathcal{F}_t^{\text{pop}}$ -measurable,

$$\mathbb{E}[\langle \xi_{t+1}, \zeta_t \rangle] = \mathbb{E}[\langle \mathbb{E}[\xi_{t+1} \mid \mathcal{F}_t^{\text{pop}}], \zeta_t \rangle] = 0.$$

Moreover, the preceding total-variation estimate gives

$$\begin{aligned} \|\zeta_t\| &\leq \|\zeta_t\|_1 = 2d_{\text{TV}}(\Phi_t^\theta(\hat{\mu}_t^M, \hat{\mu}_t^M), \Phi_t^\theta(\mu_t^\theta, \mu_t^\theta)) \\ &\leq 2(1 + L_K)d_{\text{TV}}(\hat{\mu}_t^M, \mu_t^\theta) = (1 + L_K)\|\hat{\mu}_t^M - \mu_t^\theta\|_1 \\ &\leq \sqrt{N}(1 + L_K)\|\hat{\mu}_t^M - \mu_t^\theta\|, \end{aligned}$$

where the first and second equalities use the total variation definition of [Levin and Peres \(2026\)](#) and the last inequality follows from Cauchy-Schwarz. Consequently,

$$\begin{aligned} \mathbb{E}[\|\hat{\mu}_{t+1}^M - \mu_{t+1}^\theta\|^2] &= \mathbb{E}[\|\xi_{t+1}\|^2] + \mathbb{E}[\|\zeta_t\|^2] \\ &\leq \frac{1}{M} + N(1 + L_K)^2 \mathbb{E}[\|\hat{\mu}_t^M - \mu_t^\theta\|^2]. \end{aligned}$$

The initial particles are i.i.d. with law μ_0 , so the same variance calculation gives $\mathbb{E}[\|\hat{\mu}_0^M - \mu_0\|^2] \leq 1/M$. Induction over the finite horizon gives

$$\max_{s \in \mathcal{T}} \mathbb{E}[\|\hat{\mu}_s^M - \mu_s^\theta\|^2] \leq \frac{C}{M}.$$

Since $\hat{z}_s$ and z_s^θ contain the first $N - 1$ coordinates of these probability vectors,

$$\mathbb{E}[\|\hat{z} - z^\theta\|^2] \leq \sum_{s=1}^T \mathbb{E}[\|\hat{\mu}_s^M - \mu_s^\theta\|^2] \leq \frac{C}{M}. \quad (40)$$

Sensitivity error. Let $\mathcal{B}_t$ collect the random inputs used to simulate auxiliary batch t . These inputs are independent across batches and independent of the population particle system. Set $\mathcal{F}_{t-1}^{\text{off}} := \mathcal{F}_T^{\text{pop}} \vee \sigma(\mathcal{B}_1, \dots, \mathcal{B}_{t-1})$. Conditionally on $\mathcal{F}_{t-1}^{\text{off}}$, simulate n independent rollouts of length t . For rollout $r \in \llbracket 1, n \rrbracket$, draw independent randomizers $\bar{Q}_s^{(t,r)}$, $s \in \llbracket 1, t-1 \rrbracket$, with the same law as Q_s , and set

$$\bar{m}_0^{(t,r)} = \mu_0, \quad \bar{m}_s^{(t,r)} = (1 - \eta)\hat{\mu}_s^M + \eta\bar{Q}_s^{(t,r)}, \quad s \in \llbracket 1, t-1 \rrbracket.$$

Define

$$\bar{X}_0^{(t,r)} \sim \mu_0, \quad \bar{\alpha}_s^{(t,r)} \sim \pi_s^\theta(\cdot \mid \bar{X}_s^{(t,r)}, \bar{m}_s^{(t,r)}), \quad \bar{X}_{s+1}^{(t,r)} \sim P_s(\cdot \mid \bar{X}_s^{(t,r)}, \bar{m}_s^{(t,r)}, \bar{\alpha}_s^{(t,r)}), \quad s \in \llbracket 0, t-1 \rrbracket.$$

Thus $\bar{X}_t^{(t,r)}$ is the terminal state of rollout r in auxiliary batch t . Set $\hat{D}_0 = 0$. For $t \in \llbracket 1, T \rrbracket$, define $\hat{D}_t \in \mathbb{R}^{(N-1) \times d_\theta}$ recursively by its rows of size d_θ :

$$(\hat{D}_t)_{k,:} := \frac{1}{n} \sum_{r=1}^n \mathbf{1}_{\{\bar{X}_t^{(t,r)} = x_k\}} \left[\sum_{s=0}^{t-1} \nabla_\theta \log p_s^\theta(\bar{\alpha}_s^{(t,r)} \mid \bar{X}_s^{(t,r)}, \bar{m}_s^{(t,r)}) - \frac{1-\eta}{\eta} \sum_{s=1}^{t-1} \hat{D}_s^\top H(\bar{Q}_s^{(t,r)}) \right]^\top, \quad k \in \llbracket 1, N-1 \rrbracket.$$

The policy derivatives hold the action, state, and population argument fixed. We write $\hat{D} = (\hat{D}_t)_{t=1}^T$.

For $k \in \llbracket 1, N-1 \rrbracket$, define $F_{t,k}(\theta, z) := v_t^\theta(z)(k)$ for $z \in \mathcal{Z}_N^{T+1}$, where $v_t^\theta(z)$ is the prescribed-flow state law from (25). This is the objective with horizon t , zero running reward, and terminal reward $x \mapsto \mathbf{1}_{\{x=x_k\}}$. It depends only on $z_0, \dots, z_{t-1}$ and $F_{t,k}(\theta, \bar{z}^\theta) = \mu_t^\theta(k)$ for $\bar{z}^\theta = (z_0, z^\theta)$. The same finite recursion used for F_N gives bounded first derivatives of $F_{t,k}$ that are Lipschitz in z , uniformly in θ .

Conditionally on $\mathcal{F}_{t-1}^{\text{off}}$, the population coordinates and earlier sensitivity estimates are fixed. Applying the conditional-mean identity and (39) to $F_{t,k}$, and combining the $N - 1$ rows, gives

$$\left\| \mathbb{E}[\hat{D}_t \mid \mathcal{F}_{t-1}^{\text{off}}] - D_t^\theta \right\|_{\text{F}} \leq C \left(\eta + \|\hat{z} - z^\theta\| + \sum_{s=1}^{t-1} \|\hat{D}_s - D_s^\theta\|_{\text{F}} \right). \quad (41)$$

For the conditional fluctuation, let $Y_{t,r}$ be the matrix contributed by rollout r in the definition of $\widehat{D}_t$, so that $\widehat{D}_t = n^{-1} \sum_{r=1}^n Y_{t,r}$. Write the policy scores as $\overline{\Psi}_s^{(t,r)}$, as defined in (65). Since at most one row of $Y_{t,1}$ is nonzero,

$$\begin{aligned} \|Y_{t,1}\|_{\mathbb{F}}^2 &\leq \left\| \sum_{s=0}^{t-1} \overline{\Psi}_s^{(t,1)} - \frac{1-\eta}{\eta} \sum_{s=1}^{t-1} \widehat{D}_s^\top H(\overline{Q}_s^{(t,1)}) \right\|^2 \\ &\leq 2 \left\| \sum_{s=0}^{t-1} \overline{\Psi}_s^{(t,1)} \right\|^2 + \frac{2(1-\eta)^2}{\eta^2} \left\| \sum_{s=1}^{t-1} \widehat{D}_s^\top H(\overline{Q}_s^{(t,1)}) \right\|^2 \\ &\leq 2t \sum_{s=0}^{t-1} \|\overline{\Psi}_s^{(t,1)}\|^2 + \frac{2(t-1)}{\eta^2} \sum_{s=1}^{t-1} \|\widehat{D}_s\|_{\mathbb{F}}^2 \|H(\overline{Q}_s^{(t,1)})\|^2. \end{aligned}$$

The second inequality uses $\|a-b\|^2 \leq 2\|a\|^2 + 2\|b\|^2$. The last uses $\|\sum_{j=1}^L v_j\|^2 \leq L \sum_{j=1}^L \|v_j\|^2$ for sums of t and $t-1$ terms, respectively, together with $\|D^\top h\| \leq \|D\|_{\mathbb{F}} \|h\|$ and $(1-\eta)^2 \leq 1$. Taking conditional expectations gives

$$\begin{aligned} \mathbb{E}[\|Y_{t,1}\|_{\mathbb{F}}^2 \mid \mathcal{F}_{t-1}^{\text{off}}] &\leq 2t \sum_{s=0}^{t-1} \mathbb{E}[\|\overline{\Psi}_s^{(t,1)}\|^2 \mid \mathcal{F}_{t-1}^{\text{off}}] + \frac{2(t-1)}{\eta^2} \sum_{s=1}^{t-1} \|\widehat{D}_s\|_{\mathbb{F}}^2 \mathbb{E}[\|H(\overline{Q}_s^{(t,1)})\|^2] \\ &\leq \frac{C}{\eta^2} \left(1 + \sum_{s=1}^{t-1} \|\widehat{D}_s\|_{\mathbb{F}}^2 \right). \end{aligned}$$

The last inequality follows from Assumption B.8 and $0 < \eta < 1$: the policy-score bound is uniform, the earlier $\widehat{D}_s$ are fixed under the conditioning, and the new randomizers are independent of $\mathcal{F}_{t-1}^{\text{off}}$. Since the $Y_{t,r}$ are conditionally i.i.d.,

$$\begin{aligned} \mathbb{E}[\|\widehat{D}_t - \mathbb{E}[\widehat{D}_t \mid \mathcal{F}_{t-1}^{\text{off}}]\|_{\mathbb{F}}^2 \mid \mathcal{F}_{t-1}^{\text{off}}] &= \frac{1}{n} \mathbb{E}[\|Y_{t,1} - \mathbb{E}[Y_{t,1} \mid \mathcal{F}_{t-1}^{\text{off}}]\|_{\mathbb{F}}^2 \mid \mathcal{F}_{t-1}^{\text{off}}] \\ &\leq \frac{1}{n} \mathbb{E}[\|Y_{t,1}\|_{\mathbb{F}}^2 \mid \mathcal{F}_{t-1}^{\text{off}}] \\ &\leq \frac{C}{n\eta^2} \left(1 + \sum_{s=1}^{t-1} \|\widehat{D}_s\|_{\mathbb{F}}^2 \right). \end{aligned} \tag{42}$$

Set $e_t := \mathbb{E}[\|\widehat{D}_t - D_t^\theta\|_{\mathbb{F}}^2]$, with $e_0 = 0$. Write $\widehat{D}_t - D_t^\theta = A_t + B_t$, with the fluctuation $A_t := \widehat{D}_t - \mathbb{E}[\widehat{D}_t \mid \mathcal{F}_{t-1}^{\text{off}}]$ and the conditional error $B_t := \mathbb{E}[\widehat{D}_t \mid \mathcal{F}_{t-1}^{\text{off}}] - D_t^\theta$. Since B_t is $\mathcal{F}_{t-1}^{\text{off}}$ -measurable and $\mathbb{E}[A_t \mid \mathcal{F}_{t-1}^{\text{off}}] = 0$, the cross term vanishes and $e_t = \mathbb{E}[\|A_t\|_{\mathbb{F}}^2] + \mathbb{E}[\|B_t\|_{\mathbb{F}}^2]$. By $\|a+b\|^2 \leq 2\|a\|^2 + 2\|b\|^2$ and (27),

$$\mathbb{E}[\|\widehat{D}_s\|_{\mathbb{F}}^2] \leq 2e_s + 2\|D_s^\theta\|_{\mathbb{F}}^2 \leq 2e_s + C.$$

Squaring (41), taking expectations with (40), and taking expectations in (42), we obtain

$$e_t \leq C \left(\eta^2 + \frac{1}{M} + \frac{1}{n\eta^2} \right) + C \left(1 + \frac{1}{n\eta^2} \right) \sum_{s=1}^{t-1} e_s.$$

Since $n\eta^2 \geq 1$, induction yields

$$\max_{t \in [1, T]} e_t \leq C \left(\eta^2 + \frac{1}{M} + \frac{1}{n\eta^2} \right), \quad \max_{t \in [1, T]} \mathbb{E}[\|\widehat{D}_t\|_{\mathbb{F}}^2] \leq C. \tag{43}$$

Bias and mean-square error of the online estimator. The online stage uses new simulation randomness. Define $\widehat{G}_{B,M,n}^{\lambda,\eta}$ by equation (12) and set for simplicity $\widehat{G} := \widehat{G}_{B,M,n}^{\lambda,\eta}$. Conditional on $(\widehat{z}, \widehat{D})$, its B contributions are i.i.d., and $\widehat{G} := \mathbb{E}[\widehat{G} \mid \widehat{z}, \widehat{D}] = \Gamma^\lambda(\widehat{z}, \widehat{D})$. Squaring (39) with $\kappa = \lambda$, taking expectations, and applying (40) and (43) gives

$$\mathbb{E}[\|\widehat{G} - \nabla_\theta J(\theta)\|^2] \leq C \left(\lambda^2 + \eta^2 + \frac{1}{n\eta^2} + \frac{1}{M} \right).$$

Since $\mathbb{E}[\widehat{G}] = \mathbb{E}[\overline{G}]$, Jensen's inequality and $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$ imply

$$\|\mathbb{E}[\widehat{G}] - \nabla_{\theta} J(\theta)\| \leq C \left(\lambda + \eta + \frac{1}{\eta\sqrt{n}} + \frac{1}{\sqrt{M}} \right).$$

Finally, write $Y_b := \widehat{\mathcal{R}}^{(b)}(\widehat{\mathcal{S}}_{\text{pol}}^{(b)} + \widehat{\mathcal{S}}_{\text{mf}}^{(b)})$, so that $\widehat{G} = \frac{1}{B} \sum_{b=1}^B Y_b$. The returns are bounded by $T\overline{R} + \overline{g}$ (Assumption B.1). The same sum-of-squares and matrix bounds as above, together with Assumption B.8, give

$$\begin{aligned} \mathbb{E}[\|\widehat{\mathcal{S}}_{\text{pol}}^{(1)}\|^2 \mid \widehat{z}, \widehat{D}] &\leq C, \\ \mathbb{E}[\|\widehat{\mathcal{S}}_{\text{mf}}^{(1)}\|^2 \mid \widehat{z}, \widehat{D}] &\leq \frac{T(1-\lambda)^2}{\lambda^2} \sum_{t=1}^T \|\widehat{D}_t\|_{\text{F}}^2 \mathbb{E}[\|H(Q_t^{(1)})\|^2] \leq C\lambda^{-2} \sum_{t=1}^T \|\widehat{D}_t\|_{\text{F}}^2. \end{aligned}$$

Since the online randomizers are independent of $(\widehat{z}, \widehat{D})$ and the Y_b are conditionally i.i.d.,

$$\begin{aligned} \mathbb{E}[\|\widehat{G} - \overline{G}\|^2 \mid \widehat{z}, \widehat{D}] &= \frac{1}{B} \mathbb{E}[\|Y_1 - \mathbb{E}[Y_1 \mid \widehat{z}, \widehat{D}]\|^2 \mid \widehat{z}, \widehat{D}] \\ &\leq \frac{1}{B} \mathbb{E}[\|Y_1\|^2 \mid \widehat{z}, \widehat{D}] \\ &\leq \frac{C}{B} \left(1 + \lambda^{-2} \sum_{t=1}^T \|\widehat{D}_t\|_{\text{F}}^2 \right). \end{aligned}$$

Taking expectations and using (43) bounds this term by $C(1 + \lambda^{-2})/B$. Since $\widehat{G} - \overline{G}$ has conditional mean zero and $\overline{G}$ is $\sigma(\widehat{z}, \widehat{D})$ -measurable, the cross term vanishes and

$$\begin{aligned} \mathbb{E}[\|\widehat{G} - \nabla_{\theta} J(\theta)\|^2] &= \mathbb{E}[\|\widehat{G} - \overline{G}\|^2] + \mathbb{E}[\|\overline{G} - \nabla_{\theta} J(\theta)\|^2] \\ &\leq C \left(\lambda^2 + \frac{1 + \lambda^{-2}}{B} + \eta^2 + \frac{1}{n\eta^2} + \frac{1}{M} \right). \end{aligned}$$

This proves the theorem. $\square$

B.3 Continuous-state space

This appendix gives the mixture construction, assumptions, and proofs for Section 3.2.

B.3.1 Encoder, decoder, and numerical fitting

Fix K . Let $\mathcal{K}_K \subset \mathcal{Z}_K$ be a fixed non-empty compact fitting set. In practice, $\mathcal{K}_K$ imposes lower and upper bounds on the weights and covariance eigenvalues, bounds the means, and keeps the projected means separated so that component labels remain ordered. The decoder

$$\mathcal{D}_K : \widetilde{\mathcal{Z}}_K \longrightarrow \mathcal{P}_2(\mathbb{R}^d), \quad \mathcal{D}_K(p, m, \Sigma) = \sum_{j=1}^K p_j \mathcal{N}(m_j, \Sigma_j),$$

is continuous in $\mathcal{W}_2$ and satisfies

$$M_2(\mathcal{D}_K(p, m, \Sigma)) = \sum_{j=1}^K p_j (\|m_j\|^2 + \text{tr } \Sigma_j).$$

Thus the compact fitting constraints give the uniform moment bound required by the perturbation estimates. No inverse identity $\mathcal{D}_K \circ \mathcal{E}_K = \text{Id}$ is needed.

Assumption B.9 (Encoder regularity). *The encoder $\mathcal{E}_K : \mathcal{P}_2(\mathbb{R}^d) \rightarrow \mathcal{Z}_K$ is Borel measurable and satisfies $z_{K,t}^{\theta} = \mathcal{E}_K(\mu_t^{\theta}) \in \mathcal{K}_K$. The paths $\theta \mapsto z_{K,t}^{\theta}$ are continuous for objective consistency, $\mathcal{C}^1$ on an open neighborhood of Θ for the policy-gradient representation, and $\mathcal{C}^3$ with bounded derivatives when the centered-difference bias is estimated.*

Practical encoder and numerical fit. Let f_z denote the density of $\mathcal{D}_K(z)$. A standard choice is the constrained likelihood encoder

$$\mathcal{E}_K(\mu) \in \operatorname{argmax}_{z \in \mathcal{K}_K} \ell_K(\mu, z), \quad \ell_K(\mu, z) := \int_{\mathbb{R}^d} \log f_z(x) \mu(dx),$$

with a fixed measurable tie-breaking rule. Replacing μ by an empirical law gives the constrained EM problem described by [Dempster et al. \(1977\)](#). The compact constraints make this fitting problem well posed; the continuity and differentiability required by Assumption B.9 are imposed on the resulting fitted path.

For the numerical algorithm, let $\hat{\mathcal{E}}_{K,L} : \mathcal{P}_2(\mathbb{R}^d) \rightarrow \mathcal{K}_K$ denote the fixed constrained EM implementation with L particles, including its initialization, stopping rule, and label ordering. We summarize its statistical and optimization error by the following assumption.

Assumption B.10 (Accuracy of fitted coordinates). *The numerical fitting maps are measurable. For the interacting particle laws $\hat{\mu}_t^{\theta,L}$ used in the offline stage, the computed coordinates $\hat{z}_{K,t}^{\theta,L} := \hat{\mathcal{E}}_{K,L}(\hat{\mu}_t^{\theta,L})$ satisfy*

$$\sup_{\theta \in \Theta} \max_{t \in \mathcal{T}} \mathbb{E}[\|\hat{z}_{K,t}^{\theta,L} - z_{K,t}^{\theta}\|^2] \leq a_L^2, \quad a_L \rightarrow 0.$$

The error a_L includes both particle and optimization error. The same bound is required at the shifted parameters $\theta \pm \eta e_\ell$. Under the $\mathcal{C}^3$ part of Assumption B.9, for the centered differences in (67), fix a coordinate direction e_ℓ and define the exact and fitted endpoint errors

$$z_{t,\pm}^\ell := z_{K,t}^{\theta \pm \eta e_\ell}, \quad \hat{z}_{t,\pm}^\ell := \hat{z}_{K,t}^{\theta \pm \eta e_\ell}, \quad \varepsilon_{t,\pm}^\ell := \hat{z}_{t,\pm}^\ell - z_{t,\pm}^\ell.$$

The column of the estimator in direction e_ℓ is

$$\hat{D}_t e_\ell = \frac{\hat{z}_{t,+}^\ell - \hat{z}_{t,-}^\ell}{2\eta}.$$

Therefore,

$$\hat{D}_t e_\ell - \partial_\ell z_{K,t}^\theta = \underbrace{\frac{z_{t,+}^\ell - z_{t,-}^\ell}{2\eta} - \partial_\ell z_{K,t}^\theta}_{\text{central-difference error}} + \underbrace{\frac{\varepsilon_{t,+}^\ell - \varepsilon_{t,-}^\ell}{2\eta}}_{\text{fitting error}}. \quad (44)$$

The $\mathcal{C}^3$ assumption gives, uniformly in ℓ ,

$$\left\| \frac{z_{t,+}^\ell - z_{t,-}^\ell}{2\eta} - \partial_\ell z_{K,t}^\theta \right\| \leq C_K \eta^2,$$

while Assumption B.10 gives

$$\mathbb{E}[\|\varepsilon_{t,+}^\ell\|^2 + \|\varepsilon_{t,-}^\ell\|^2] \leq 2a_{n_0}^2.$$

Using $(u+v)^2 \leq 2u^2 + 2v^2$ in (44), and summing over the d_θ columns, yields

$$\mathbb{E}[\|\hat{D}_t - \nabla_\theta z_{K,t}^\theta\|_F^2] \leq C_K \left(\eta^4 + \frac{a_{n_0}^2}{\eta^2} \right), \quad n_0 = \frac{n}{2d_\theta}. \quad (45)$$

Hence consistency requires

$$\eta \rightarrow 0 \quad \text{and} \quad \frac{a_{n_0}}{\eta} \rightarrow 0.$$

Finally, the mixture approximation error is measured by ε_K in (15). Its convergence as K grows is a separate approximation question and is not needed for the fixed- K results.

B.3.2 Mixture coordinates and transport map

The fitted coordinate space is $\mathcal{Z}_K = \Delta_K^\circ \times \mathbb{M}_K^e \times (\mathbb{S}_{++}^d)^K$, and randomized coordinates belong to $\tilde{\mathcal{Z}}_K = \Delta_K^\circ \times (\mathbb{R}^d)^K \times (\mathbb{S}_{++}^d)^K$. A point (p, m, Σ) represents the mixture $\sum_j p_j \mathcal{N}(m_j, \Sigma_j)$, with Σ_j a covariance matrix. The randomizer is $R_t = (Q_t, A_t, B_t)$ and the perturbation is Ψ_λ from (13).

Let ϕ_i be the density of $\mathcal{N}(m_i, \Sigma_i)$ and set

$$r_i(x) = \frac{p_i \phi_i(x)}{\sum_{k=1}^K p_k \phi_k(x)}.$$

Given $a \in \Delta_K^\circ$, we define the map

$$[0, 1] \ni u \mapsto F_a(u) := \min \left\{ i \in \llbracket 1, K \rrbracket : u \leq \sum_{k=1}^i a_k \right\}.$$

We also define the mapping

$$\begin{cases} \mathbb{R}^d \times [0, 1] \ni (x, u) & \mapsto \iota(x, u) = F_{r(x)}(u), \\ \mathbb{R}^d \times [0, 1] \ni (x, u) & \mapsto v(x, u) = \frac{u - \sum_{k < i(x, u)} r_k(x)}{r_{i(x, u)}(x)}, \\ \mathbb{R}^d \times [0, 1] \ni (x, u) & \mapsto j(\omega, x, u) = \iota(x, u) \mathbf{1}_{v(x, u) \leq 1-\lambda} + F_{Q_t(\omega)}\left(\frac{(v(x, u) - (1-\lambda))}{\lambda}\right) \mathbf{1}_{v(x, u) > 1-\lambda} \end{cases}$$

We now define the transport map $T_{K,t}^{\lambda, \theta}$ over $\Omega \times \mathbb{R}^d \times [0, 1]$ as

$$T_{K,t}^{\lambda, \theta}(\omega, x, u) = m_{j(\omega, x, u), K, t}^{\lambda, \theta}(\omega) + \Gamma_{j(\omega, x, u), K, t}^{\lambda, \theta}(\omega) (\Gamma_{i(x, u), K, t}^\theta)^{-1}(x - m_{i(x, u), K, t}^\theta). \quad (46)$$

Proof of Proposition 3.6. All maps in the definition of $T_{K,t}^{\lambda, \theta}$ in (46) are jointly measurable by construction. First, at $\lambda = 0$, by construction it is clear that $T_{K,t}^{0, \theta}(\omega, x, u) = x$.

Fix $\omega \in \Omega$ and let $X \sim \nu_{K,t}^\theta$ and $U \sim \mathcal{U}([0, 1])$ be independent. Set $I = \iota(X, U)$ and $V = v(X, U)$. Let $f : \llbracket 1, K \rrbracket \times \mathbb{R}^d \times [0, 1] \rightarrow \mathbb{R}$ be a bounded measurable test map. We hence have that

$$\begin{aligned} \int_{\mathbb{R}^d} \int_0^1 f(\iota(x, u), x, v(x, u)) du \nu_{K,t}^\theta(dx) &= \sum_{k=1}^K \int_{\mathbb{R}^d} \int_{s_{k-1}(x)}^{s_k(x)} f(k, x, \frac{u - s_{k-1}(x)}{r_k(x)}) du \nu_{K,t}^\theta(dx), \\ &= \sum_{k=1}^K \int_{\mathbb{R}^d} \int_{s_{k-1}(x)}^{s_k(x)} f(k, x, \frac{u - s_{k-1}(x)}{r_k(x)}) du \nu_{K,t}^\theta(dx), \end{aligned}$$

where we denoted $s_k(x) := \sum_{i=1}^k r_i(x)$, $s_0(x) := 0$. Now, using the $\mathcal{C}^1$ diffeomorphism: $[s_{k-1}(x), s_k(x)] \ni u \mapsto v := \frac{u - s_{k-1}(x)}{r_k(x)} \in [0, 1]$. for fixed $x \in \mathbb{R}^d$ such that it follows that $du = r_k(x)dv$, we can write

$$\sum_{k=1}^K \int_{\mathbb{R}^d} \int_{s_{k-1}(x)}^{s_k(x)} f(k, x, \frac{u - s_{k-1}(x)}{r_k(x)}) du \nu_{K,t}^\theta(dx) = \sum_{k=1}^K \int_{\mathbb{R}^d} \int_0^1 f(k, x, v) r_k(x) dv \nu_{K,t}^\theta(dx),$$

Hence, by identification of the law it follows

$$\mathcal{L}(I, X, V)(dk, dx, dv) = \sum_{i=1}^K r_k(x) \nu_{K,t}^\theta(dx) \delta_i(dk) dv = \sum_{i=1}^K p_k \phi_k(x) \delta_i(dk) dx dv \quad (47)$$

It follows directly from (47) that $\mathbb{P}_I = \sum_{k=1}^K p_k \delta_k$ that the conditional law of $X|I = k$ for any $k \in \llbracket 1, K \rrbracket$ is given by $\mathcal{L}(X|I = k)(dx) = \phi_k(x)dx = \mathcal{N}(m_k, \Sigma_k)(dx)$ and that $V \sim \mathcal{U}([0, 1])$ while being independent of (I, X) . Hence, it follows that for fixed $\omega \in \Omega$ the law of the target index $J := j^\lambda(\omega, X, U)$ satisfies

$$\mathbb{P}(J = j|I = i) = (1 - \lambda) \mathbf{1}_{j=i} + \lambda q_{j,t}(\omega).$$

Conditionally on $I = i$ and $J = j$, the vector $(\Gamma_{i,K,t}^\theta)^{-1}(X - m_{i,K,t}^\theta)$ is standard Gaussian. Consequently, the transported vector has law $\mathcal{N}(m_{j,K,t}^{\lambda, \theta}(\omega), \Sigma_{j,K,t}^{\lambda, \theta}(\omega))$ under this conditioning. Summing over i gives the weight $(1 - \lambda)p_j + \lambda q_{j,t}(\omega)$, proving the claim. $\square$

B.3.3 Perturbation estimate

Assumption B.11 (Moment bounds). *There exists a uniform constant $C_K < \infty$ in $(t, \theta, j) \in \mathcal{T} \times \Theta \times \llbracket 1, K \rrbracket$ such that the following holds*

$$\begin{cases} \|m_{j,K,t}^\theta\|^2 + \|\Sigma_{j,K,t}^\theta\|_F^2 & \leq C_K, \\ \mathbb{E} \left[\max_{j \in \llbracket 1, K \rrbracket} (\|A_{j,t}\|^2 + \|B_{j,t}\|_F^2) \right] & \leq C_K. \end{cases}$$

Proof of Proposition 3.7. Fix t, θ and suppress these indices. We use a coupling of the same source and target mixtures; it need not be the symmetric-square-root transport used in Proposition 3.6. Independently of R_t , draw I with weights p , a standard Gaussian vector Z , and $V \sim \mathcal{U}([0, 1])$. If $V \leq 1 - \lambda$, set $J = I$; otherwise draw J with weights Q_t , using additional independent randomness. Put

$$X = m_I + \Sigma_I^{1/2} Z, \quad Y = m_J^\lambda + \Sigma_J^{1/2} ((1 - \lambda)I_d + \lambda B_{J,t}) Z.$$

Conditionally on R_t , this couples $\nu_{K,t}^\theta$ with $M_{K,t}^{\lambda,\theta}$. Indeed, J has weights $(1 - \lambda)p + \lambda Q_t$, and its conditional component covariance is precisely the one in (13).

On $\{V \leq 1 - \lambda\}$,

$$Y - X = \lambda(A_{I,t} - m_I) + \lambda \Sigma_I^{1/2} (B_{I,t} - I_d) Z.$$

The moment bounds and $\|\Sigma_I^{1/2} (B_{I,t} - I_d)\|_F^2 \leq \|\Sigma_I\|_{\text{op}} \|B_{I,t} - I_d\|_F^2$ give a conditional mean squared displacement bounded by $C\lambda^2$. Here and below C depends only on the stated moment bounds and the fixed dimension. On $\{V > 1 - \lambda\}$, those same bounds, using the maximum over components for A_t, B_t , give $\mathbb{E}[\|X\|^2 + \|Y\|^2 \mid V > 1 - \lambda] \leq C$. Hence

$$\mathbb{E}[\mathcal{W}_2^2(M_{K,t}^{\lambda,\theta}, \nu_{K,t}^\theta)] \leq \mathbb{E}[\|Y - X\|^2] \leq C((1 - \lambda)\lambda^2 + \lambda) \leq C\lambda.$$

The Wasserstein triangle inequality followed by Minkowski's inequality yields

$$(\mathbb{E}[\mathcal{W}_2^2(M_{K,t}^{\lambda,\theta}, \mu_t^\theta)])^{1/2} \leq \mathcal{W}_2(\nu_{K,t}^\theta, \mu_t^\theta) + (\mathbb{E}[\mathcal{W}_2^2(M_{K,t}^{\lambda,\theta}, \nu_{K,t}^\theta)])^{1/2},$$

which proves the claim. When $K = 1$, the component-switching construction is unnecessary, and the same coupling gives the bound $(\mathbb{E}[\mathcal{W}_2^2(M_{1,t}^{\lambda,\theta}, \nu_{1,t}^\theta)])^{1/2} \leq C\lambda$. $\square$

B.3.4 Objective perturbation consistency

Assumption B.12 (Objective regularity). *We make the following assumptions*

- (1) *The randomizers $(R_t)_{t \in \mathcal{T}}$ are independent across time and independent of the simulation randomness*
- (2) *The initial law satisfies $\mu_0 \in \mathcal{P}_2(\mathbb{R}^d)$. Furthermore, for any $t \in \mathcal{T}, \theta \in \Theta$, and $\eta, m \in \mathcal{P}_2(\mathbb{R}^d)$, the measure $\Phi_t^\theta(\eta, m)$ belongs to $\mathcal{P}_2(\mathbb{R}^d)$.*
- (3) *There exists a positive constant L such that*

$$\begin{cases} \mathcal{W}_2(\Phi_t^\theta(\eta, m), \Phi_t^\theta(\eta', m')) & \leq L(\mathcal{W}_2(\eta, \eta') + \mathcal{W}_2(m, m')), \\ |\bar{R}_t^\theta(\eta, m) - \bar{R}_t^\theta(\eta', m')| & \leq L(\mathcal{W}_2(\eta, \eta') + \mathcal{W}_2(m, m')), \\ |\bar{g}(\eta, m) - \bar{g}(\eta', m')| & \leq L(\mathcal{W}_2(\eta, \eta') + \mathcal{W}_2(m, m')). \end{cases}$$

uniformly in $\theta, t, \eta, \eta', m, m'$.

Note that the construction in Appendix A.1 ensures (1).

Proof of Theorem 3.8. Fix $\theta \in \Theta$, let $\eta_{K,t}^{\lambda,\theta} := \mathcal{L}(X_t^{K,\lambda,\theta})$, and set $e_t := \mathcal{W}_2(\eta_{K,t}^{\lambda,\theta}, \eta_{K,t}^{0,\theta})$. Independence of the current randomizer and state gives

$$\begin{cases} \eta_{K,t+1}^{\lambda,\theta} &= \mathbb{E}[\Phi_t^\theta(\eta_{K,t}^{\lambda,\theta}, M_{K,t}^{\lambda,\theta})], \\ \eta_{K,t+1}^{0,\theta} &= \Phi_t^\theta(\eta_{K,t}^{0,\theta}, \nu_{K,t}^\theta) \end{cases}$$

The squared Wasserstein distance is convex under mixtures. To verify this, let γ_ω be an optimal coupling between $\Phi_t^\theta(\eta_{K,t}^{\lambda,\theta}, M_{K,t}^{\lambda,\theta}(\omega))$ and $\eta_{K,t+1}^{0,\theta}$ for a given realization ω . Taking the expectation $\mathbb{E}[\gamma_\omega]$ yields a valid coupling between the expected measure $\eta_{K,t+1}^{\lambda,\theta}$ and the deterministic measure $\eta_{K,t+1}^{0,\theta}$. Because the Wasserstein distance is the infimum over all valid couplings, it is bounded by the cost of this averaged coupling:

$$\mathcal{W}_2^2(\eta_{K,t+1}^{\lambda,\theta}, \eta_{K,t+1}^{0,\theta}) \leq \mathbb{E}[\mathcal{W}_2^2(\Phi_t^\theta(\eta_{K,t}^{\lambda,\theta}, M_{K,t}^{\lambda,\theta}), \Phi_t^\theta(\eta_{K,t}^{0,\theta}, \nu_{K,t}^\theta))].$$

Using the Lipschitz continuity of Φ_t^θ (Assumption B.12), this implies:

$$e_{t+1}^2 \leq L^2 \mathbb{E}[(e_t + \mathcal{W}_2(M_{K,t}^{\lambda,\theta}, \nu_{K,t}^\theta))^2].$$

Taking the square root and applying Minkowski's inequality with $p = 2$ separates the terms:

$$e_{t+1} \leq L(e_t + (\mathbb{E}[\mathcal{W}_2^2(M_{K,t}^{\lambda,\theta}, \nu_{K,t}^\theta)])^{1/2}).$$

Setting $\delta := C_K \sqrt{\lambda}$, the perturbation estimate from Proposition 3.7 gives $e_{t+1} \leq L(e_t + \delta)$. Since $e_0 = 0$, a direct induction yields $e_t \leq \delta \sum_{s=1}^T L^s$ for any $t \geq 1$. Finally, applying the Lipschitz bounds on the rewards $\bar{R}_t^\theta$ and terminal cost $\bar{g}$ from Assumption B.12, and summing over time, yields:

$$|J_K^\lambda(\theta) - J_K(\theta)| \leq L \sum_{t=0}^T (e_t + \mathbb{E}[\mathcal{W}_2(M_{K,t}^{\lambda,\theta}, \nu_{K,t}^\theta)]) \leq L \sum_{t=0}^T (e_t + \delta) \leq C_{K,T} \sqrt{\lambda},$$

uniformly in θ , for some constant $C_{K,T} > 0$ depending only on L, T , and C_K . $\square$

Projection error and maximizers. To prove (15), set $a_t := \mathcal{W}_2(\mathcal{L}(X_t^{K,0,\theta}), \mu_t^\theta)$ and $\delta_t := \mathcal{W}_2(\nu_{K,t}^\theta, \mu_t^\theta)$. The original and projected dynamics give

$$a_{t+1} \leq L(a_t + \delta_t), \quad a_0 = 0.$$

Since $\delta_t \leq \varepsilon_K$, induction yields $a_t \leq \varepsilon_K \sum_{s=1}^t L^s$. Applying the reward bounds gives

$$|J_K(\theta) - J(\theta)| \leq L \sum_{t=0}^T (a_t + \delta_t) \leq C_T^{\text{proj}} \varepsilon_K, \quad C_T^{\text{proj}} := L \sum_{t=0}^T \left(1 + \sum_{s=1}^t L^s\right).$$

The second bound in (15) follows by the triangle inequality and Theorem 3.8.

Proof of Proposition 3.9. For any $\theta \in \Theta$, we have

$$\begin{aligned} J(\theta) - J(\theta_{K,\lambda}^*) &= J(\theta) - J_K^\lambda(\theta) + J_K^\lambda(\theta) - J_K^\lambda(\theta_{K,\lambda}^*) + J_K^\lambda(\theta_{K,\lambda}^*) - J(\theta_{K,\lambda}^*) \\ &\leq 2 \sup_{\theta \in \Theta} |J(\theta) - J_K^\lambda(\theta)|, \end{aligned}$$

where we used that $J_K^\lambda(\theta) \leq J_K^\lambda(\theta_{K,\lambda}^*)$. Hence, by taking the supremum over $\theta \in \Theta$, in the left hand-side, it follows that

$$0 \leq \sup_{\theta \in \Theta} J(\theta) - J(\theta_{K,\lambda}^*) \leq 2(C_T^{\text{proj}} \varepsilon_K + C_{K,T} \sqrt{\lambda}).$$

$\square$

B.3.5 Objective gradient convergence

Fix K . For a prescribed coordinate flow $z = (z_1, \dots, z_T) \in \tilde{\mathcal{Z}}_K^T$, set $m_0 = \mu_0$ and $m_t = \mathcal{D}_K(z_t)$ for $t \geq 1$. Define $v_t : \Theta^\circ \times \tilde{\mathcal{Z}}_K^T \rightarrow \mathcal{P}_2(\mathbb{R}^d)$ recursively by

$$v_0^\theta(z) = \mu_0, \quad v_{t+1}^\theta(z) = \Phi_t^\theta(v_t^\theta(z), m_t).$$

The objective for this prescribed population flow is $F_K : \Theta^\circ \times \tilde{\mathcal{Z}}_K^T \rightarrow \mathbb{R}$, defined by

$$F_K(\theta, z) = \sum_{t=0}^{T-1} \bar{R}_t^\theta(v_t^\theta(z), m_t) + \bar{g}(v_T^\theta(z), m_T).$$

Write $z^\theta = (z_{K,t}^\theta)_{t=1}^T$, $Z^{\lambda,\theta} = (\Psi_\lambda(z_{K,t}^\theta, R_t))_{t=1}^T$, and $D^\theta = \nabla_\theta z^\theta$. Conditioning on the randomizers gives

$$J_K(\theta) = F_K(\theta, z^\theta), \quad J_K^\lambda(\theta) = \mathbb{E}[F_K(\theta, Z^{\lambda,\theta})].$$

Throughout, $\nabla_\theta F_K$ denotes differentiation with the prescribed flow held fixed.

Assumption B.13 (Continuous-state gradient regularity). *Assume that:*

- (1) Regularity of the fitted flow. *The parameter set Θ is compact, and $\theta \mapsto z^\theta = (\mathcal{E}_K(\mu_t^\theta))_{t=1}^T$ extends to a $\mathcal{C}^1$ map on an open neighborhood Θ° of Θ , with values in $\mathcal{Z}_K^T$. The initial law μ_0 is independent of θ .*
- (2) Regularity of the prescribed-flow objective. *The map F_K is $\mathcal{C}^1$ on the open set $\Theta^\circ \times \tilde{\mathcal{Z}}_K^T \subset \mathbb{R}^{d_\theta + Tq_K}$. There exist $L < \infty$ and $r \geq 0$ such that, uniformly in $\theta \in \Theta^\circ$ and $z, z' \in \tilde{\mathcal{Z}}_K^T$,*

$$\begin{aligned} \|\nabla F_K(\theta, z)\| &\leq L(1 + \|z\|)^r, \\ \|\nabla F_K(\theta, z) - \nabla F_K(\theta, z')\| &\leq L(1 + \|z\| + \|z'\|)^r \|z - z'\|. \end{aligned}$$

Here ∇F_K contains the derivatives with respect to both θ and z .

- (3) Randomizer moments. *Assume Assumption B.11 and condition (1) of Assumption B.12. The randomizers have a common law ρ , independent of θ and λ . For some $\lambda_0 \in (0, 1)$, this law satisfies condition (1) of Assumption B.14 for every $0 < \lambda \leq \lambda_0$. Moreover, for each $t \in \llbracket 1, T \rrbracket$,*

$$\mathbb{E} \left[\max_{1 \leq j \leq K} (\|A_{j,t}\|^{r+1} + \|B_{j,t}\|_F^{2(r+1)}) \right] < \infty.$$

Condition (2) is an additional regularity assumption on the prescribed-flow objective; it does not follow from the Wasserstein Lipschitz bounds of Assumption B.12. It requires polynomial control at infinity and uniform control as weights or covariance eigenvalues approach zero. The case $r = 0$ is the bounded, Lipschitz derivative assumption used in the discrete proof.

Proof of Theorem 3.10. By compactness and the $\mathcal{C}^1$ regularity of the fitted flow, its coordinates and first derivatives are uniformly bounded. Moreover, because its covariance matrices are positive definite, there exists $c_K > 0$ such that $\Sigma_{j,K,t}^\theta \succeq c_K I_d$ uniformly in θ, t, j . These bounds also hold on a sufficiently small neighborhood of Θ .

Set

$$W := 1 + \sum_{t=1}^T \sum_{j=1}^K (\|A_{j,t}\| + \|B_{j,t}\|_F^2).$$

The moment assumption gives $\mathbb{E}[W^{r+1}] < \infty$. We first establish

$$\|Z^{\lambda,\theta} - z^\theta\| + \|\nabla_\theta Z^{\lambda,\theta} - D^\theta\| \leq C\lambda W. \quad (48)$$

For the weights and means, this follows directly from their affine perturbations. For the covariance block (j, t) , recall $\Gamma_{j,K,t}^\theta = (\Sigma_{j,K,t}^\theta)^{1/2}$ and set

$$V_{j,t}^\lambda := 2\lambda(B_{j,t} - I_d) + \lambda^2(B_{j,t} - I_d)^2.$$

By (13) and $\Sigma_{j,K,t}^\theta = \Gamma_{j,K,t}^\theta I_d \Gamma_{j,K,t}^\theta$,

$$\begin{aligned}\Sigma_{j,K,t}^{\lambda,\theta} - \Sigma_{j,K,t}^\theta &= \Gamma_{j,K,t}^\theta \left(((1-\lambda)I_d + \lambda B_{j,t})^2 - I_d \right) \Gamma_{j,K,t}^\theta \\ &= \Gamma_{j,K,t}^\theta \left(2\lambda(B_{j,t} - I_d) + \lambda^2(B_{j,t} - I_d)^2 \right) \Gamma_{j,K,t}^\theta \\ &= \Gamma_{j,K,t}^\theta V_{j,t}^\lambda \Gamma_{j,K,t}^\theta.\end{aligned}$$

Since $\lambda^2 \leq \lambda$ and $\|(B_{j,t} - I_d)^2\|_F \leq \|B_{j,t} - I_d\|_F^2$,

$$\|V_{j,t}^\lambda\|_F \leq C\lambda(1 + \|B_{j,t}\|_F^2).$$

For each parameter coordinate θ_ℓ , differentiating $(\Gamma_{j,K,t}^\theta)^2 = \Sigma_{j,K,t}^\theta$ gives

$$\Gamma_{j,K,t}^\theta \partial_{\theta_\ell} \Gamma_{j,K,t}^\theta + (\partial_{\theta_\ell} \Gamma_{j,K,t}^\theta) \Gamma_{j,K,t}^\theta = \partial_{\theta_\ell} \Sigma_{j,K,t}^\theta.$$

The covariance lower bound implies

$$\|\partial_{\theta_\ell} \Gamma_{j,K,t}^\theta\|_F \leq \frac{1}{2\sqrt{c_K}} \|\partial_{\theta_\ell} \Sigma_{j,K,t}^\theta\|_F.$$

Consequently, both $\Gamma_{j,K,t}^\theta$ and its first derivatives are uniformly bounded. Since $B_{j,t}$, and hence $V_{j,t}^\lambda$, does not depend on θ , the product rule gives

$$\partial_{\theta_\ell} \Sigma_{j,K,t}^{\lambda,\theta} - \partial_{\theta_\ell} \Sigma_{j,K,t}^\theta = (\partial_{\theta_\ell} \Gamma_{j,K,t}^\theta) V_{j,t}^\lambda \Gamma_{j,K,t}^\theta + \Gamma_{j,K,t}^\theta V_{j,t}^\lambda (\partial_{\theta_\ell} \Gamma_{j,K,t}^\theta),$$

so that, using $\|XYX'\|_F \leq \|X\|_F \|Y\|_F \|X'\|_F$,

$$\|\partial_{\theta_\ell} \Sigma_{j,K,t}^{\lambda,\theta} - \partial_{\theta_\ell} \Sigma_{j,K,t}^\theta\|_F \leq 2\|\partial_{\theta_\ell} \Gamma_{j,K,t}^\theta\|_F \|\Gamma_{j,K,t}^\theta\|_F \|V_{j,t}^\lambda\|_F.$$

Together with $\|\Sigma_{j,K,t}^{\lambda,\theta} - \Sigma_{j,K,t}^\theta\|_F \leq \|\Gamma_{j,K,t}^\theta\|_F^2 \|V_{j,t}^\lambda\|_F$ and the bound on $V_{j,t}^\lambda$, this yields

$$\|\Sigma_{j,K,t}^{\lambda,\theta} - \Sigma_{j,K,t}^\theta\|_F + \|\nabla_{\theta} \Sigma_{j,K,t}^{\lambda,\theta} - \nabla_{\theta} \Sigma_{j,K,t}^\theta\| \leq C\lambda(1 + \|B_{j,t}\|_F^2).$$

Summing over the finitely many blocks proves (48).

In particular,

$$\|Z^{\lambda,\theta}\| + \|\nabla_{\theta} Z^{\lambda,\theta}\| \leq CW.$$

By the chain rule and the growth assumption in Assumption B.13(2), using $W \geq 1$,

$$\begin{aligned}\|\nabla_{\theta} [F_K(\theta, Z^{\lambda,\theta})]\| &= \|\nabla_{\theta} F_K(\theta, Z^{\lambda,\theta}) + (\nabla_{\theta} Z^{\lambda,\theta})^\top \nabla_z F_K(\theta, Z^{\lambda,\theta})\| \\ &\leq (1 + \|\nabla_{\theta} Z^{\lambda,\theta}\|) L(1 + \|Z^{\lambda,\theta}\|)^r \leq L(1 + CW)^{r+1} \leq CW^{r+1},\end{aligned}$$

which is integrable. This bound is uniform in θ in a neighborhood of Θ . By the mean value theorem, the difference quotients of $\theta \mapsto F_K(\theta, Z^{\lambda,\theta})$ are therefore also dominated by CW^{r+1} . Dominated convergence then justifies differentiation under the expectation, giving

$$\begin{aligned}\nabla_{\theta} J_K^\lambda(\theta) &= \mathbb{E}[\nabla_{\theta} F_K(\theta, Z^{\lambda,\theta}) + (\nabla_{\theta} Z^{\lambda,\theta})^\top \nabla_z F_K(\theta, Z^{\lambda,\theta})], \\ \nabla_{\theta} J_K(\theta) &= \nabla_{\theta} F_K(\theta, z^\theta) + (D^\theta)^\top \nabla_z F_K(\theta, z^\theta).\end{aligned}$$

Subtracting and adding $(D^\theta)^\top \mathbb{E}[\nabla_z F_K(\theta, Z^{\lambda,\theta})]$ gives

$$\begin{aligned}\nabla_{\theta} J_K^\lambda - \nabla_{\theta} J_K &= \mathbb{E}[\nabla_{\theta} F_K(\theta, Z^{\lambda,\theta}) - \nabla_{\theta} F_K(\theta, z^\theta)] + (D^\theta)^\top \mathbb{E}[\nabla_z F_K(\theta, Z^{\lambda,\theta}) - \nabla_z F_K(\theta, z^\theta)] \\ &\quad + \mathbb{E}[(\nabla_{\theta} Z^{\lambda,\theta} - D^\theta)^\top \nabla_z F_K(\theta, Z^{\lambda,\theta})].\end{aligned}$$

Using (48), the derivative bounds, and the uniform bound on D^θ , each term is bounded by $C\lambda\mathbb{E}[W^{r+1}]$. The estimate is uniform in θ , proving the result. $\square$

B.3.6 Density and score for the randomized perturbation

Assumption B.14 (Assumptions on ρ the law of the randomizer $(R_t)_{t \in \llbracket 1, T \rrbracket}$). Let $\mathcal{B}_\lambda := \{B \in \mathbb{S}_d : B \succ -\frac{1-\lambda}{\lambda} I_d\}$. We make the following assumptions.

$$(1) \quad \rho(\Delta_K^\circ \times (\mathbb{R}^d)^K \times (\mathcal{B}_\lambda)^K) = 1.$$

(2) ρ admits a density $f_{(Q,A,B)}$ with respect the Lebesgue measure over $\Delta_K^\circ \times (\mathbb{R}^d)^K \times (\mathcal{B}_\lambda)^K$. Moreover, we assume the following factorization

$$f_{(Q,A,B)}(q, a, b) = f_Q(q) f_A(a) f_B(b),$$

for some maps f_Q, f_A, f_B implying that (Q_t, A_t, B_t) are mutually independent.

(3) For $K \geq 2$, $Q_t = \varphi(U_t)$, where the density of U_t satisfies Assumption B.4 with $N = K$. For $K = 1$, set $Q_t = 1$.

Lemma B.15 (Jacobian determinant). Fix $\Sigma \in \mathbb{S}_{++}^d$ and $\lambda \in (0, 1)$. Define $F : \mathcal{B}_\lambda \rightarrow \mathbb{S}_{++}^d$ by

$$F(B) := \Sigma^{1/2} ((1 - \lambda) I_d + \lambda B)^2 \Sigma^{1/2}, \quad B \in \mathcal{B}_\lambda.$$

For $B \in \mathcal{B}_\lambda$, set $C := (1 - \lambda) I_d + \lambda B$. The differential of F at B is the linear map

$$DF(B) : \mathbb{S}^d \rightarrow \mathbb{S}^d, \quad H \mapsto \lambda \Sigma^{1/2} (CH + HC) \Sigma^{1/2}.$$

Its determinant, computed in the same independent-entry coordinates on the domain and codomain, is

$$\det DF(B) = \lambda^{d(d+1)/2} (\det \Sigma)^{(d+1)/2} \prod_{i=1}^d (2c_i) \prod_{1 \leq i < j \leq d} (c_i + c_j).$$

where $c_1, \dots, c_d > 0$ are the eigenvalues of C .

Proof. Set $S := \Sigma^{1/2}$. For symmetric H small enough that $B + H \in \mathcal{B}_\lambda$, expansion gives

$$F(B + H) - F(B) = \lambda S(CH + HC)S + \lambda^2 SH^2 S.$$

The last term is $O(\|H\|_{\mathbb{F}}^2)$, which proves the formula for $DF(B)$. Define the linear maps $\mathcal{L}_C, \mathcal{C} : \mathbb{S}^d \rightarrow \mathbb{S}^d$ by $\mathcal{L}_C(H) := CH + HC$ and $\mathcal{C}(H) := SHS$. Then

$$DF(B)(H) = \lambda \mathcal{C}(\mathcal{L}_C(H)), \quad \det DF(B) = \lambda^{d(d+1)/2} \det \mathcal{C} \det \mathcal{L}_C,$$

since $\dim \mathbb{S}^d = d(d+1)/2$. Let $(u_1, \dots, u_d)$ be an orthonormal eigenbasis of C , with $Cu_i = c_i u_i$, and set $E_{i,j} := u_i u_j^\top$. The matrices $E_{i,i}$ and $E_{i,j} + E_{j,i}$, for $i < j$, form a basis of $\mathbb{S}^d$. Since $CE_{i,j} = c_i E_{i,j}$ and $E_{i,j}C = c_j E_{i,j}$,

$$\mathcal{L}_C(E_{i,i}) = 2c_i E_{i,i}, \quad \mathcal{L}_C(E_{i,j} + E_{j,i}) = (c_i + c_j)(E_{i,j} + E_{j,i}).$$

Thus $\mathcal{L}_C$ is diagonal in this basis, and, since the determinant of a linear map does not depend on the basis,

$$\det \mathcal{L}_C = \prod_{i=1}^d (2c_i) \prod_{1 \leq i < j \leq d} (c_i + c_j).$$

For $\mathcal{C}$, let $a_1, \dots, a_d > 0$ be the eigenvalues of S , and define $E_{i,j}$ as above from an orthonormal eigenbasis of S . Since $SE_{i,j}S = a_i a_j E_{i,j}$,

$$\mathcal{C}(E_{i,i}) = a_i^2 E_{i,i}, \quad \mathcal{C}(E_{i,j} + E_{j,i}) = a_i a_j (E_{i,j} + E_{j,i}).$$

Hence

$$\det \mathcal{C} = \prod_{i=1}^d a_i^2 \prod_{1 \leq i < j \leq d} a_i a_j = \prod_{i=1}^d a_i^{d+1} = (\det S)^{d+1} = (\det \Sigma)^{(d+1)/2},$$

since each index appears twice in the first product and $d-1$ times in the second. Substituting the two determinants into the composition formula proves the result. $\square$

Lemma B.16 (Explicit perturbation density). *Under the conditions of Lemma 3.11, fix $z = (p, m, \Sigma) \in \tilde{\mathcal{Z}}_K$ and $\lambda \in (0, 1)$. The inverse coordinates are given by (16). For $b \in (\mathcal{B}_\lambda)^K$, let $c_{j,1}, \dots, c_{j,d}$ denote the eigenvalues of $(1 - \lambda)I_d + \lambda b_j$. Define the Jacobian map $\mathcal{J}^\lambda : \tilde{\mathcal{Z}}_K \times (\mathcal{B}_\lambda)^K \rightarrow (0, \infty)$ by*

$$\mathcal{J}^\lambda(z, b) := \lambda^{q_K} \prod_{j=1}^K \left[(\det \Sigma_j)^{(d+1)/2} \prod_{i=1}^d (2c_{j,i}) \prod_{1 \leq i < \ell \leq d} (c_{j,i} + c_{j,\ell}) \right]. \quad (49)$$

Then, for $y \in Y_\lambda(z)$,

$$h^\lambda(y | z) = \frac{f_Q(q^\lambda(y | z)) f_A(a^\lambda(y | z)) f_B(b^\lambda(y | z))}{\mathcal{J}^\lambda(z, b^\lambda(y | z))}, \quad (50)$$

and the density is zero elsewhere. For $K = 1$, the factor f_Q and the free weight coordinates are omitted. In particular, $Z_{K,t}^{\lambda,\theta}$ has density $h^\lambda(\cdot | z_{K,t}^\theta)$.

Proof. For the fixed $z = (p, m, \Sigma)$, consider the map $\Psi_\lambda(z, \cdot)$ from (13):

$$\Psi_\lambda(z, (q, a, b)) = \left((1 - \lambda)p + \lambda q, (1 - \lambda)m + \lambda a, (\Sigma_j^{1/2}((1 - \lambda)I_d + \lambda b_j)^2 \Sigma_j^{1/2})_{j=1}^K \right).$$

For $b_k \in \mathcal{B}_\lambda$, the matrix $(1 - \lambda)I_d + \lambda b_k$ is positive definite. The square map is a smooth diffeomorphism of $\mathbb{S}_{++}^d$ onto itself, with inverse given by the positive definite square root. Thus the displayed map is a smooth diffeomorphism from $\Delta_K^\circ \times (\mathbb{R}^d)^K \times (\mathcal{B}_\lambda)^K$ onto $\mathcal{Y}_t^{\lambda,\theta} \times (\mathbb{R}^d)^K \times (\mathbb{S}_{++}^d)^K$, with the inverse given by (16).

For each j , let F_j be the covariance map F of Lemma B.15 with $\Sigma = \Sigma_j$. Each output block depends only on its corresponding input block. Hence, in the free weight, mean, and independent covariance coordinates, the Jacobian matrix is block diagonal:

$$D_{(q,a,b)} \Psi_\lambda(z, (q, a, b)) = \text{diag}(\lambda I_{K-1}, \lambda I_{Kd}, DF_1(b_1), \dots, DF_K(b_K)).$$

Its determinant is therefore the product of the block determinants. By Lemma B.15,

$$\begin{aligned} \det D_{(q,a,b)} \Psi_\lambda(z, (q, a, b)) &= \lambda^{K-1} \lambda^{Kd} \prod_{j=1}^K \det DF_j(b_j) \\ &= \lambda^{(K-1)+Kd+Kd(d+1)/2} \prod_{j=1}^K \left[(\det \Sigma_j)^{(d+1)/2} \prod_{i=1}^d (2c_{j,i}) \prod_{1 \leq i < \ell \leq d} (c_{j,i} + c_{j,\ell}) \right] \\ &= \mathcal{J}^\lambda(z, b) > 0, \end{aligned}$$

since $q_K = (K - 1) + Kd + Kd(d + 1)/2$. For $K = 1$, the weight block is omitted. The change-of-variables formula, together with $f_{(Q,A,B)} = f_Q f_A f_B$, now gives (50). $\square$

B.3.7 Policy-gradient representation

Assumption B.17 (Likelihood-ratio representation).

- (1) For every $t \in \llbracket 0, T - 1 \rrbracket$ and for every $(x, \mu, a) \in \mathbb{R}^d \times \mathcal{P}_2(\mathbb{R}^d) \times \mathcal{A}$, and every admissible $z \in \tilde{\mathcal{Z}}_K$, the maps $\theta \mapsto p_t^\theta(a | x, \mu)$ and $\theta \mapsto h^\lambda(y | z_{K,t+1}^\theta)$ are of class $\mathcal{C}^1(\Theta^\circ)$, where $\Theta \subset \Theta^\circ$, is an open set and strictly positive. This requires the map $\theta \mapsto z_{K,t}^\theta \in \tilde{\mathcal{Z}}_K$ to be $\mathcal{C}^1(\Theta^\circ)$. Moreover, they satisfy sufficient regularity conditions to justify differentiation under the integral sign and the corresponding score-function identities, namely existence of an integrable function G_θ such that the relevant derivatives are uniformly dominated by G_θ .

- (2) Moreover, we assume that

$$\sup_{\theta \in \Theta} \mathbb{E} \left[\sum_{t=0}^{T-1} \|\nabla_\theta \log p_t^\theta(X_t, \mathcal{D}_K(Q_t, M_t, \Sigma_t), A_t)\|^2 + \sum_{t=1}^T \|\nabla_\theta \log h^\lambda(Q_t, M_t, \Sigma_t | z_{K,t}^\theta)\|^2 \right] < \infty$$

(3) Denote by S the trajectory random variable

$$S := (X_0^{K,\lambda,\theta}, \alpha_0^{K,\lambda,\theta}, X_1^{K,\lambda,\theta}, \dots, \alpha_{T-1}^{K,\lambda,\theta}, X_T^{K,\lambda,\theta}, Z_{K,1}^{\lambda,\theta}, \dots, Z_{K,T}^{\lambda,\theta}),$$

and define its law by $\mathbb{P}^{\lambda,\theta} := \mathcal{L}(S)$. We assume that $\mathcal{R}(S) \in L^2(\Omega, \mathcal{F}, \mathbb{P}^{\lambda,\theta})$.

Proof of Theorem 3.12. Fix $\lambda \in (0, 1)$. Write a trajectory as $s = (x_0, a_0, x_1, \dots, a_{T-1}, x_T, z_1, \dots, z_T)$. For $t \in \mathcal{T}$, define the population-argument map $m_t : \tilde{\mathcal{Z}}_K^T \rightarrow \mathcal{P}_2(\mathbb{R}^d)$ by

$$m_0(z_1, \dots, z_T) = \mu_0, \quad m_t(z_1, \dots, z_T) = \mathcal{D}_K(z_t), \quad t \in \llbracket 1, T \rrbracket$$

We abbreviate these arguments as m_t . In particular, the initial population argument is kept exact and is not projected onto the Gaussian-mixture family.

Recalling the policy reference measure ν_A introduced in Section 2, define

$$\nu(ds) := \mu_0(dx_0) \prod_{t=1}^T dz_t \prod_{t=0}^{T-1} [\nu_A(da_t) P_t(dx_{t+1} \mid x_t, m_t, a_t)].$$

By independence of the randomizers, the trajectory law is

$$\mathbb{P}^{\lambda,\theta}(ds) := \mathcal{L}(S)(ds) = \ell^{\lambda,\theta}(s) \nu(ds),$$

where $\ell^{\lambda,\theta}(s) = \prod_{t=0}^{T-1} p_t^\theta(a_t \mid x_t, m_t) \prod_{t=1}^T h^\lambda(z_t \mid z_{K,t}^\theta)$. Here the return uses the same population arguments,

$$\mathcal{R}(s) = \sum_{t=0}^{T-1} r_t(x_t, m_t, a_t) + g(x_T, m_T),$$

and is independent of θ when s is held fixed. The reference measure ν is also independent of θ . Hence, we have $J_K^\lambda(\theta) = \int \mathcal{R}(s) \mathbb{P}^{\lambda,\theta}(ds) = \int \mathcal{R}(s) \ell^{\lambda,\theta}(s) \nu(ds)$. Now, under the Assumptions B.14, B.17, we get with dominated convergence theorem that

$$\nabla_\theta J_K^\lambda(\theta) = \int \mathcal{R}(s) \nabla_\theta \ell^{\lambda,\theta}(s) \nu(ds),$$

and we end using the (17) term and $\nabla_\theta \log h^\lambda(z_t \mid z_{K,t}^\theta) = (D_t^\theta)^\top s^\lambda(z_t, z_{K,t}^\theta)$. The probabilistic representation follows by definition of the quantities. $\square$

B.3.8 Sensitivity estimation and statistical bounds

Assumption B.18 (Statistical regularity). Assume Assumptions B.11, B.12, and B.13. Write $z^\theta = (z_{K,t}^\theta)_{t=1}^T \in \mathbb{R}^{Tq_K}$ and $D^\theta = \nabla_\theta z^\theta \in \mathbb{R}^{Tq_K \times d_\theta}$. Assume additionally:

- (1) Regular fitted coordinates. There exists a compact set $\mathcal{C}_K \subset \tilde{\mathcal{Z}}_K$ containing all exact and fitted coordinates used below. The fitted coordinates use the same component ordering as the exact coordinates. The map $\theta \mapsto z^\theta$ is $\mathcal{C}^3$, with uniformly bounded third derivatives on an admissible neighborhood containing all shifted parameters $\theta \pm \eta e_\ell$.
- (2) Fitting accuracy. Let $\hat{z}^{N,u} = (\hat{z}_{K,t}^{N,u})_{t=1}^T \in \mathcal{C}_K^T$ denote the coordinates fitted from a system of N interacting particles under parameter u . For a deterministic sequence $a_N \rightarrow 0$,

$$\sup_u \mathbb{E}[\|\hat{z}^{N,u} - z^u\|^2] \leq Ca_N^2.$$

The supremum is over all parameters at which a fit is computed, namely the base parameter $u = \theta$ (fitted with $N = M$ particles) and the shifted parameters $u = \theta \pm \eta e_\ell$, $\ell \in \llbracket 1, d_\theta \rrbracket$ (fitted with $N = n_0$ particles in the centered differences (67)), for $\theta \in \Theta$ and $0 < \eta \leq \eta_0$.

(3) Score identities and moments. *The likelihood-ratio identities hold for every prescribed coordinate flow $c \in \mathcal{C}_K^T$, both for differentiation in θ with c fixed and for differentiation in c , with no additional boundary term. For one rollout using population arguments $m_0 = \mu_0$ and $m_t = \mathcal{D}_K(\Psi_\lambda(c_t, R_t))$, denote its return and policy score by $\mathcal{R}^\lambda(c)$ and $\mathcal{S}_{\text{pol}}^\lambda(c)$. Uniformly in θ , c and $0 < \lambda \leq \lambda_0$,*

$$\mathbb{E}[|\mathcal{R}^\lambda(c)|^2 \|\mathcal{S}_{\text{pol}}^\lambda(c)\|^2] \leq C,$$

and

$$\mathbb{E}\left[|\mathcal{R}^\lambda(c)|^2 \sum_{t=1}^T \|s^\lambda(\Psi_\lambda(c_t, R_t), c_t)\|^2\right] \leq \frac{C}{\lambda^2}.$$

(4) Online simulations. *Conditional on the offline estimates, the B online trajectories are independent and identically distributed, using fresh simulation randomness.*

Proof of Theorem 3.13. Fix θ and set $n_0 := n/(2d_\theta)$. For each offline system of N particles simulated under parameter u , denote its states by $(X_t^{i,N,u})_{i=1}^N$ and define

$$\hat{\mu}_t^{N,u} := \frac{1}{N} \sum_{i=1}^N \delta_{X_t^{i,N,u}}, \quad \hat{z}^{N,u} := (\mathcal{E}_K(\hat{\mu}_t^{N,u}))_{t=1}^T \in \mathcal{C}_K^T.$$

The base system uses $(N, u) = (M, \theta)$, and the shifted systems use $(N, u) = (n_0, \theta \pm \eta e_\ell)$, where e_ℓ is the ℓ -th canonical basis vector of $\mathbb{R}^{d_\theta}$. Set

$$\hat{z} := \hat{z}^{M,\theta}, \quad \hat{D} := \begin{pmatrix} \hat{D}_1 \\ \vdots \\ \hat{D}_T \end{pmatrix} \in \mathbb{R}^{Tq_K \times d_\theta}, \quad \hat{D}e_\ell := \frac{\hat{z}^{n_0, \theta + \eta e_\ell} - \hat{z}^{n_0, \theta - \eta e_\ell}}{2\eta}, \quad \ell \in \llbracket 1, d_\theta \rrbracket,$$

where $\hat{D}_t \in \mathbb{R}^{q_K \times d_\theta}$ is the time- t block, as in (67). Write $\hat{G} := \hat{G}_{B,M,n}^{\lambda,\eta}(\theta)$. Throughout, the time-zero population argument remains fixed at μ_0 .

Conditional mean and its stability. For a prescribed coordinate flow c , define the randomized flow by

$$Z^\lambda : \tilde{\mathcal{Z}}_K^T \times \Omega \longrightarrow \tilde{\mathcal{Z}}_K^T, \quad Z^\lambda(c, \omega) = (\Psi_\lambda(c_t, R_t(\omega)))_{t=1}^T.$$

We omit ω for simplicity. Using the prescribed-flow objective F_K , define $\Gamma^\lambda : \mathcal{C}_K^T \times \mathbb{R}^{Tq_K \times d_\theta} \rightarrow \mathbb{R}^{d_\theta}$ by

$$\Gamma^\lambda(c, D) = \mathbb{E}[\nabla_\theta F_K(\theta, Z^\lambda(c))] + D^\top \mathbb{E}[(\nabla_c Z^\lambda(c))^\top \nabla_z F_K(\theta, Z^\lambda(c))]. \quad (51)$$

Here $\nabla_c Z^\lambda(c)$ is the Jacobian with respect to the prescribed coordinates.

The policy-score identity identifies the first term in (51). For the population score, differentiation with respect to c , followed by the chain rule, gives

$$\begin{aligned} \mathbb{E} \left[\mathcal{R}^\lambda(c) \begin{pmatrix} s^\lambda(\Psi_\lambda(c_1, R_1), c_1) \\ \vdots \\ s^\lambda(\Psi_\lambda(c_T, R_T), c_T) \end{pmatrix} \right] &= \nabla_c \mathbb{E}[F_K(\theta, Z^\lambda(c))] \\ &= \mathbb{E}[(\nabla_c Z^\lambda(c))^\top \nabla_z F_K(\theta, Z^\lambda(c))]. \end{aligned}$$

Consequently, $\Gamma^\lambda(c, D)$ is the expectation of the score estimator using coordinates c and sensitivity matrix D .

The same covariance calculation as in the gradient convergence proof gives, uniformly for $c \in \mathcal{C}_K^T$,

$$\|Z^\lambda(c) - c\| + \|\nabla_c Z^\lambda(c) - I\| \leq C\lambda W, \quad (52)$$

where

$$W = 1 + \sum_{t=1}^T \sum_{j=1}^K (\|A_{j,t}\| + \|B_{j,t}\|_{\mathbb{F}}^2), \quad \mathbb{E}[W^{r+1}] < \infty.$$

The constant C in (52) depends on c only through an upper bound on $\|c\|$ and a lower bound on the covariance eigenvalues of c , and both are uniform over the compact set $\mathcal{C}_K$. Indeed, $c \mapsto \|c\|$ is continuous, hence bounded on $\mathcal{C}_K$. Moreover, $\lambda_{\min}(\Sigma) = \min_{\|u\|=1} u^\top \Sigma u$ is 1-Lipschitz for $\|\cdot\|_{\text{op}}$, so $c \mapsto \min_j \lambda_{\min}(\Sigma_j)$ is continuous and positive on $\mathcal{C}_K \subset \tilde{\mathcal{Z}}_K$. It therefore attains a positive minimum c_K on $\mathcal{C}_K$.

Since $\|c\| \leq C$ on $\mathcal{C}_K^T$, $\lambda \leq 1$ and $W \geq 1$, the bound (52) gives $\|Z^\lambda(c)\| \leq \|c\| + C\lambda W \leq CW$ and $\|\nabla_c Z^\lambda(c)\| \leq \|I\| + C\lambda W \leq CW$. The growth bound in Assumption B.13(2) gives $\|\nabla_z F_K(\theta, Z^\lambda(c))\| \leq L(1 + \|Z^\lambda(c)\|)^r \leq CW^r$. Hence

$$\|\mathbb{E}[(\nabla_c Z^\lambda(c))^\top \nabla_z F_K(\theta, Z^\lambda(c))]\| \leq \mathbb{E}[\|\nabla_c Z^\lambda(c)\| \|\nabla_z F_K(\theta, Z^\lambda(c))\|] \leq C \mathbb{E}[W^{r+1}] \leq C.$$

Since Γ^λ is linear in D ,

$$\|\Gamma^\lambda(c, D) - \Gamma^\lambda(c, D^\theta)\| \leq C\|D - D^\theta\|_F. \quad (53)$$

At $\lambda = 0$,

$$\Gamma^0(c, D^\theta) = \nabla_\theta F_K(\theta, c) + (D^\theta)^\top \nabla_z F_K(\theta, c).$$

Since $Z^0(c) = c$ and $\nabla_c Z^0(c) = I$, subtracting and adding $(D^\theta)^\top \mathbb{E}[\nabla_z F_K(\theta, Z^\lambda(c))]$ gives

$$\begin{aligned} \Gamma^\lambda(c, D^\theta) - \Gamma^0(c, D^\theta) &= \mathbb{E}[\nabla_\theta F_K(\theta, Z^\lambda(c)) - \nabla_\theta F_K(\theta, c)] + (D^\theta)^\top \mathbb{E}[\nabla_z F_K(\theta, Z^\lambda(c)) - \nabla_z F_K(\theta, c)] \\ &\quad + (D^\theta)^\top \mathbb{E}[(\nabla_c Z^\lambda(c) - I)^\top \nabla_z F_K(\theta, Z^\lambda(c))]. \end{aligned}$$

By the Lipschitz bound in Assumption B.13(2) and (52), $\|\nabla F_K(\theta, Z^\lambda(c)) - \nabla F_K(\theta, c)\| \leq L(1 + \|Z^\lambda(c)\| + \|c\|)^r \|Z^\lambda(c) - c\| \leq C\lambda W^{r+1}$. The last term is bounded by $\|D^\theta\| C\lambda W \cdot CW^r$ using (52) and the growth bound. Since D^θ is uniformly bounded and $\mathbb{E}[W^{r+1}] < \infty$,

$$\|\Gamma^\lambda(c, D^\theta) - \Gamma^0(c, D^\theta)\| \leq C\lambda. \quad (54)$$

Moreover, the Lipschitz derivative bounds on the compact set of base coordinates give

$$\|\Gamma^0(c, D^\theta) - \Gamma^0(z^\theta, D^\theta)\| \leq C\|c - z^\theta\|. \quad (55)$$

Since $\Gamma^0(z^\theta, D^\theta) = \nabla_\theta J_K(\theta)$, combining (53), (54), and (55) yields

$$\|\Gamma^\lambda(c, D) - \nabla_\theta J_K(\theta)\| \leq C(\lambda + \|c - z^\theta\| + \|D - D^\theta\|_F). \quad (56)$$

Population-coordinate error. The fitting-accuracy assumption directly gives

$$\mathbb{E}[\|\hat{z} - z^\theta\|^2] \leq Ca_M^2. \quad (57)$$

Sensitivity error. Define the exact centered difference $D^\eta \in \mathbb{R}^{Tq_K \times d_\theta}$ column-wise by

$$D^\eta e_\ell = \frac{z^{\theta+\eta e_\ell} - z^{\theta-\eta e_\ell}}{2\eta}.$$

Taylor's formula along the ℓ -th coordinate gives, with Taylor remainder $R_\ell^\pm$,

$$\begin{aligned} z^{\theta \pm \eta e_\ell} &= z^\theta \pm \eta \partial_\ell z^\theta + \frac{\eta^2}{2} \partial_\ell^2 z^\theta + R_\ell^\pm, \\ \|R_\ell^\pm\| &\leq \frac{\eta^3}{6} \sup_{|u| \leq \eta} \|\partial_\ell^3 z^{\theta+ue_\ell}\| \leq C\eta^3, \end{aligned}$$

by Assumption B.18(1). Subtracting the two expansions cancels the constant and quadratic terms. Since $D^\theta e_\ell = \partial_\ell z^\theta$,

$$\|(D^\eta - D^\theta)e_\ell\| = \left\| \frac{R_\ell^+ - R_\ell^-}{2\eta} \right\| \leq C\eta^2.$$

Squaring and summing over the d_θ columns yields

$$\|D^\eta - D^\theta\|_F \leq C\eta^2. \quad (58)$$

For each shifted fit, set $\epsilon_\ell^\pm := \widehat{z}^{n_0, \theta \pm \eta e_\ell} - z^{\theta \pm \eta e_\ell} \in \mathbb{R}^{TqK}$. Subtracting the fitted and exact centered differences gives

$$\begin{aligned} (\widehat{D} - D^\eta)_{e_\ell} &= \frac{(\widehat{z}^{n_0, \theta + \eta e_\ell} - z^{\theta + \eta e_\ell}) - (\widehat{z}^{n_0, \theta - \eta e_\ell} - z^{\theta - \eta e_\ell})}{2\eta} \\ &= \frac{\epsilon_\ell^+ - \epsilon_\ell^-}{2\eta}. \end{aligned}$$

The squared Frobenius norm is the sum of the squared column norms. Using $\|a - b\|^2 \leq 2\|a\|^2 + 2\|b\|^2$ and the fitting bound $\mathbb{E}[\|\epsilon_\ell^\pm\|^2] \leq Ca_{n_0}^2$ from Assumption B.18(2),

$$\begin{aligned} \mathbb{E}[\|\widehat{D} - D^\eta\|_F^2] &= \sum_{\ell=1}^{d_\theta} \mathbb{E}\left[\left\|\frac{\epsilon_\ell^+ - \epsilon_\ell^-}{2\eta}\right\|^2\right] \leq \frac{1}{2\eta^2} \sum_{\ell=1}^{d_\theta} (\mathbb{E}[\|\epsilon_\ell^+\|^2] + \mathbb{E}[\|\epsilon_\ell^-\|^2]) \\ &\leq C \frac{a_{n_0}^2}{\eta^2}. \end{aligned}$$

Decomposing $\widehat{D} - D^\theta = (\widehat{D} - D^\eta) + (D^\eta - D^\theta)$ and using $\|A + B\|_F^2 \leq 2\|A\|_F^2 + 2\|B\|_F^2$, together with (58), proves

$$\mathbb{E}[\|\widehat{D} - D^\theta\|_F^2] \leq C\left(\eta^4 + \frac{a_{n_0}^2}{\eta^2}\right). \quad (59)$$

Because D^θ is uniformly bounded and $a_{n_0} \leq \eta$,

$$\mathbb{E}[\|\widehat{D}\|_F^2] \leq C.$$

Bias and mean-square error. Define the sigma-algebra generated by the offline particle systems by

$$\mathcal{F}^{\text{off}} := \sigma\left((X_t^{i, M, \theta})_{t \in \llbracket 0, T \rrbracket, i \in \llbracket 1, M \rrbracket}, (X_t^{i, n_0, \theta + \varepsilon \eta e_\ell})_{t \in \llbracket 0, T \rrbracket, i \in \llbracket 1, n_0 \rrbracket, \ell \in \llbracket 1, d_\theta \rrbracket, \varepsilon \in \{-1, 1\}}\right).$$

Here $\vee$ denotes the sigma-algebra generated by the indicated sigma-algebras. Since the fitted coordinates are functions of these particles, $\widehat{z}$ and $\widehat{D}$ are $\mathcal{F}^{\text{off}}$ -measurable. Set $\overline{G} := \mathbb{E}[\widehat{G} \mid \mathcal{F}^{\text{off}}]$. The online simulation randomness gives $\overline{G} = \Gamma^\lambda(\widehat{z}, \widehat{D})$.

Since $\overline{G} = \Gamma^\lambda(\widehat{z}, \widehat{D})$, the stability estimate (56) and $(a + b + c)^2 \leq 3(a^2 + b^2 + c^2)$ give

$$\begin{aligned} \mathbb{E}[\|\overline{G} - \nabla_\theta J_K(\theta)\|^2] &\leq C \mathbb{E}[(\lambda + \|\widehat{z} - z^\theta\| + \|\widehat{D} - D^\theta\|_F)^2] \\ &\leq C(\lambda^2 + \mathbb{E}[\|\widehat{z} - z^\theta\|^2] + \mathbb{E}[\|\widehat{D} - D^\theta\|_F^2]) \\ &\leq C\left(\lambda^2 + a_M^2 + \eta^4 + \frac{a_{n_0}^2}{\eta^2}\right), \end{aligned} \quad (60)$$

where the last inequality uses the population-coordinate bound (57) and the sensitivity bound (59). Since $\mathbb{E}[\widehat{G}] = \mathbb{E}[\overline{G}]$, Jensen's inequality proves the bias bound in the Theorem.

Conditional on $\mathcal{F}^{\text{off}}$, the online estimator averages B independent contributions. Applying the same conditional-variance argument as in the paragraph *Bias and mean-square error of the online estimator* of the discrete state space bias and mean-square error proof in Appendix B.2.5, together with the score moment bounds of Assumption B.18(3), gives

$$\mathbb{E}[\|\widehat{G} - \overline{G}\|^2 \mid \mathcal{F}^{\text{off}}] \leq \frac{C}{B}(1 + \lambda^{-2}\|\widehat{D}\|_F^2). \quad (61)$$

Write $\widehat{G} - \nabla_\theta J_K(\theta) = (\widehat{G} - \overline{G}) + (\overline{G} - \nabla_\theta J_K(\theta))$. Since $\overline{G}$ is $\mathcal{F}^{\text{off}}$ -measurable and $\mathbb{E}[\widehat{G} - \overline{G} \mid \mathcal{F}^{\text{off}}] = 0$, the cross term vanishes:

$$\begin{aligned} \mathbb{E}[\langle \widehat{G} - \overline{G}, \overline{G} - \nabla_\theta J_K(\theta) \rangle] &= \mathbb{E}[\langle \mathbb{E}[\widehat{G} - \overline{G} \mid \mathcal{F}^{\text{off}}], \overline{G} - \nabla_\theta J_K(\theta) \rangle] \\ &= 0. \end{aligned}$$

Expanding the squared norm and taking expectations therefore gives

$$\mathbb{E}[\|\widehat{G} - \nabla_\theta J_K(\theta)\|^2] = \mathbb{E}[\|\widehat{G} - \overline{G}\|^2] + \mathbb{E}[\|\overline{G} - \nabla_\theta J_K(\theta)\|^2].$$

Combining (60) and (61) proves the result. $\square$

C LINEAR DYNAMICS: A FULLY SOLVABLE FRAMEWORK

We present an application of the continuous-state setting developed in Section 3.2 in the linear dynamics setting where the problem remains fully tractable. We will assume that the underlying dynamics is linear in its parameters with Gaussian noise. The literature on this kind of problem is extensive, see [Yong \(2013\)](#); [Elliott et al. \(2013\)](#); [Carmona et al. \(2019\)](#) among others, but its tractability makes it well-suited for benchmarking. For simplicity and readability, we provide the computations in the case $d = 1$ but it can easily be adapted to the multi-dimensional setting. Hence, we make the following assumptions

Assumption C.1. For any $(t, x, a, m) \in \llbracket 0, T - 1 \rrbracket \times \mathbb{R} \times \mathbb{R} \times \mathcal{P}_2(\mathbb{R})$, we assume

$$\begin{cases} P_t(x, a, m) &= \mathcal{N}(a_t + b_t x + \bar{b}_t \bar{m} + c_t a, d_t), \\ X_0 &\sim \mathcal{N}(m_0, \Sigma_0), \\ \pi_t^\theta(\cdot | x, m) &= \mathcal{N}(e_t^\theta x + \bar{e}_t^\theta m, f_t^\theta) \end{cases}$$

for some $\mathbb{R}$ -valued families $(a_t)_t, (b_t)_t, (\bar{b}_t)_t, (c_t)_t, (e_t^\theta)_t, (\bar{e}_t^\theta)_t$ and $(d_t)_t, (f_t^\theta)_t \in \mathbb{R}_+^*$ and some $m_0 \in \mathbb{R}$ and $\Sigma_0 \in \mathbb{R}_+$ and where we denoted $\bar{m} := \mathbb{E}_{X \sim m}[X]$. Moreover, we impose sufficient regularity conditions on the maps $\theta \mapsto (e_t^\theta, \bar{e}_t^\theta, f_t^\theta)$ to justify the score-function identity underlying the policy-gradient representation.

Remark C.2. Even if we are assuming the underlying transition kernel dynamics, we are still relatively in a model-free setting as we do not have access to its parameters.

Lemma C.3. (Stability of the process $(X_t)_{t \in \mathcal{T}}$).

Let $t \in \llbracket 0, T \rrbracket$ and $\theta \in \Theta$. Under Assumptions C.1, we have

$$X_t^\theta \sim \mathcal{N}(m_t^\theta, \Sigma_t^\theta),$$

where the sequences $(m_t^\theta)_{t \in \mathcal{T}}$ and $(\Sigma_t^\theta)_{t \in \mathcal{T}}$ are solution to the linear system

$$\begin{cases} m_{t+1}^\theta &= a_t + (b_t + \bar{b}_t)m_t^\theta + c_t(e_t^\theta + \bar{e}_t^\theta)m_t^\theta, \\ \Sigma_{t+1}^\theta &= (b_t + c_t e_t^\theta)^2 \Sigma_t^\theta + c_t^2 (f_t^\theta) + d_t \end{cases} \quad (62)$$

starting from $m_0^\theta = m_0$ and $\Sigma_0^\theta = \Sigma_0$.

Proof. The proof is straightforward recalling that $X_{t+1} | (X_t, \alpha_t, \mathbb{P}_{X_t}) \sim P_t(X_t, \alpha_t, \mathbb{P}_{X_t})$ $\mathbb{P}$ -a.s. Hence, it implies that $\mathbb{P}$ -a.s.

$$X_{t+1} = a_t + b_t X_t + \bar{b}_t \mathbb{E}[X_t] + c_t(e_t^\theta X_t + \bar{e}_t^\theta \mathbb{E}[X_t]) + \sqrt{f_t^\theta} \epsilon_{t+1} + \sqrt{d_t} \tilde{\epsilon}_{t+1}, \quad (63)$$

where ϵ_{t+1} and $\tilde{\epsilon}_{t+1}$ are independent standard Gaussian random variables independent of X_t . The result follows taking the expectation and the variance in (63). $\square$

Hence, under Assumption C.1 and the specific Gaussian parametrization, it follows that the process X^θ remains in the manifold of Gaussian random variables, namely its law is characterized by a finite-dimensional vector, here the couple $(\mathbb{E}[X_t^\theta], \mathbb{V}[X_t^\theta])$ for any $t \in \mathcal{T}$.

Proposition C.4. (Transport map representation). *According to the transport map construction in 3.6, the following holds*

$$M_t^{\lambda,\theta}(\omega) = \mathcal{N}(m_t^{\lambda,\theta}(\omega), \Sigma_t^{\lambda,\theta}(\omega)), \quad \omega \in \Omega, \quad t \in \llbracket 1, T \rrbracket.$$

where we set

$$\begin{cases} m_t^{\lambda,\theta}(\omega) &:= (1-\lambda)m_t^\theta + \lambda A_t(\omega), \\ \Sigma_t^{\lambda,\theta}(\omega) &:= ((1-\lambda) + \lambda B_t(\omega))^2 \Sigma_t^\theta \end{cases}$$

Proof. This is a direct application of Proposition 3.6. $\square$

Lemma C.5. (Law of the random variables $(m_t^{\lambda,\theta}, \Sigma_t^{\lambda,\theta})_{t \in \llbracket 1, T \rrbracket}$).

Let $U_\lambda = \mathbb{R} \times (-\frac{1-\lambda}{\lambda}, +\infty)$. For any $t \in \llbracket 1, T \rrbracket$, $\theta \in \Theta$ and $\lambda \in (0, 1)$, suppose that $(A_t, B_t) \in U_\lambda$ $\mathbb{P}$ -a.s and admits a strictly positive $\mathcal{C}^1$ density map $f_{(A_t, B_t)}$ with respect to the Lebesgue measure over U_λ . Then the random variable $(m_t^{\lambda,\theta}, \Sigma_t^{\lambda,\theta})$ admits the following density

$$h^\lambda((m, \Sigma) \mid z_{1,t}^\theta) = \frac{f_{(A_t, B_t)}(a_t^{\lambda,\theta}(m, \Sigma), b_t^{\lambda,\theta}(m, \Sigma))}{2\lambda^2 \sigma_t^\theta \sqrt{\Sigma}},$$

where $z_{1,t}^\theta := (m_t^\theta, \Sigma_t^\theta) = (m_t^\theta, (\sigma_t^\theta)^2) \in \mathbb{R} \times \mathbb{R}_+^*$, and the maps $a_t^{\lambda,\theta}$ and $b_t^{\lambda,\theta}$ are defined as

$$\begin{cases} a_t^{\lambda,\theta}(m, \Sigma) &= \frac{m - (1-\lambda)m_t^\theta}{\lambda}, \\ b_t^{\lambda,\theta}(m, \Sigma) &= \frac{1}{\lambda} \left(\frac{\sqrt{\Sigma}}{\sigma_t^\theta} - (1-\lambda) \right) \end{cases}$$

and where the quantities $(m_t^\theta, \sigma_t^\theta)$ are given by (62).

Proof. This is a direct application of Lemma B.16 specified to our setting where $K = 1$. $\square$

Theorem C.6. (Policy gradient in the L-Q setting).

For any $\theta \in \Theta$, the map J^λ defined in (5) is differentiable with respect to θ with gradient map given by

$$\nabla_\theta J^\lambda(\theta) = \mathbb{E} \left[\sum_{t=0}^{T-1} R_t^{\lambda,\theta} \nabla_\theta \log \pi_t^\theta(X_t^{\lambda,\theta}, \mathcal{N}(m_t^{\lambda,\theta}, \Sigma_t^{\lambda,\theta}), a_t^{\lambda,\theta}) + \sum_{t=1}^T R_t^{\lambda,\theta} \nabla_\theta \log h^\lambda((m_t^{\lambda,\theta}, \Sigma_t^{\lambda,\theta}) \mid z_{1,t}^\theta) \right]$$

where we denoted the random variables $(R_s^{\lambda,\theta})_{s \in \mathcal{T}}$ as

$$\begin{cases} R_t^{\lambda,\theta} &:= \sum_{s=t}^{T-1} r_s(X_s^{\lambda,\theta}, \mathcal{N}(m_s^{\lambda,\theta}, \Sigma_s^{\lambda,\theta}), a_s^{\lambda,\theta}) + g(X_T^{\lambda,\theta}, \mathcal{N}(m_T^{\lambda,\theta}, \Sigma_T^{\lambda,\theta})), \quad t \in \llbracket 0, T-1 \rrbracket, \\ R_T^{\lambda,\theta} &:= g(X_T^{\lambda,\theta}, \mathcal{N}(m_T^{\lambda,\theta}, \Sigma_T^{\lambda,\theta})) \end{cases}$$

Proof. The proof follows also as a direct application of Theorem 3.12. $\square$

Corollary C.7. (Score function of the mean-field density).

For any $\lambda \in (0, 1)$, $\theta \in \Theta$ and $t \in \llbracket 1, T \rrbracket$, we have

$$\begin{aligned} \nabla_\theta \log h^\lambda((m, \Sigma) \mid z_{1,t}^\theta) &= -\frac{1-\lambda}{\lambda} \partial_a \log f_{(A_t, B_t)}(a_t^{\lambda,\theta}(m, \Sigma), b_t^{\lambda,\theta}(m, \Sigma)) \nabla_\theta m_t^\theta, \\ &\quad -\frac{1}{\lambda} \partial_b \log f_{(A_t, B_t)}(a_t^{\lambda,\theta}(m, \Sigma), b_t^{\lambda,\theta}(m, \Sigma)) \frac{\sqrt{\Sigma}}{\sigma_t^\theta} \nabla_\theta \log \sigma_t^\theta \\ &\quad - \nabla_\theta \log(\sigma_t^\theta) \end{aligned}$$

Proof. The proof is straightforward noticing that

$$\nabla_\theta \left(\frac{1}{\sigma_t^\theta} \right) = -\frac{\nabla_\theta \sigma_t^\theta}{(\sigma_t^\theta)^2} = -\frac{\nabla_\theta \log(\sigma_t^\theta)}{\sigma_t^\theta}$$

and recalling (C.5). $\square$

From the computation of the score function of the mean-field density in Corollary C.7, it appears that we need to get estimators of the quantities $(\nabla_\theta m_t^\theta)_{t \in \llbracket 1, T \rrbracket}$ and $((\nabla_\theta \sigma_t^\theta)^2)_{t \in \llbracket 1, T \rrbracket}$ similarly as we had to compute estimators of $(D_t^\theta)_{t \in \mathcal{T}}$. Unlike the approach developed in the main body, we can look for a probabilistic approximation of these quantities in the spirit of the discrete-time setting. Indeed, by noticing that

$$\begin{cases} m_t^\theta &= \mathbb{E} \left[\sum_{s=0}^{T-1} \mathbf{1}_{t=s} X_s^\theta + \mathbf{1}_{t=T} X_T^\theta \right], \\ (\sigma_t^\theta)^2 &= \mathbb{E} \left[\sum_{s=0}^{T-1} \mathbf{1}_{s=t} (X_s^\theta)^2 + \mathbf{1}_{t=T} (X_T^\theta)^2 \right] - (m_t^\theta)^2, \end{cases}$$

We hence are able to provide the following probabilistic approximations of the required gradient quantities.

$$\begin{cases} \nabla_\theta m_t^\theta &\approx \mathbb{E} \left[X_t^{\lambda, \theta} \left(\sum_{\tau=0}^{t-1} \nabla_\theta \log \pi_\tau^\theta(X_\tau^{\lambda, \theta}, \mathcal{N}(M_\tau, \Sigma_\tau), \alpha_\tau^{\lambda, \theta}) + \sum_{\tau=1}^{t-1} \nabla_\theta \log h^\lambda((M_\tau, \Sigma_\tau) \mid z_{1, \tau}^\theta) \right) \right. \\ &\quad \left. + \mathbf{1}_{t=T} X_T^{\lambda, \theta} \nabla_\theta \log h^\lambda((M_T, \Sigma_T) \mid z_{1, T}^\theta) \right], \\ \nabla_\theta (\sigma_t^\theta)^2 &\approx \mathbb{E} \left[(X_t^{\lambda, \theta})^2 \left(\sum_{\tau=0}^{t-1} \nabla_\theta \log \pi_\tau^\theta(X_\tau^{\lambda, \theta}, \mathcal{N}(M_\tau, \Sigma_\tau), \alpha_\tau^{\lambda, \theta}) + \sum_{\tau=1}^{t-1} \nabla_\theta \log h^\lambda((M_\tau, \Sigma_\tau) \mid z_{1, \tau}^\theta) \right) \right. \\ &\quad \left. + \mathbf{1}_{t=T} (X_T^{\lambda, \theta})^2 \nabla_\theta \log h^\lambda((M_T, \Sigma_T) \mid z_{1, T}^\theta) \right] - 2 \nabla_\theta m_t^\theta m_t^\theta. \end{cases}$$

The approximations follow by similar arguments as in (Meunier et al., 2026, Section 3.1) so we refer to the references therein for full details. Hence, it follows by noticing that $\nabla_\theta \log(\sigma_t^\theta) = \frac{\nabla_\theta (\sigma_t^\theta)^2}{2(\sigma_t^\theta)^2}$.

The probabilistic representations of the moment sensitivities $\nabla_\theta m_t^\theta$ and $\nabla_\theta \sigma_t^\theta$ provide an alternative model-free approach to evaluate the mean-field score. By mirroring the likelihood-ratio technique of the finite-state setting, this formulation bypasses the need for the centered policy differences used in the general continuous-state algorithm. We denote this variant as Transport-Proba. Following the structure of Algorithm 1, we compare Transport-Proba directly against both standard REINFORCE and the default Transport estimator. The results are presented in Appendix E.3.3.

D ALGORITHMIC DETAILS

D.1 Sampling details

Each gradient step at the current θ has an offline stage, which estimates the population coordinates and their sensitivities, and an online stage, which simulates the perturbed trajectories entering the estimator (12) or (18). All stages use the same rollout: given population arguments $(m_t)_{t \in \mathcal{T}}$,

$$X_0 \sim \mu_0, \quad \alpha_t \sim \pi_t^\theta(\cdot \mid X_t, m_t), \quad X_{t+1} \sim P_t(\cdot \mid X_t, m_t, \alpha_t), \quad t \in \llbracket 0, T-1 \rrbracket, \quad (64)$$

with policy scores $\nabla_\theta \log p_t^\theta(\alpha_t \mid X_t, m_t)$. The procedure is summarized in Algorithm 1.

Offline stage: population coordinates. Run M interacting particles $(X_t^\ell)_{\ell=1}^M$ through (64), with X_0^ℓ i.i.d. with law μ_0 and population arguments given by their empirical law $m_t = \hat{\mu}_t^M := \frac{1}{M} \sum_{\ell=1}^M \delta_{X_t^\ell}$. The estimated coordinates are

$$\hat{z}_t := \begin{cases} (\hat{\mu}_t^M(1), \dots, \hat{\mu}_t^M(N-1)), & \mathcal{X} \text{ finite,} \\ \mathcal{E}_K(\hat{\mu}_t^M), & \mathcal{X} = \mathbb{R}^d, \end{cases}$$

where, in continuous state space, $\mathcal{E}_K$ is computed by constrained Expectation-Maximization on the particles.

Offline stage: sensitivities in finite state space. Set $\hat{D}_0^{\eta, M, n} := 0$. For $t = 1, \dots, T$, run a batch of n rollouts (64) of length t , indexed by $r \in \llbracket 1, n \rrbracket$, with population arguments

$$\begin{aligned} \overline{m}_0^{(t, r)} &= \mu_0, \quad \overline{m}_s^{(t, r)} = (1 - \eta) \hat{\mu}_s^M + \eta \overline{Q}_s^{(t, r)}, \\ \overline{Q}_s^{(t, r)} &= \varphi(\overline{U}_s^{(t, r)}), \quad s \in \llbracket 1, t-1 \rrbracket, \end{aligned}$$

where the $\overline{U}_s^{(t, r)}$ are i.i.d. with density ρ . Denote the states and actions of these rollouts by $\overline{X}_s^{(t, r)}$ and $\overline{\alpha}_s^{(t, r)}$, respectively, and define their policy scores by

$$\overline{\Psi}_s^{(t, r)} := \nabla_\theta \log p_s^\theta(\overline{\alpha}_s^{(t, r)} \mid \overline{X}_s^{(t, r)}, \overline{m}_s^{(t, r)}) \in \mathbb{R}^{d_\theta}, \quad s \in \llbracket 0, t-1 \rrbracket, \quad (65)$$

where the derivative holds the action, state, and population argument fixed. The k -th row of $\widehat{D}_t := \widehat{D}_t^{\eta, M, n} \in \mathbb{R}^{(N-1) \times d_\theta}$, written $\widehat{D}_t^{\eta, M, n}$ in (12) for $k \in \llbracket 1, N-1 \rrbracket$, is

$$(\widehat{D}_t)_{k,:} := \frac{1}{n} \sum_{r=1}^n \mathbf{1}_{\{\overline{X}_t^{(t,r)} = x_k\}} \left[\sum_{s=0}^{t-1} \overline{\Psi}_s^{(t,r)} - \frac{1-\eta}{\eta} \sum_{s=1}^{t-1} \widehat{D}_s^\top H(\overline{Q}_s^{(t,r)}) \right]^\top. \quad (66)$$

Each row is the score-function estimator of Theorem 3.4, at scale η and horizon t , for the terminal reward $\mathbf{1}_{\{x=x_k\}}$, with $\widehat{\mu}^M$ and the earlier estimates $\widehat{D}_1, \dots, \widehat{D}_{t-1}$ in place of μ^θ and $D_1^\theta, \dots, D_{t-1}^\theta$. It estimates the sensitivity of the η -perturbed state law.

Offline stage: sensitivities in continuous state space. For each $\ell \in \llbracket 1, d_\theta \rrbracket$, run two independent systems of n_0 interacting particles as above, at the shifted parameters $\theta \pm \eta e_\ell$, where e_ℓ is the ℓ -th canonical basis vector of $\mathbb{R}^{d_\theta}$ and $\theta \pm \eta e_\ell \in \Theta$. Denote by $\widehat{z}_t^{\theta \pm \eta e_\ell}$ their fitted coordinates, which stack the weights, means and independent covariance entries. Set

$$\widehat{D}_t e_\ell := \frac{\widehat{z}_t^{\theta + \eta e_\ell} - \widehat{z}_t^{\theta - \eta e_\ell}}{2\eta}, \quad t \in \llbracket 1, T \rrbracket, \quad \ell \in \llbracket 1, d_\theta \rrbracket. \quad (67)$$

These centered differences use $n = 2d_\theta n_0$ particles in total.

Online stage. Run B rollouts (64) with population arguments $\widehat{M}_0^{(b)} = \mu_0$ and, for $t \in \llbracket 1, T \rrbracket$,

$$\widehat{M}_t^{(b)} = \begin{cases} (1-\lambda)\widehat{\mu}_t^M + \lambda\varphi(U_t^{(b)}), & \mathcal{X} \text{ finite,} \\ \mathcal{D}_K(\Psi_\lambda(\widehat{z}_t, R_t^{(b)})), & \mathcal{X} = \mathbb{R}^d, \end{cases}$$

where the randomizers $U_t^{(b)}$ or $R_t^{(b)}$ are i.i.d. with law ρ . Their returns $\widehat{\mathcal{R}}^{(b)}$ and policy scores $\widehat{\mathcal{S}}_{\text{pol}}^{(b)}$, together with $\widehat{D}$, give the estimator (12) or (18), and θ is updated by $\theta \leftarrow \theta + \gamma \widehat{G}_{B, M, n}^{\lambda, \eta}(\theta)$.

D.2 Computational Cost

The finite-state score estimator uses the representation of population probabilities as expectations of state indicators. The continuous-state centered-difference estimator avoids differentiating the nonlinear mixture-fitting map.

Proposition D.1 (Simulation cost). *One policy update uses*

$$C_{\text{sim}} = \begin{cases} T(M+B) + nT(T+1)/2, & \text{finite states, fresh auxiliary batches,} \\ T(M+n+B), & \text{finite states, shared auxiliary batch,} \\ T(M+n+B), & \text{continuous states.} \end{cases}$$

These counts exclude the arithmetic cost of mixture fitting and score evaluation. The finite-state MSE theorem covers fresh auxiliary batches; the experiments use a shared batch.

Proof. The population and online stages use MT and BT transitions, respectively. Fresh finite-state auxiliary batches require $n \sum_{t=1}^T t = nT(T+1)/2$ transitions, whereas a shared batch of n trajectories of length T requires nT transitions. In continuous state space, the $2d_\theta$ auxiliary systems contain $n = 2d_\theta n_0$ particles in total and require nT transitions. $\square$

Simulator calls. In continuous state space, for a fixed total n , increasing d_θ decreases the number $n_0 = n/(2d_\theta)$ of particles per shifted system. In finite state space, drawing a fresh auxiliary batch of length t for each t , the variant covered by Theorem 3.5, raises the auxiliary cost from nT to $nT(T+1)/2$ transitions. For reference, MF-REINFORCE (Meunier et al., 2026), with two auxiliary trajectories resampled at each terminal time, uses $C_{\text{MF}} = B_{\text{MF}}T + n_{\text{MF}}T(T+1)$ transitions.

Arithmetic and storage. Let $q = N - 1$ in finite state space and $q = q_K$ in continuous state space. Beyond simulation, policy evaluation and mixture fitting, one update costs $O(BTqd_\theta)$ operations for the main scores, plus $O(nT^2qd_\theta)$ for the finite-state auxiliary recursion or $O(Tqd_\theta)$ for the centered differences, and $O(Tq(1 + d_\theta))$ memory for $(\hat{z}_t, \hat{D}_t)_{t=1}^T$. The $2d_\theta$ shifted particle systems are independent and are simulated as a single batch, whereas the finite-state sensitivities are computed sequentially in t .

E EXPERIMENT DETAILS

E.1 Protocol and settings

All configurations use five paired seeds and Adam (Kingma and Ba, 2014). Within each benchmark, methods share the policy class, the initialization, the training initial laws and the validation schedule. On the finite benchmarks, the initial law is resampled during training and validation uses a fixed initial law, as in Meunier et al. (2026); the continuous benchmarks use a fixed initial law throughout. Every ten updates, the frozen policy is evaluated with $M_{\text{val}} = 10^5$ interacting particles, except on bimodal allocation, where validation is exact.

Evaluation of the optimality gap. For seed s , let $\hat{J}_s : \Theta \rightarrow \mathbb{R}$ denote the validation estimate of the objective. The learned parameter $\hat{\theta}_s$ and the reference parameter θ^* are evaluated from the same initial distribution, with the same validation seed and particle budget. The reported gap for that seed is

$$|\hat{J}_s(\hat{\theta}_s) - \hat{J}_s(\theta^*)|.$$

We report the mean and standard deviation of these five gaps. Table 4 reports $\hat{J}_s(\hat{\theta}_s)$, while the $J(\theta^*)$ column of Table 1 lists the separately computed reference value. The gap calculation uses the sampled reference $\hat{J}_s(\theta^*)$. On bimodal allocation, both objectives are evaluated analytically.

Reference policies. Two-state control uses the closed-form policy in Appendix E.2.1; advertising uses the optimal feedback of Motte (2021). The linear-quadratic reference follows from the Riccati equations, and the portfolio reference combines the optimal feedback gain with a concave quadratic program for the intercepts. Bimodal allocation solves the scalar equation $J(\theta) = 0$ by bisection. Distribution planning uses gradient ascent on the exact population recursion to obtain a reference value of -0.056991 ; global optimality is not certified. Cybersecurity is evaluated by its objective because no optimum is available.

For Table 1, Transport’s selected scales are $\lambda_\star/4$ on two-state control, linear-quadratic control and the portfolio; $\lambda_\star/2$ on distribution planning; and $\lambda_\star/8$ on bimodal allocation. Transport-Proba selects $\lambda_\star/2$ on both benchmarks.

Budgets. The implemented Transport update uses $C_{\text{sim}} = T(M + n + B)$ simulated transitions (Proposition D.1), using a shared auxiliary batch in finite state space, and every method within a benchmark receives the same simulation budget. In finite state space, the experiments estimate $\hat{D}_1, \dots, \hat{D}_T$ from a single auxiliary batch of n rollouts of length T ; Appendix E.3.4 compares it with fresh batches of length t for each t , the variant covered by Theorem 3.5, which costs $nT(T + 1)/2$ auxiliary transitions. REINFORCE simulates $M + n + B$ trajectories. MF-REINFORCE uses $T(M + B_{\text{MF}}) + n_{\text{MF}}T(T + 1)$ transitions and reads the population flow off the same M particles as Transport.

The split between n and B is selected by the mean-square error of the gradient estimator against the exact gradient, over 120 replications. On the continuous benchmarks, the auxiliary stage gives 128 particles to each of the $2d_\theta$ shifted systems, so $n = 256d_\theta$. The bimodal benchmark gives 4000 particles to each of its two systems. Table 3 collects the allocations.

All training was performed on the Runpod GPU cloud, using an RTX A4000. The code was run in parallel on the 16 available vCPUs, with at most 14 workers at a time.

Table 3: Transport allocations and optimization settings. M , n and B are the population, auxiliary and main sample sizes, and the budget is the resulting number of simulated transitions per update, $T(M + n + B)$, which every method of a benchmark receives. Transport is trained at $\lambda \in \lambda_\star\{1/8, 1/4, 1/2, 1, 2\}$, with $\lambda_\star = B^{-1/4}$; σ is the finite-state simplex randomizer scale and K the number of mixture components.

Benchmark	T	M	n	B	Budget	Updates	Step size	$\lambda_\star$	η	σ	K
Two-state	5	200	12	248	2 300	20 000	10^{-3}	0.252	0.98	0.75	—
Cybersecurity	3	200	51	153	1 212	20 000	10^{-3}	0.284	0.98	0.75	—
Distribution	5	500	280	280	5 300	30 000	10^{-4}	0.244	0.98	0.5	—
Advertising	5	200	65	195	2 300	10 000	10^{-3}	0.268	0.95	1	—
Linear-quadratic	20	150	10 240	111	210 020	10 000	10^{-3}	0.308	0.95	—	1
Portfolio	10	100	5 120	211	54 310	50 000	10^{-2}	0.262	0.95	—	1
Bimodal allocation	1	2 000	8 000	100 000	110 000	1 000	10^{-3}	0.0562	0.04	—	1,2

Perturbation scales. The main scale is taken in $\lambda_\star\{1/8, 1/4, 1/2, 1, 2\}$, with $\lambda_\star = B^{-1/4}$ the order suggested by Theorem 3.5.

On the finite benchmarks, $Q_t = \varphi(U_t)$ with $U_t \sim \mathcal{N}(0, \sigma^2 I_{N-1})$. The auxiliary scale and the randomizer scale are selected jointly in $\eta \in \{0.85, 0.95, 0.98\}$ and $\sigma \in \{0.5, 0.75, 1\}$, before training and at $\lambda_\star$, by the same mean-square error criterion as the budget split, over three warm-started reference policies. This gives $(\eta, \sigma) = (0.98, 0.75)$, $(0.98, 0.75)$, $(0.98, 0.5)$ and $(0.95, 1)$ on two-state control, cybersecurity, distribution planning and targeted advertising. On two-state control, Transport is also trained at $\lambda_\star$ with $\eta \in \{\eta_\star/2, \eta_\star\}$, where $\eta_\star = n^{-1/4}$ suggested by Theorem 3.5, and $2\eta_\star > 1$. MF-REINFORCE uses $\varepsilon = 0.2, 1, 2$ on two-state control, cybersecurity and distribution planning, the best scales reported by Meunier et al. (2026), and $\varepsilon = 1$ on targeted advertising, the best screened scale over $\{0.5, 1, 2\}$ over three seeds at the matched budget.

The continuous benchmarks use $\eta = 0.95$, selected against the exact sensitivities, except the bimodal benchmark, which uses $\eta = 0.04$ (Appendix E.2.7). On the continuous benchmarks, the randomizer (13) uses $A_{j,t} \sim \mathcal{N}(0, I_d)$, $B_{j,t} = e^{G_{j,t}}$ with $G_{j,t} \sim \mathcal{N}(0, 1)$ in dimension one, and, for $K = 2$, the same simplex randomizer for the weights. Linear-quadratic control and the portfolio use $K = 1$; only the bimodal benchmark compares $K = 1$ and $K = 2$. Transport-Proba is trained on the same grid of λ as Transport and has no auxiliary scale.

E.2 Benchmarks

E.2.1 Two-state control

Following Meunier et al. (2026); Gu et al. (2023), states are $\{0, 1\}$ and actions are stay (ST) or move (MV). A move from x succeeds with probability λ_x , where $(\lambda_0, \lambda_1) = (0.5, 0.8)$. Writing $m = \mu(1)$, the running and terminal rewards are $r(x, \mu) = g(x, \mu) = \mathbf{1}_{\{x=1\}} - m^2 - 10|m - 0.4|$. The stationary policy uses two logits, $\pi^\theta(\text{MV} | x) = (1 + e^{-\theta_x})^{-1}$, initialized at zero. The horizon is $T = 5$. Training samples $m_0 \sim \mathcal{U}([0.1, 0.9])$, and validation uses $m_0 = 0.8$. The optimal policy stays with probabilities $(0.2, 0.25)$ in states $(0, 1)$; it sends every initial law to $(0.6, 0.4)$ in one step.

E.2.2 Cybersecurity

We use the model of Meunier et al. (2026); Kolokoltsov and Bensoussan (2016); Carmona et al. (2023), with states DI, DS, UI, US denoting defended/undefended and infected/susceptible computers. Action $a \in \{0, 1\}$ switches protection at rate $0.8a$. Infection rates are

$$\iota_D(\mu) = 0.24 + 0.4\mu(\text{DI}) + 0.4\mu(\text{UI}), \quad \iota_U(\mu) = 0.18 + 0.3\mu(\text{DI}) + 0.3\mu(\text{UI}).$$

Recovery rates are 0.5 for defended and 0.4 for undefended computers. These rates define a generator $Q^{\mu,a}$, and the transition kernel is $\exp(0.2Q^{\mu,a})$. The reward at time t is $-0.2\gamma^t[0.3\mathbf{1}_{\{\text{DI}, \text{DS}\}}(x) + 0.5\mathbf{1}_{\{\text{DI}, \text{UI}\}}(x)]$, with $\gamma = 0.5^{0.2}$. The policy is a tanh MLP with two hidden layers of width 32, taking (t, μ) as input. Training uses $T = 3$ and $\mu_0 \sim \text{Dirichlet}(\mathbf{1}_4)$; validation uses the uniform law and $T = 50$.

E.2.3 Distribution planning

Following Meunier et al. (2026); Carmona et al. (2023), states lie on $\mathbb{Z}/10\mathbb{Z}$, actions are $a \in \{-1, 0, 1\}$, and $X_{t+1} = X_t + a \pmod{10}$. The rewards are $r(x, a, \mu) = -0.01|a| - \|\mu - \mu_{\text{target}}\|^2$ and $g(x, \mu) = -\|\mu - \mu_{\text{target}}\|^2$, with

$$\mu_{\text{target}} = (0.02, 0.04, 0.09, 0.16, 0.19, 0.19, 0.16, 0.09, 0.04, 0.02).$$

The policy is a tanh MLP with two hidden layers of width 64, taking (t, μ) as input. The horizon is $T = 5$. Training samples $\mu_0 \sim \text{Dirichlet}(\mathbf{1}_{10})$; validation starts from the uniform law.

E.2.4 Targeted advertising

Adapting Motte (2021), let $x \in \{0, 1\}$ indicate customer status and $a \in \{0, 1\}$ indicate advertising. With $p = \mu(1)$,

$$P(1 \mid x, a, \mu) = \min\{p + 0.2a, 1\}, \quad r_t(x, a, \mu) = 0.5^t(x - 0.15a), \quad g = 0.$$

The policy is a tanh MLP taking (t, μ) as input. The horizon is $T = 5$. Training samples $p_0 \sim \mathcal{U}([0.05, 0.95])$; validation uses $p_0 = 0.5$.

E.2.5 Linear-quadratic control

For $\bar{x}(m) = \int x m(dx)$, the policy and dynamics are

$$A_t \mid X_t, m_t \sim \mathcal{N}(\theta_t^1 X_t + \theta_t^2 \bar{x}(m_t), \tau^2), \quad X_{t+1} = aX_t + bA_t + c\bar{x}(m_t) + \sigma\varepsilon_{t+1},$$

with independent standard Gaussian noise. The objective is $J = -\mathbb{E}[\sum_{t < T} (qX_t^2 + rA_t^2 + \gamma\bar{x}(\mu_t^\theta)^2) + q_T X_T^2 + \gamma_T \bar{x}(\mu_T^\theta)^2]$, with $T = 20$, $a = 0.9$, $b = 0.5$, $c = 0.05$, $q = q_T = 1$, $r = 0.1$, $\gamma = \gamma_T = 5$, $\tau = 0.2$ and $\sigma = 0.1$. Initially $X_0 = 1 + V$, with $V \sim \text{Laplace}(0, 1/(2\sqrt{2}))$, so the population law is not Gaussian, and $\theta = 0$. The optimal policy is given by the Riccati equations of the mean-field linear-quadratic problem. Transport uses $K = 1$.

E.2.6 Mean-variance portfolio

The wealth dynamics and Gaussian policy are

$$X_{t+1} = X_t + \alpha_t R_{t+1}, \quad \alpha_t = k_t(X_t - \bar{x}(m_t)) + \ell_t + \tau\varepsilon_t,$$

with $r(x, a, m) = -\gamma\bar{x}(m)^2$ and $g(x, m) = x - \chi(x - \bar{x}(m))^2$. Returns $R_{t+1} \sim \mathcal{N}(0.02, 0.08^2)$ are independent, and independent of the exploration noises. We use $T = 10$, $X_0 \sim \mathcal{N}(1, 0.04)$, $\chi = 10$, $\gamma = 2$ and $\tau = 0.2$. The optimal feedback gain is $k_t^* = -\bar{r}/(\bar{r}^2 + \sigma_R^2)$, with $\bar{r} = 0.02$ and $\sigma_R = 0.08$, and the optimal (ℓ_t^*) solves a T -dimensional concave quadratic program. Transport uses $K = 1$.

E.2.7 Bimodal population allocation

This benchmark tests the effect of the number of Gaussian components. The population has two modes, and its mean and variance do not depend on the policy parameter. A single Gaussian therefore loses the information needed to optimize the objective.

Model and objective. Take $T = 1$, $\mathcal{X} = \mathcal{A} = \mathbb{R}$, $\Theta = [0.75, 0.95]$, and $X_0 \sim \frac{1}{2}\mathcal{N}(-1, 0.01) + \frac{1}{2}\mathcal{N}(1, 0.01)$. The policy and transition kernel are

$$\pi^\theta(\cdot \mid x, m) = \mathcal{N}(\theta x, 0.99 - 1.01\theta^2), \quad P(\cdot \mid x, m, a) = \mathcal{N}(a, 0.01).$$

The policy variance is positive throughout Θ . The terminal law is

$$\mu_1^\theta = \frac{1}{2}\mathcal{N}(-\theta, 1 - \theta^2) + \frac{1}{2}\mathcal{N}(\theta, 1 - \theta^2).$$

Every such law is bimodal, with mean zero and variance one.

Set $r_0 \equiv 0$ and define the terminal reward $g : \mathbb{R} \times \mathcal{P}_2(\mathbb{R}) \rightarrow \mathbb{R}$ by

$$g(x, m) = -100 \left(\int_{\mathbb{R}} \exp \left[-\frac{y^2}{2(0.2)^2} \right] m(dy) - 0.1 \right)^2.$$

The integral measures how concentrated the population is near zero, which cannot be determined from its mean and variance alone. Since the reward depends only on the population law, the expected REINFORCE gradient is zero.

Exact validation. Integrating the terminal reward against the Gaussian mixture gives

$$J(\theta) = -100 \left(\frac{0.2}{\sqrt{1.04 - \theta^2}} \exp \left[-\frac{\theta^2}{2(1.04 - \theta^2)} \right] - 0.1 \right)^2.$$

The unique maximizer is $\theta^* \simeq 0.868437$, with $J(\theta^*) = 0$. We initialize at $\theta = 0.8$ and use the exact optimality gap $-J(\theta)$ for validation.

Effect of the mixture size. We compare $K = 1$ and $K = 2$ using maximum-likelihood mixture fitting. At the population level, the fitted single Gaussian matches the mean and variance of μ_1^θ . Since these are zero and one for every θ , this fit is always $\mathcal{N}(0, 1)$. Evaluating the reward on this fitted distribution gives

$$J_1(\theta) = -100 \left(\int_{\mathbb{R}} e^{-y^2/(2(0.2)^2)} \mathcal{N}(0, 1)(dy) - 0.1 \right)^2 = -100 \left(\frac{0.2}{\sqrt{1.04}} - 0.1 \right)^2 \simeq -0.923831.$$

Thus J_1 is constant: changing θ changes the actual population, but not its single-Gaussian approximation. The $K = 1$ projected objective therefore provides no gradient for learning. With $K = 2$, the terminal law is represented exactly, so $J_2 = J$.

E.3 Additional results

E.3.1 Learning curves and population dynamics

The population trajectories in Figure 1 identify the effects missed by REINFORCE. In distribution planning, the reward charges for movement while the benefit depends on the resulting population distribution. REINFORCE consequently remains close to the initial law. Transport learns a terminal law closer to the target, whereas MF-REINFORCE concentrates too much mass near its center. In the portfolio, REINFORCE increases mean wealth despite the running penalty on its square. The population correction substantially reduces this excess.

The advantage is smaller on the other finite-state benchmarks (Figure 3). Cybersecurity gives similar final objectives for Transport and REINFORCE. On advertising, both population-corrected methods improve on REINFORCE, which learns to avoid advertising. However, neither recovers the optimal policy of advertising until the customer share reaches 0.9. MF-REINFORCE attains the higher mean objective, with substantial variation across seeds.

Table 4: Final validation objective on the two benchmarks absent from Table 1 (mean $\pm$ standard deviation over five seeds; larger is better). Transport uses the scale with the highest mean final objective. Reference values are computed from the benchmark models; gap evaluation is described in Appendix E.1.

Benchmark	T	Reference	REINFORCE	MF-REINFORCE	Transport	(λ, η)
Cybersecurity	3	—	-0.4887 ± 0.0007	-0.4986 ± 0.0142	-0.4883 ± 0.0010	$(0.568666, 0.98)$
Advertising	5	1.0188	0.9681 ± 0.0036	0.9921 ± 0.0176	0.9821 ± 0.0159	$(0.133802, 0.95)$

E.3.2 Perturbation scale and mixture size

The scale sweeps in Figure 2 and Tables 5–6 show two competing effects. Larger perturbations can make the perturbed objective easier to optimize, while changing which policy it favors. Advertising gives a clear example: at the largest tested scale, the perturbed optimization error is zero to the reported precision, but the original gap remains 0.05. Further optimization of that same objective would retain this loss.

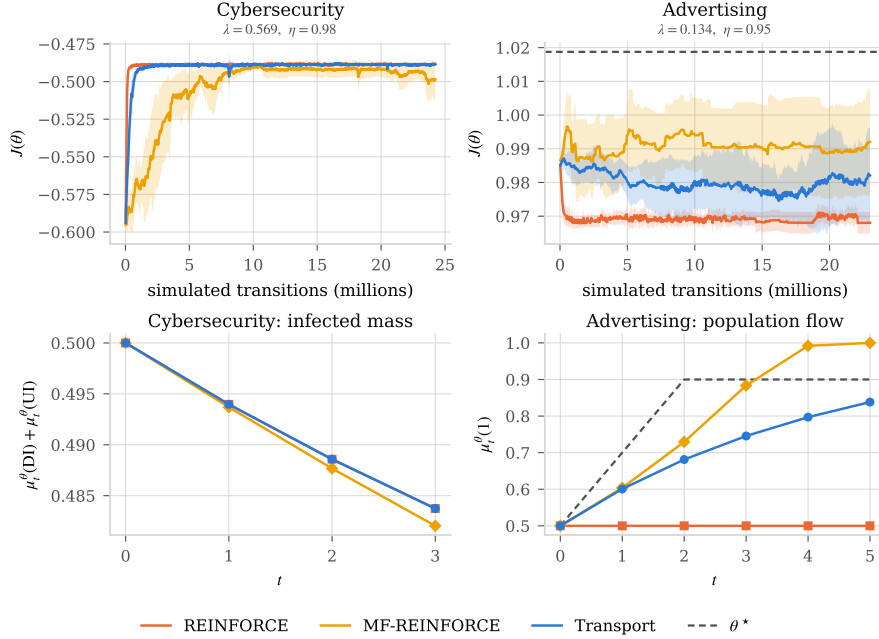


Figure 3: Cybersecurity and advertising. Top: validation objective against simulated transitions (five paired seeds). Bottom: infected mass and customer share under the final policy of the first seed. Cybersecurity has no known optimum.

Bimodal allocation separates the influence of K from that of λ . With $K = 1$, every population law is fitted by $\mathcal{N}(0, 1)$, explaining the nearly identical outcomes across scales. With $K = 2$, the fitted law retains the policy dependence, and optimization of the perturbed objective is accurate to 10^{-4} . The increasing gap with λ then follows the displacement of the learned parameter from θ^* .

Table 6 reports the change in the maximum objective value, the error in optimizing the perturbed objective, and the final gap under J . Here θ_λ^* is the perturbed reference optimizer in the benchmark’s control class; the distribution-planning reference is obtained numerically. The first column includes changes in the level of the objective, so the final gap is the relevant measure of policy quality. These quantities use the population equations, whereas Table 1 uses the validation procedure in Appendix E.1.

Table 5: Bimodal allocation: final gap and policy parameter, initialized at $\theta = 0.8$ (exact gap 0.1772). Gaps are Monte Carlo validation estimates. The optimum is $\theta^* = 0.868437$. For $K = 1$, the projected objective is constant, $J_1 = -0.923831$; for $K = 2$, the representation is exact.

Estimator	K	λ	$ J(\hat{\theta}) - J(\theta^*) $	$\hat{\theta}$
REINFORCE	—	—	0.1752 ± 0.0087	0.8000 ± 0.0000
Transport	1	0.00702927	0.1425 ± 0.1239	0.8103 ± 0.0339
Transport	1	0.0140585	0.1425 ± 0.1239	0.8103 ± 0.0339
Transport	1	0.0281171	0.1425 ± 0.1240	0.8103 ± 0.0339
Transport	1	0.0562341	0.1425 ± 0.1241	0.8102 ± 0.0339
Transport	1	0.112468	0.1426 ± 0.1245	0.8102 ± 0.0340
Transport	2	0.00702927	0.0020 ± 0.0009	0.8737 ± 0.0007
Transport	2	0.0140585	0.0068 ± 0.0017	0.8782 ± 0.0007
Transport	2	0.0281171	0.0244 ± 0.0030	0.8865 ± 0.0007
Transport	2	0.0562341	0.0903 ± 0.0053	0.9015 ± 0.0007
Transport	2	0.112468	0.3563 ± 0.0097	0.9298 ± 0.0006

Table 6: Effect of perturbation on objective values and optimization. *Value change*: $|J^\lambda(\theta_\lambda^*) - J(\theta^*)|$; *Opt. error*: $|J^\lambda(\hat{\theta}_\lambda) - J^\lambda(\theta_\lambda^*)|$; *Final gap*: $|J(\hat{\theta}_\lambda) - J(\theta^*)|$. Objectives are evaluated from the population equations; dispersion is over five seeds. MF-REINFORCE uses its own perturbed objective and scale ε .

Benchmark	Estimator	Scale	Value change	Opt. error	Final gap
Two-state	Transport	0.0315	0.1191	0.3730 ± 0.0617	0.5452 ± 0.1134
		0.063	0.2394	0.1032 ± 0.0346	0.4427 ± 0.0895
		0.126	0.4848	0.0153 ± 0.0085	0.5968 ± 0.0782
		0.252	0.9970	0.0044 ± 0.0027	1.4465 ± 0.0889
		0.504	2.1475	0.0074 ± 0.0028	3.6452 ± 0.0809
	MF-REINFORCE	0.2	2.6476	0.1892 ± 0.0445	0.7098 ± 0.0915
REINFORCE	—	—	—	29.0738 ± 0.0000	
Distribution	Transport	0.0306	0.0009	0.0371 ± 0.0010	0.0369 ± 0.0008
		0.0611	0.0021	0.0320 ± 0.0009	0.0315 ± 0.0006
		0.122	0.0054	0.0252 ± 0.0013	0.0255 ± 0.0010
		0.244	0.0159	0.0177 ± 0.0014	0.0298 ± 0.0011
		0.489	0.0603	0.0173 ± 0.0040	0.0477 ± 0.0040
	MF-REINFORCE	2	1.7877	0.0790 ± 0.0099	0.1178 ± 0.0243
REINFORCE	—	—	—	0.2046 ± 0.0000	
Advertising	Transport	0.0335	0.0044	0.0421 ± 0.0141	0.0451 ± 0.0150
		0.0669	0.0088	0.0338 ± 0.0153	0.0392 ± 0.0176
		0.134	0.0176	0.0256 ± 0.0109	0.0379 ± 0.0162
		0.268	0.0353	0.0165 ± 0.0001	0.0495 ± 0.0007
		0.535	0.0537	0.0000 ± 0.0000	0.0500 ± 0.0000
	MF-REINFORCE	1	0.0331	0.0196 ± 0.0160	0.0274 ± 0.0172
REINFORCE	—	—	—	0.0500 ± 0.0000	
Linear-quadratic	Transport	0.0385	0.2282	0.0344 ± 0.0024	0.0370 ± 0.0026
		0.077	0.1299	0.0160 ± 0.0010	0.0196 ± 0.0012
		0.154	1.0465	0.0166 ± 0.0010	0.0214 ± 0.0019
		0.308	7.3214	0.0402 ± 0.0010	0.0389 ± 0.0026
		0.616	35.5448	0.1568 ± 0.0077	0.1389 ± 0.0309
	Transport-Proba	0.0385	0.2282	0.1396 ± 0.0206	0.1435 ± 0.0207
		0.077	0.1299	0.0434 ± 0.0038	0.0471 ± 0.0042
		0.154	1.0465	0.0431 ± 0.0029	0.0469 ± 0.0029
		0.308	7.3214	0.0909 ± 0.0059	0.0868 ± 0.0029
		0.616	35.5448	0.2294 ± 0.0305	0.2644 ± 0.0697
	REINFORCE	—	—	—	0.2163 ± 0.0009
	Portfolio	Transport	0.0328	0.4970	0.6392 ± 0.1421
0.0656			0.9687	0.4231 ± 0.0796	0.6003 ± 0.1769
0.131			1.7597	0.2543 ± 0.0556	0.7145 ± 0.1265
0.262			2.4207	0.1009 ± 0.0329	1.2568 ± 0.0613
0.525			0.5683	0.1134 ± 0.0654	2.4766 ± 0.1431
Transport-Proba		0.0328	0.4970	1.6853 ± 0.4038	1.8498 ± 0.3817
		0.0656	0.9687	1.3225 ± 0.4507	1.3289 ± 0.4980
		0.131	1.7597	0.7035 ± 0.1708	0.7190 ± 0.1738
		0.262	2.4207	0.5543 ± 0.0648	1.2154 ± 0.1074
		0.525	0.5683	0.1902 ± 0.0333	2.8870 ± 0.4371
REINFORCE	—	—	—	6.8874 ± 0.0036	
Bimodal allocation	Transport ($K = 2$)	0.00703	0.0010	0.0000 ± 0.0000	0.0019 ± 0.0005
		0.0141	0.0031	0.0000 ± 0.0000	0.0066 ± 0.0010
		0.0281	0.0098	0.0000 ± 0.0000	0.0239 ± 0.0020
		0.0562	0.0303	0.0000 ± 0.0000	0.0888 ± 0.0041
		0.112	0.0988	0.0001 ± 0.0001	0.3517 ± 0.0078
	REINFORCE	—	—	—	0.1772 ± 0.0000

E.3.3 Sensitivity estimator and runtime

Transport-Proba replaces centered differences of fitted coordinates with a likelihood-ratio estimate of moment sensitivities. On linear-quadratic control, centered differences give a smaller final gap at every tested scale

Table 7: Simulator transitions per update and wall-clock time per complete training run. Each benchmark uses the same simulator budget across methods.

Benchmark	Estimator	Transitions/update	Seconds/run	Ratio
Two-state	REINFORCE	2 300	70	1.0×
	MF-REINFORCE	2 300	387	5.5×
	Transport	2 300	220	3.1×
Cybersecurity	REINFORCE	1 212	167	1.0×
	MF-REINFORCE	1 212	595	3.6×
	Transport	1 212	393	2.4×
Distribution	REINFORCE	5 300	169	1.0×
	MF-REINFORCE	5 300	2 124	12.5×
	Transport	5 300	2 706	16.0×
Advertising	REINFORCE	2 300	47	1.0×
	MF-REINFORCE	2 300	298	6.3×
	Transport	2 300	168	3.6×
Linear-quadratic	REINFORCE	210 020	349	1.0×
	Transport	210 020	719	2.1×
	Transport-Proba	210 020	2 577	7.4×
Portfolio	REINFORCE	54 310	799	1.0×
	Transport	54 310	1 796	2.2×
	Transport-Proba	54 310	1 888	2.4×
Bimodal allocation	REINFORCE	110 000	19	1.0×
	Transport	110 000	192	10.0×

(Table 6) and a shorter runtime. The portfolio comparison is closer relative to the variation across seeds. Both estimators use $K = 1$ here: the Gaussian fit retains the relevant population moments even when the true state law is non-Gaussian.

Table 7 quantifies the computational cost beyond simulation. Transport is slower than REINFORCE because each update also estimates sensitivities and evaluates population scores. The overhead is particularly large on distribution planning and bimodal allocation. Thus the improvement at matched simulator budgets comes with an additional arithmetic cost; it does not imply the same gain per second.

E.3.4 Auxiliary scale and reuse of trajectories

The auxiliary-scale sweep uses exact population flows, so its sensitivity error comes entirely from the auxiliary estimator. At small η , the factor $(1 - \eta)/\eta$ amplifies noise in the earlier sensitivities used by the recursion. The resulting error can grow substantially across time steps, explaining the large finite-budget errors near $\eta = n^{-1/4}$ in Table 8. On two-state control, the dynamics and policy are independent of the population argument, and larger radii mainly reduce this variance amplification.

Advertising shows the cost of stronger perturbations: sensitivity error is minimized at an intermediate scale, while gradient error is flatter over the larger scales. The gradient weights the sensitivities by returns and population scores, and also contains online sampling noise. Its MSE is therefore the relevant criterion for selecting η , as in the protocol.

Shared and fresh auxiliary batches give comparable errors in this sweep. Reusing a batch reduces the auxiliary simulation cost from $nT(T + 1)/2$ to nT , but introduces dependence between successive sensitivity estimates. The theorem analyzes fresh batches; Table 8 assesses the shared-batch implementation empirically. In training, two-state control at λ_* reaches final gaps of 1.93 ± 0.15 at $\eta = \eta_*/2$, and between 1.34 ± 0.13 and 1.36 ± 0.13 at $\eta \in \{\eta_*, 0.85, 0.95, 0.98\}$ (five seeds): only the smallest scale degrades the final performance.

E.3.5 Perturbation of the law and gradient

At a fixed policy, the finite-state total-variation distance is proportional to λ by construction. For a single Gaussian, the $\mathcal{W}_2$ distance also scales approximately linearly, through changes in the mean and standard deviation.

Table 8: Relative sensitivity MSE $\sum_t \mathbb{E}[\|\hat{D}_t - D_t\|_F^2] / \sum_t \|D_t\|_F^2$ (200 replications) and gradient MSE $\mathbb{E}[\|\hat{G} - \nabla_{\theta} J\|^2] / \|\nabla_{\theta} J\|^2$ (100 replications). The population flow is exact; the policy is fixed at initialization and $\lambda = B^{-1/4}$. The first scales are $\eta_*/2$, η_* and, when below one, $2\eta_*$, with $\eta_* = n^{-1/4}$.

Benchmark	n	η	Sensitivity error		Gradient error
			shared batch	fresh batches	shared batch
Two-state	12	0.269	$4.33 \cdot 10^4$	$3.93 \cdot 10^4$	$8.46 \cdot 10^3$
		0.537	31.6	32.2	15.4
		0.85	1.26	1.31	2.91
		0.95	1.05	1.07	2.75
		0.98	1.04	1.06	2.75
Cybersecurity	51	0.187	$1.13 \cdot 10^4$	$1.76 \cdot 10^4$	$1.39 \cdot 10^6$
		0.374	312	448	$4.17 \cdot 10^4$
		0.748	8.6	9.14	$1.44 \cdot 10^3$
		0.85	6.29	6.67	$1.00 \cdot 10^3$
		0.95	5.61	5.97	836
		0.98	5.58	5.92	812
Distribution	280	0.122	$2.56 \cdot 10^{10}$	$3.09 \cdot 10^{10}$	$2.85 \cdot 10^{10}$
		0.244	$3.04 \cdot 10^7$	$3.75 \cdot 10^7$	$3.60 \cdot 10^7$
		0.489	$6.45 \cdot 10^3$	$8.04 \cdot 10^3$	$9.71 \cdot 10^3$
		0.85	0.597	0.603	1.37
		0.95	0.178	0.177	0.487
		0.98	0.155	0.154	0.443
Advertising	65	0.176	20.2	12.2	$1.43 \cdot 10^4$
		0.352	1.57	1.15	$1.51 \cdot 10^3$
		0.704	0.804	0.72	496
		0.85	0.881	0.793	449
		0.95	0.913	0.866	465
		0.98	0.922	0.892	468

The two-component case additionally changes mixture weights and redistributes mass between modes, producing different behavior over the tested scales. The gradient perturbation is approximately linear on the benchmarks where it is evaluated (Figure 4).

The policy is fixed after 200 updates. Distances use 2000 randomizer draws in finite states and 200 per time step in continuous states; the latter use mixture quantiles to compute root-mean-square $\mathcal{W}_2$.

E.3.6 Bias and mean-square error

We vary one quantity at a time to distinguish sampling error from perturbation error. The inverse-sample-size slopes in M and B are consistent with Monte Carlo averaging. Sensitivity error shows the same behavior in n until systematic error becomes dominant. On advertising, the error only decreases from 0.47 to 0.38 between $n = 130$ and $n = 260$, giving a fitted slope of -0.51 . This is consistent with the error remaining in the perturbed sensitivity at fixed η .

Figure 6 isolates the continuous-state sensitivity calculation. With exact flows, centered differences have squared error of order η^4 . With fitted particle flows, dividing the endpoint fitting errors by 2η increases their contribution as η decreases. The observed curves reflect the sum of these truncation and fitting errors, as in (45). At the benchmark budgets, the fitting term dominates at every tested η : the total error decreases as η^{-2} up to $\eta = 0.95$ and remains well above the truncation error.

The continuous-state bias sweep uses exact flows and sensitivities, $B = 4096$ and 1024 replications. The bias is the norm of an average over these replications, so it is only known up to its Monte Carlo standard error. The slope in λ is therefore fitted in two steps. First, we discard every scale at which the estimated bias is smaller than twice its standard error, since such a value cannot be distinguished from noise; this removes $\lambda = 0.025$ on the portfolio. Second, among the remaining scales, we keep the smaller half, because at large λ the bias can become comparable to the gradient itself and stop growing linearly. The fit thus uses $\lambda \in \{0.025, 0.05, 0.1\}$ on linear-quadratic control and $\lambda \in \{0.05, 0.1, 0.2\}$ on the portfolio. Finite-state bias is computed exactly, so only

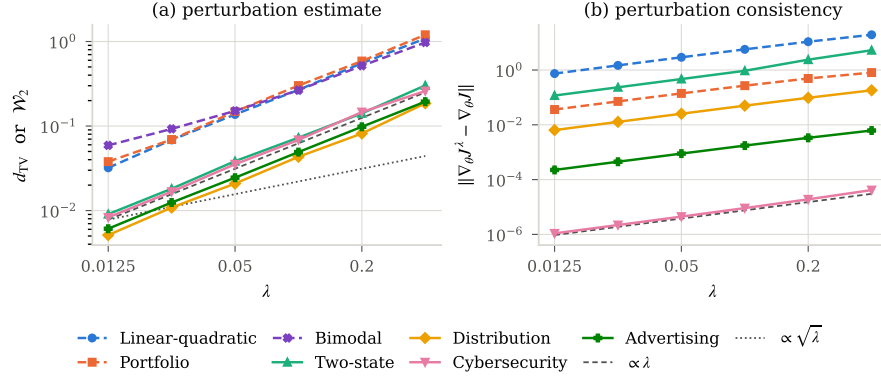


Figure 4: Perturbation at a fixed policy on all benchmarks. Left: total variation (finite states, solid) and W_2 (continuous states, dashed). Right: $\|\nabla_{\theta} J^{\lambda} - \nabla_{\theta} J\|$. Grey reference lines have slopes 1 and 1/2.

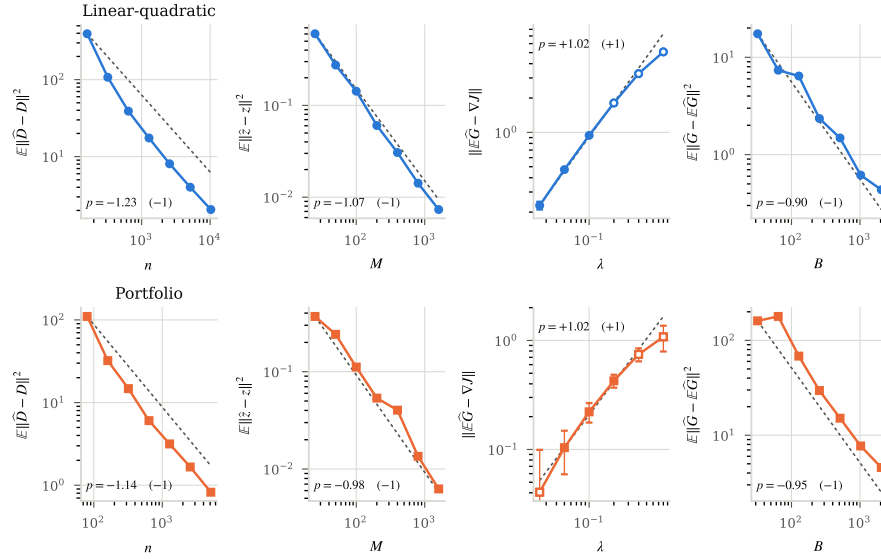


Figure 5: Continuous-state estimator errors as one quantity is varied. Dashed lines show predicted powers. Bias error bars show the Monte Carlo standard error; hollow markers are the scales excluded from the slope fit.

the second step applies, and the fit uses the three smallest scales, $\lambda \in \{0.025, 0.05, 0.1\}$. Saturation is visible on two-state control, where the bias is 0.32, 0.87 and 1.92 at these scales and then stays below the gradient norm 2.06. The continuous-state policies are $\theta = \theta^*/2$, and the finite-state policies are fixed at initialization. Full fitted slopes are reported in Table 9.

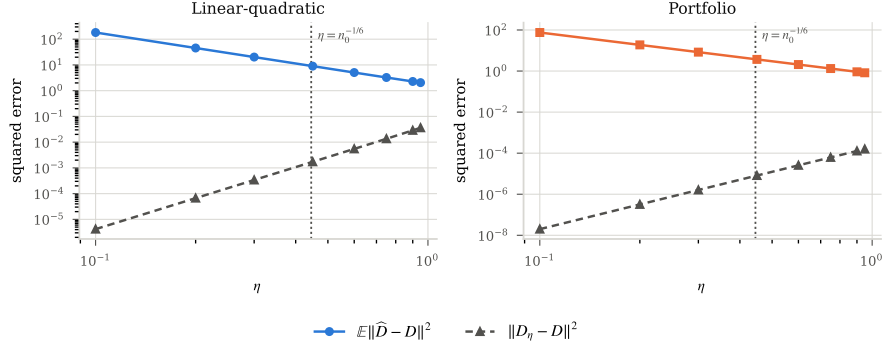


Figure 6: Continuous-state sensitivity error against η , using fitted particle flows and exact population flows. Exact flows isolate truncation error. The vertical line marks $\eta = n_0^{-1/6}$, with $n_0 = n/(2d_\theta) = 128$.

Table 9: Log-log slopes for all benchmarks. Each row varies one quantity, holding the remaining quantities at the benchmark allocation, except that the continuous-state λ sweep uses $B = 4096$. The λ slopes are fitted on the three smallest scales at which the bias is resolved, as described above.

Quantity	Swept	LQ	Portfolio	Two-state	Cyber.	Planning	Advert.	p_0
$\mathbb{E}\ \hat{D} - D\ ^2$	η (exact flows)	+4.02	+3.99	—	—	—	—	+4
$\mathbb{E}\ \hat{D} - D\ ^2$	n	-1.23	-1.14	-1.06	-1.00	-1.02	-0.51	-1
$\ \mathbb{E}\hat{G} - \nabla J\ $	λ	+1.02	+1.02	+1.28	+1.00	+1.00	+1.00	+1
$\mathbb{E}\ \hat{G} - \mathbb{E}\hat{G}\ ^2$	B	-0.90	-0.95	-0.99	-1.01	-1.03	-0.92	-1
$\mathbb{E}\ \hat{z} - z\ ^2$	M	-1.07	-0.98	-1.02	-1.02	-1.00	-1.06	-1